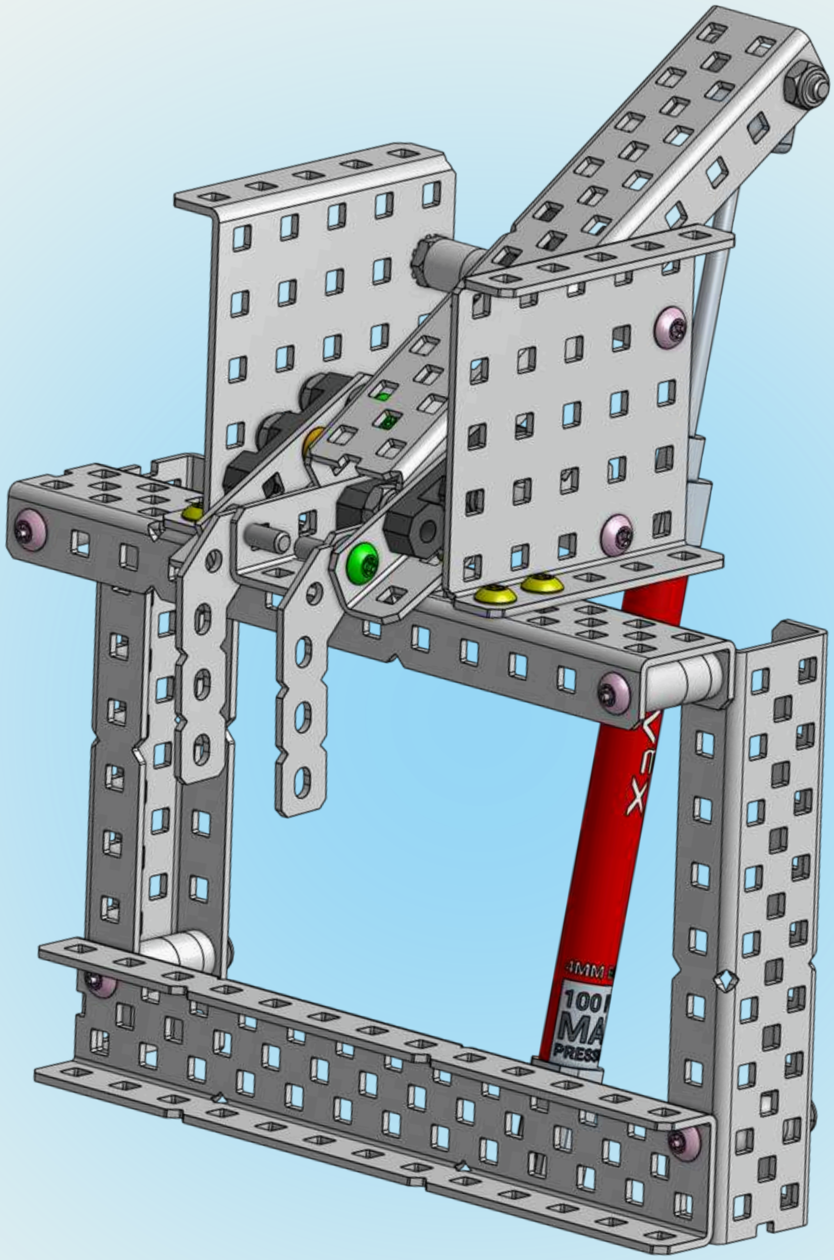
2496N

LOCKING CLAMP

001



EVANA, TOMMY, ALEXIS, PEYTON, AYAAN

IRVINE, C.A.

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WHAT IS A LOCKING CLAMP?

003

A VEX locking clamp is a **mechanism** designed to **hold game elements** with a grip that makes removing the element from the clamp **extremely difficult**.

The most effective way of constructing this subsystem is with a focus on **levers**, precise **geometry**, and **pneumatics**.

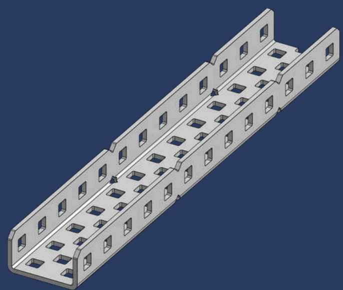
The system uses compressed air to actuate **pivoting** levers, resulting in the final lever **securing** the game element.

DID YOU KNOW?

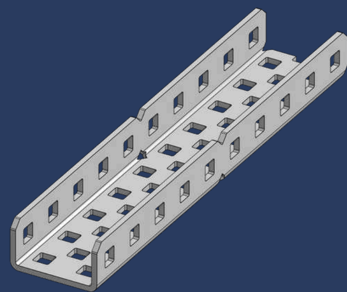
Many teams have used this subsystem in the ‘21-’22 Tipping Point season and this year’s game: High Stakes!

PARTS LIST

004



2x
1x2x1x13 Aluminum C-Channel



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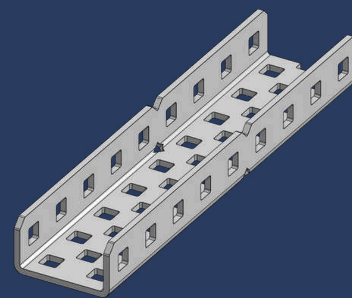

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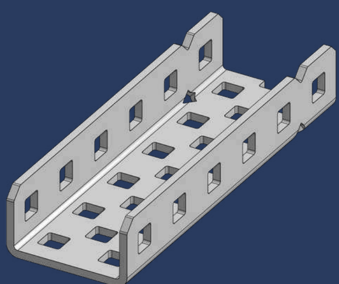
1x2x1x9 Aluminum C-Channel



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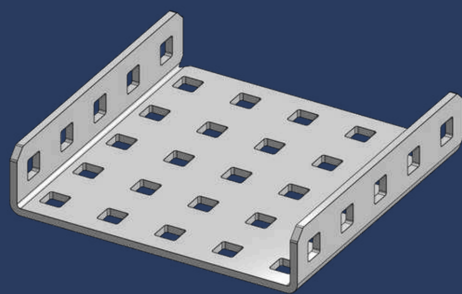
1x2x1x6 Aluminum C-Channel



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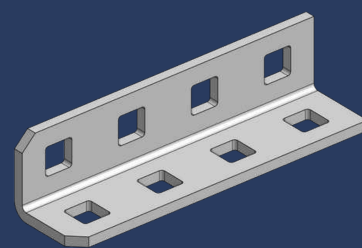
1x5x1x5 Aluminum C-Channel



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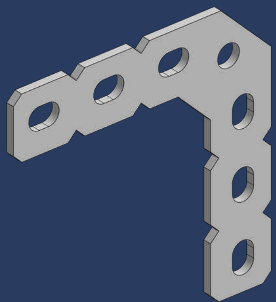
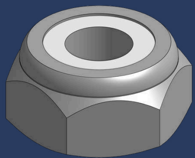
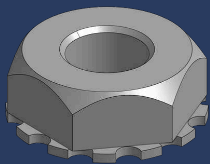
1x1x4 Aluminum Angle



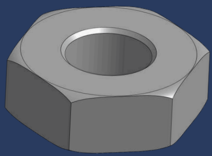
2x
90 Degree Large Flat Gusset



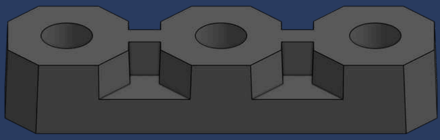
21x
Thin Nylock Nut



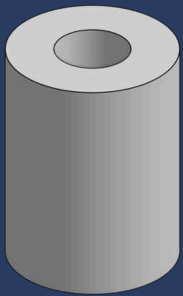
6x
Keps Nut



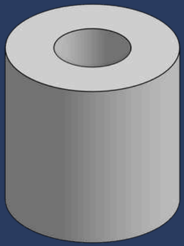
1x
Hex Nut



6x
Flat Bearing



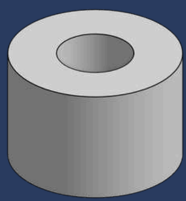
5x
1/2" x 3/8" OD x #8 Nylon Spacer



4x
1/4" x 3/8" OD x #8 Nylon Spacer

PARTS LIST

005



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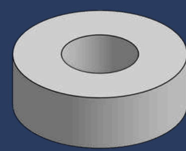
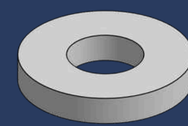
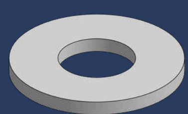
3/8" x 3/8" OD x #8 Nylon Spacer



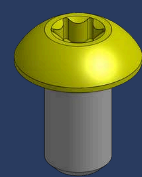
10x
1/8" x 3/8" OD x #8 Nylon Spacer



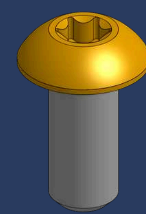
10x
1/16" x 3/8" OD x #8 Nylon Spacer



1x
1/32" x 3/8" OD x #8 Nylon Spacer



4x
#8-32 x 1/4" Star Drive Screw



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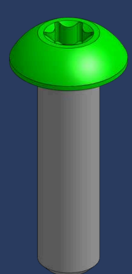
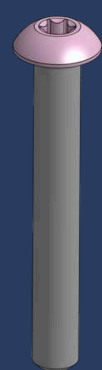
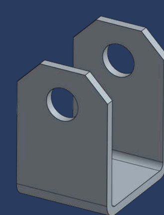
#8-32 x 3/8" Star Drive Screw



4x
#8-32 x 5/8" Star Drive Screw



9x
#8-32 x 1-1/84" Star Drive Screw

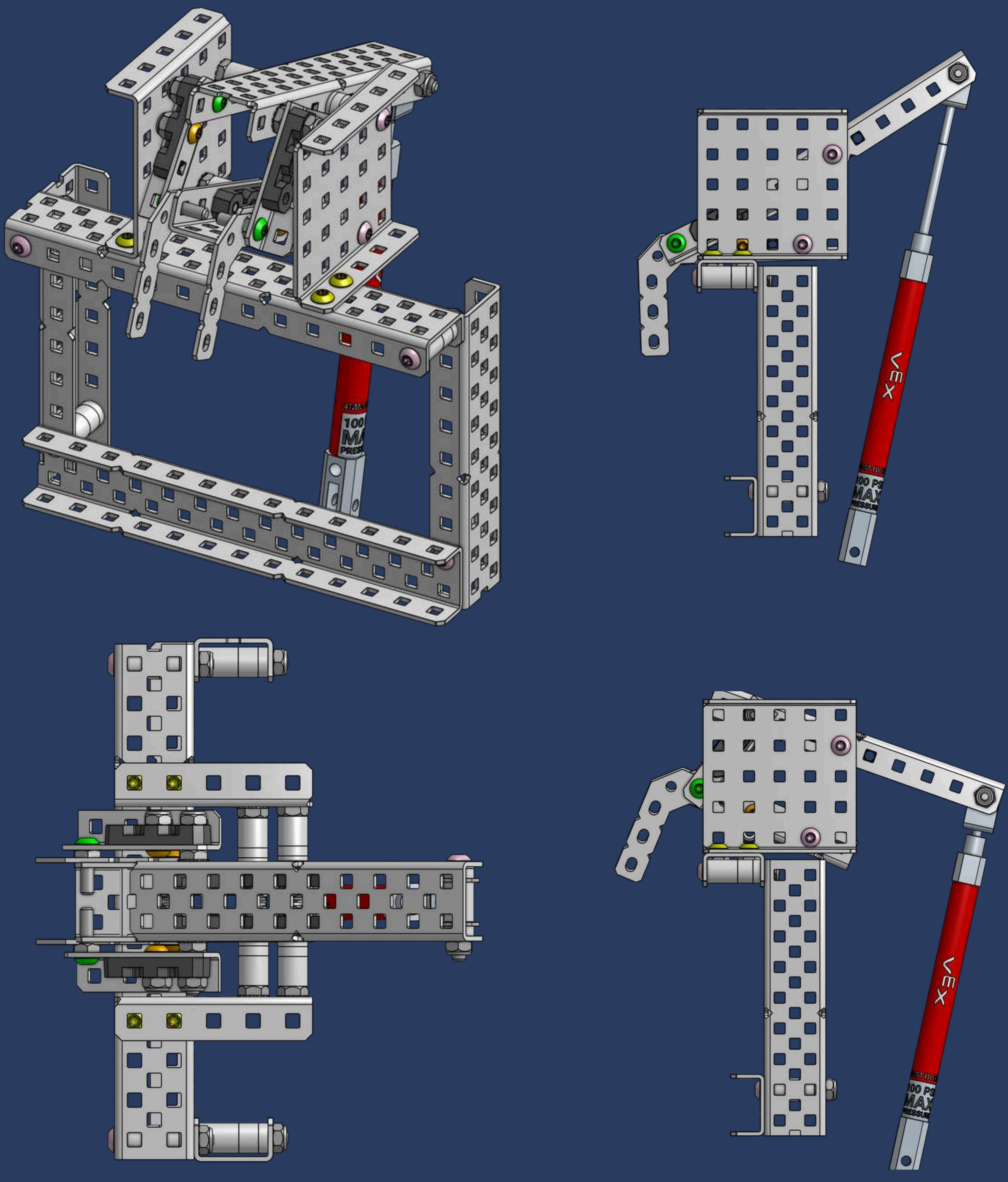


1x
Cylinder Mount

1x
75mm Stroke Pneumatic Cylinder

LOCKING CLAMP DIAGRAMS

006



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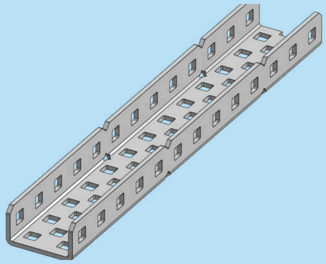
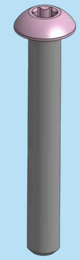
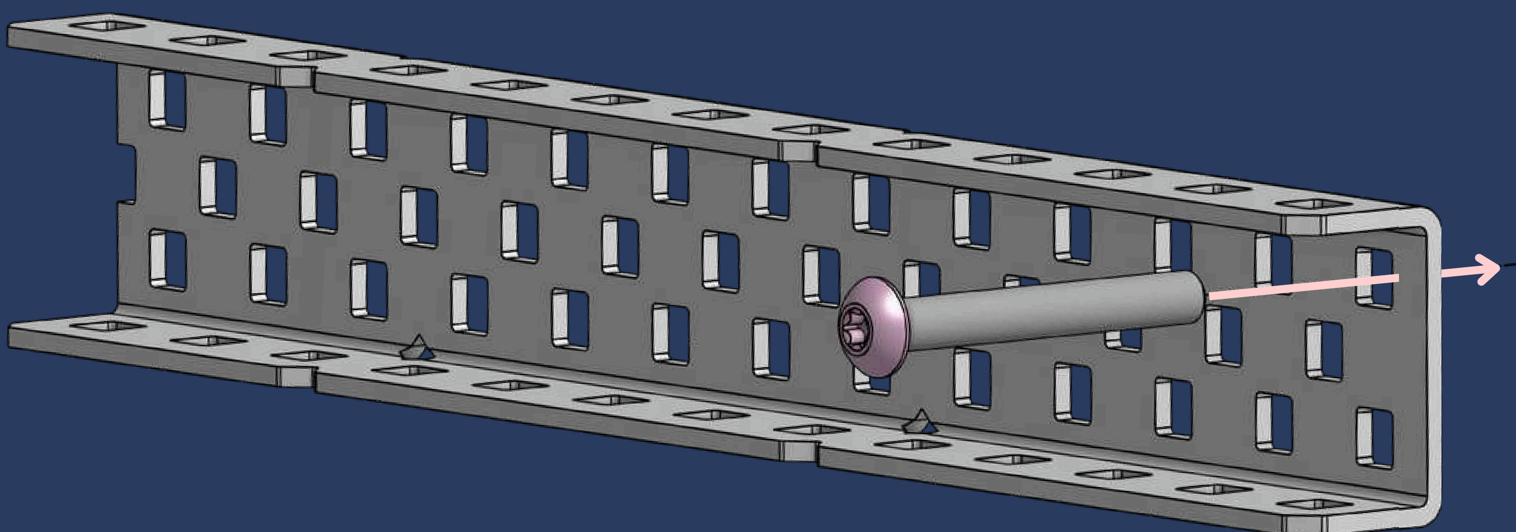


1x
1x2x1x13 Aluminum C-Channel



1x
#8-32 x 1-1/84" Star Drive Screw



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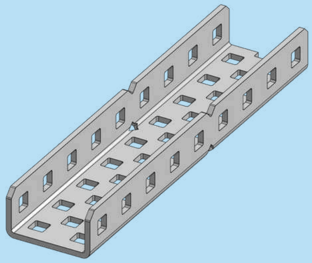
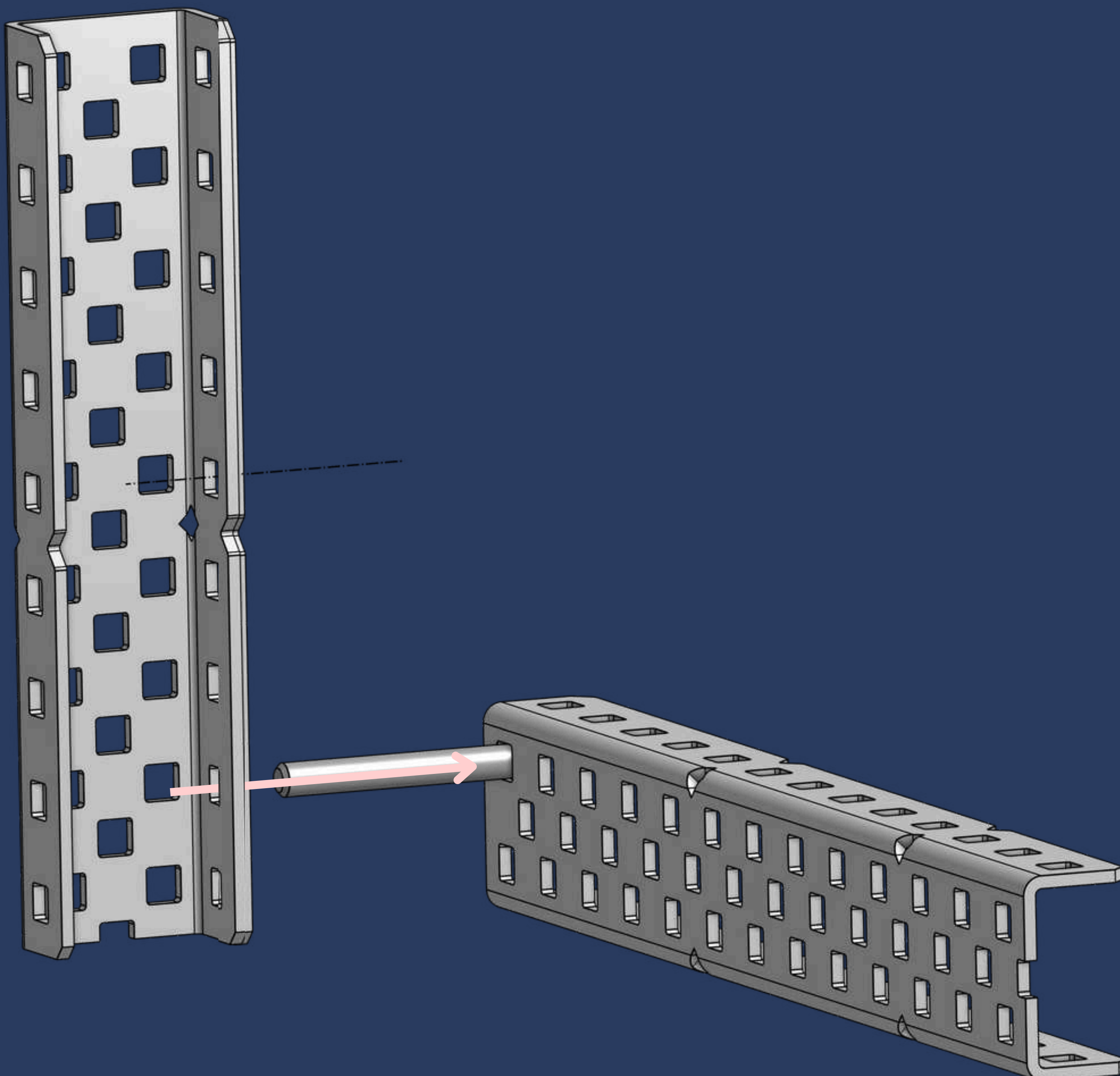


1x
1x2x1x9 Aluminum C-Channel



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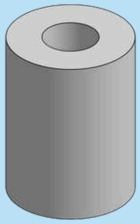
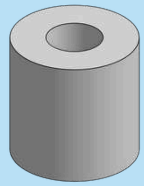


1x
1/2" x 3/8" OD x #8 Nylon Spacer



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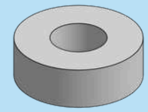
1/4" x 3/8" OD x #8 Nylon Spacer



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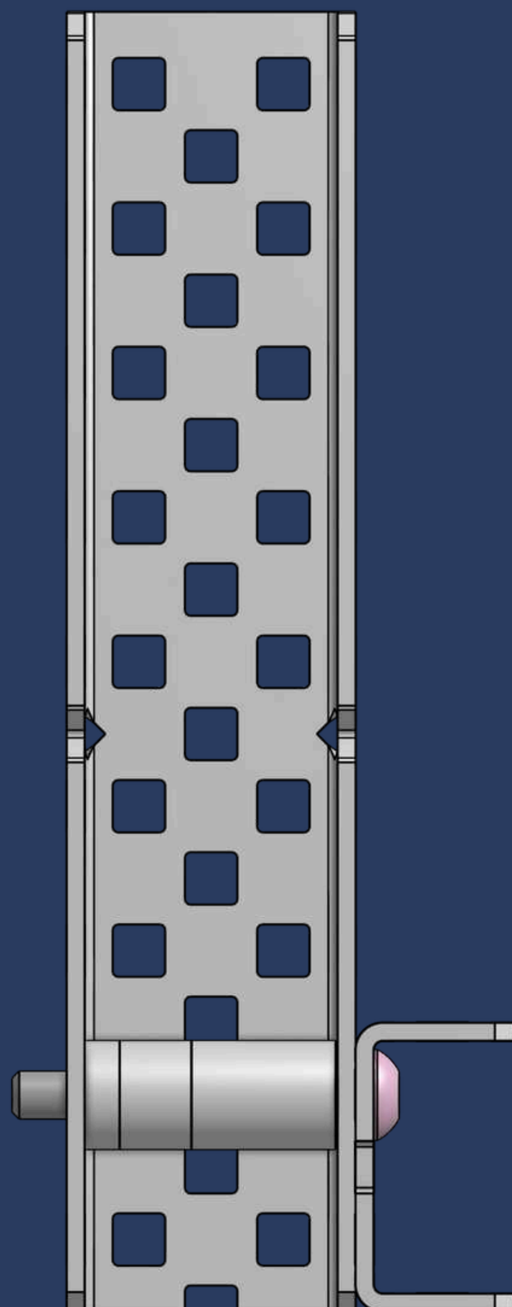
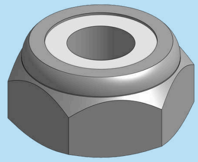


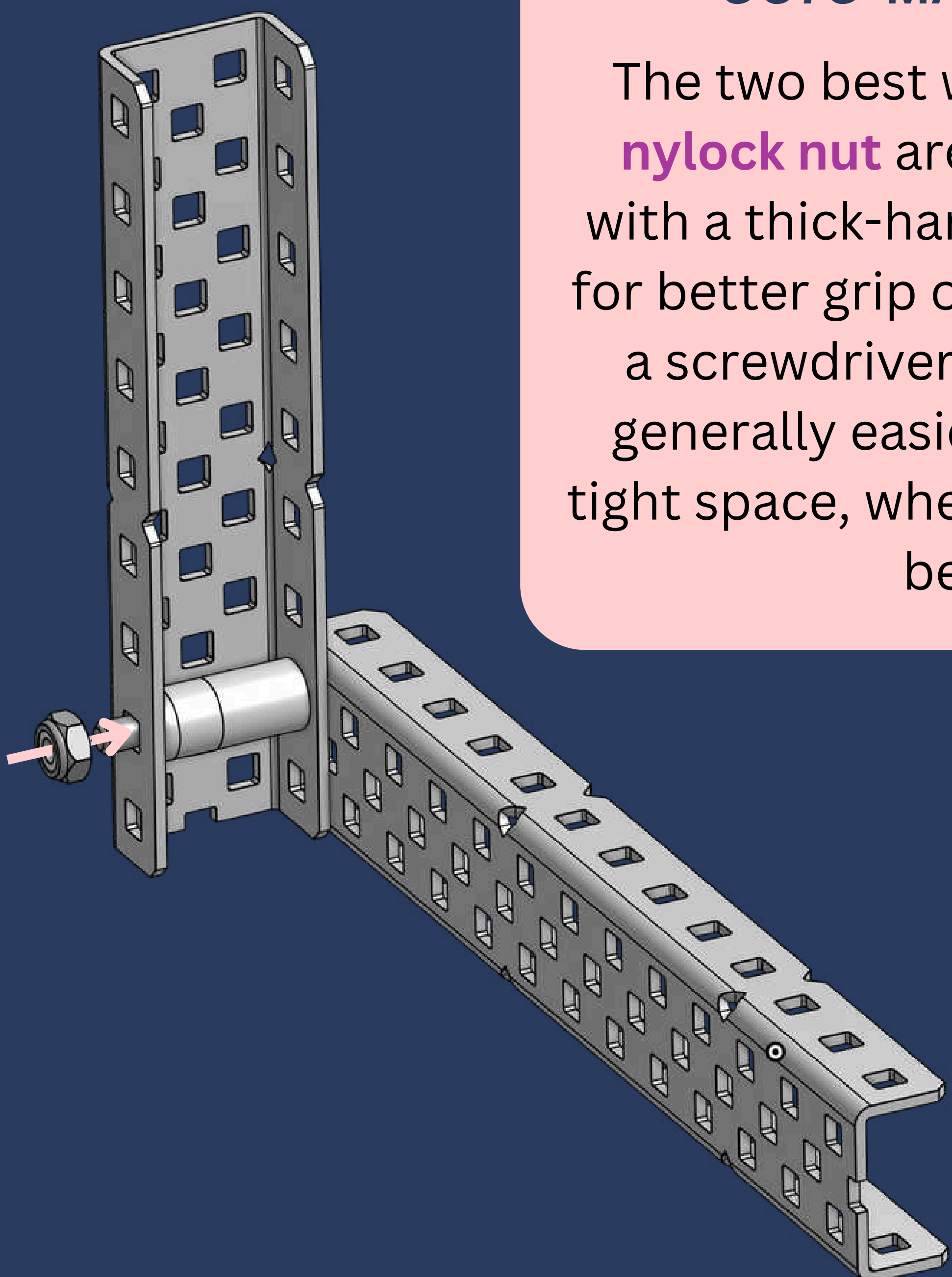
INK-CREDIBLE TIP

The three spacers between the c-channel and the pink screw is a form of bracing called **box bracing**! Box bracing strengthens the c-channel and prevents it from bending inwards.

4



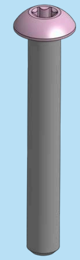
1x
Thin Nylock Nut



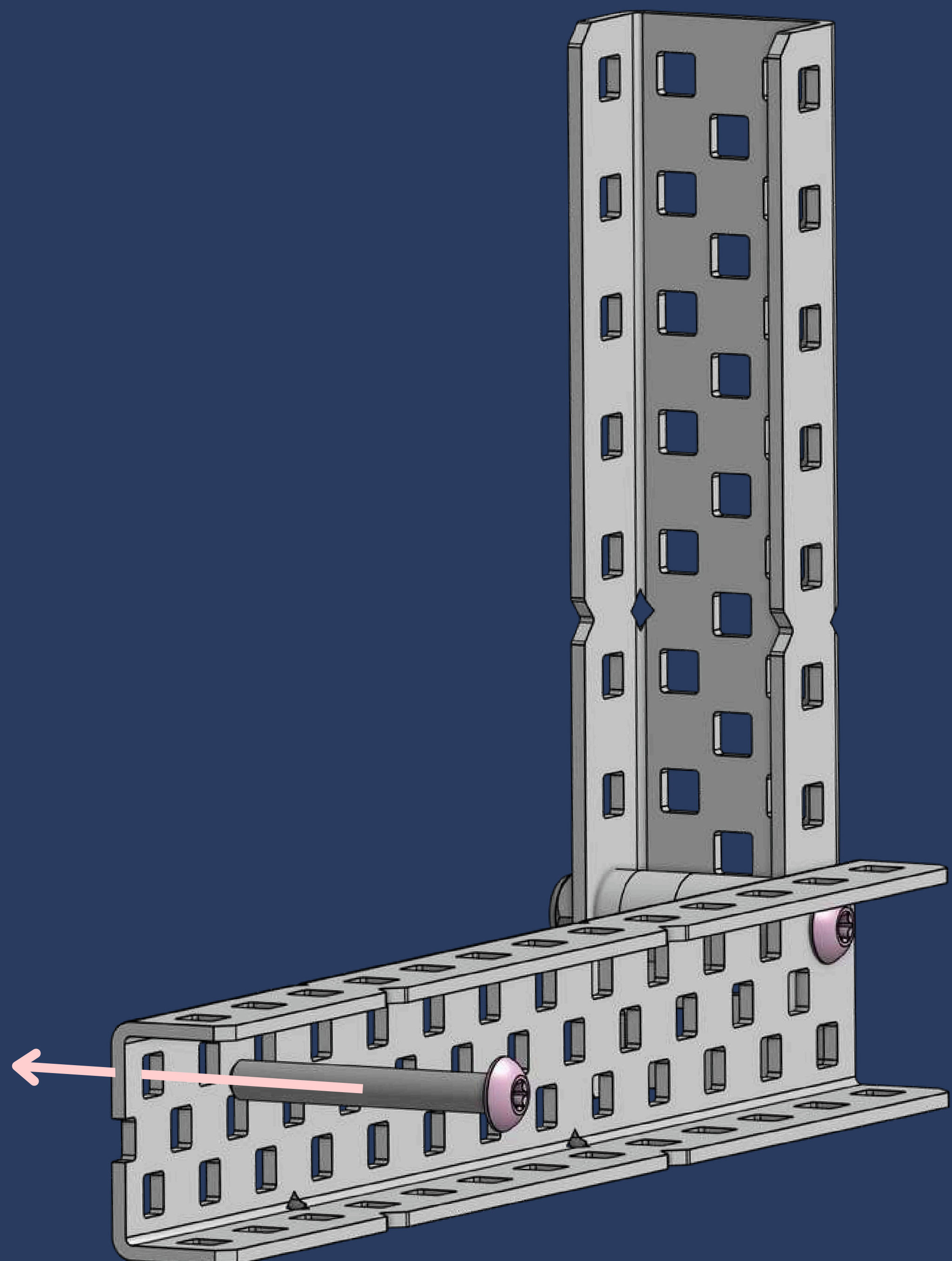
***OCTO*-MAZING TRICK**

The two best ways to **tighten a nylock nut** are using a **wrench** with a thick-handled screwdriver for better grip or a **nut driver** with a screwdriver. Nut drivers are generally easier but too big for tight space, where a wrench works better.

5



1x
#8-32 x 1-1/84" Star Drive Screw



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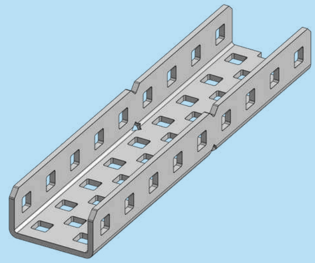
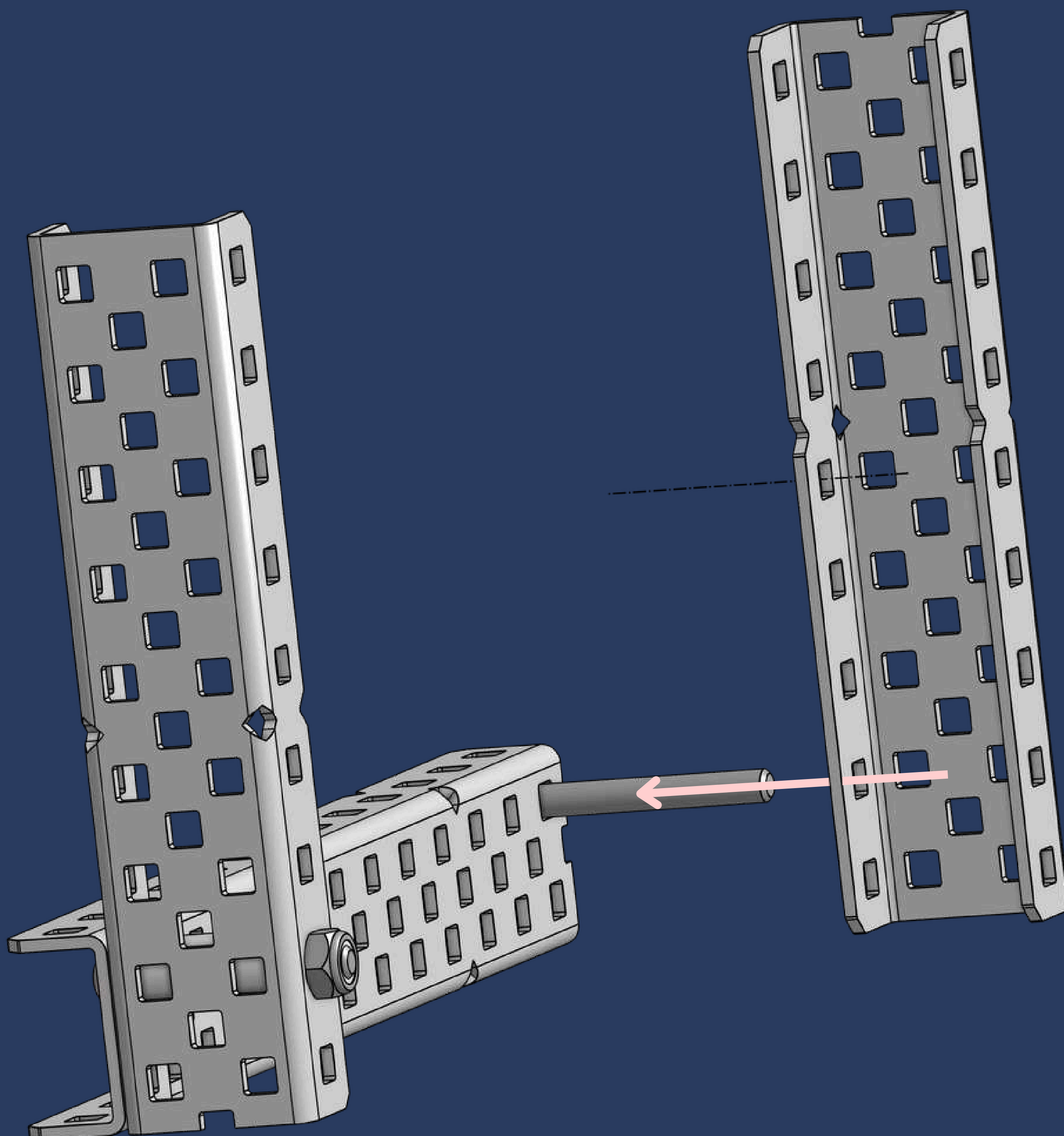


1x
1x2x1x9 Aluminum C-Channel



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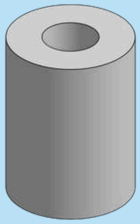

1x
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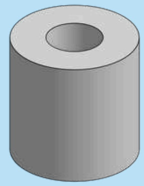

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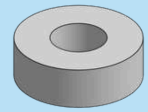
1/4" x 3/8" OD x #8 Nylon Spacer



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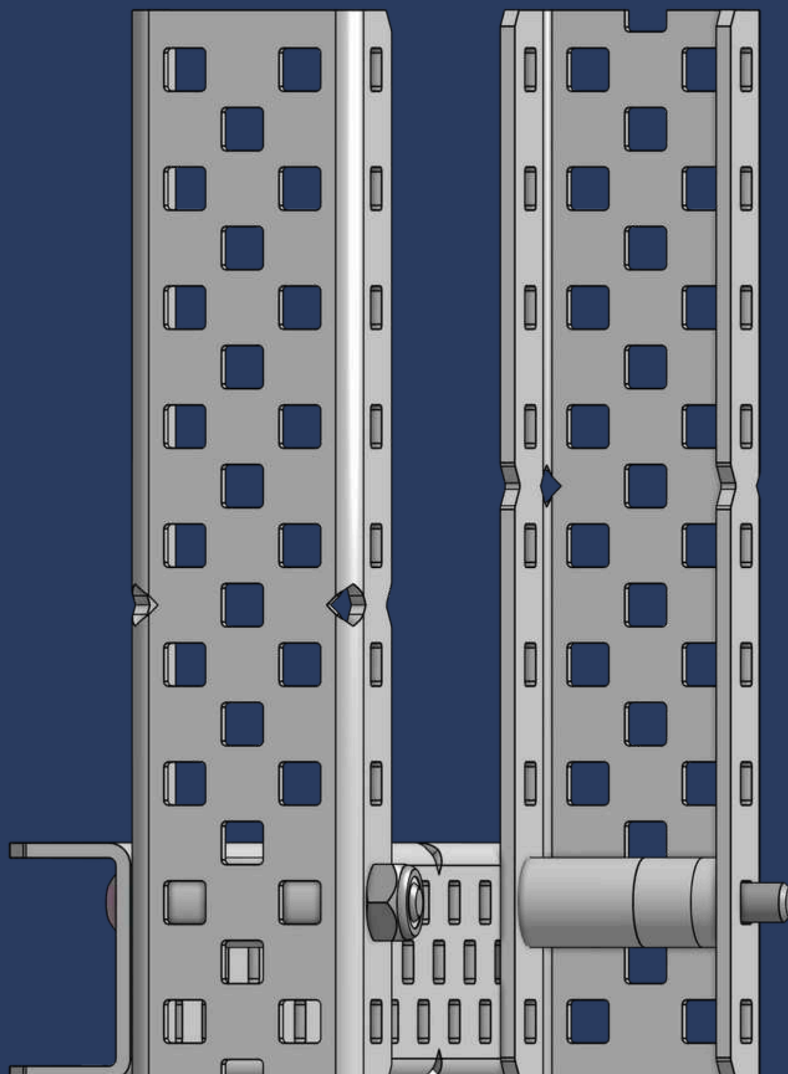
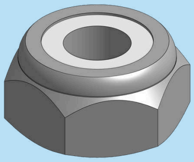


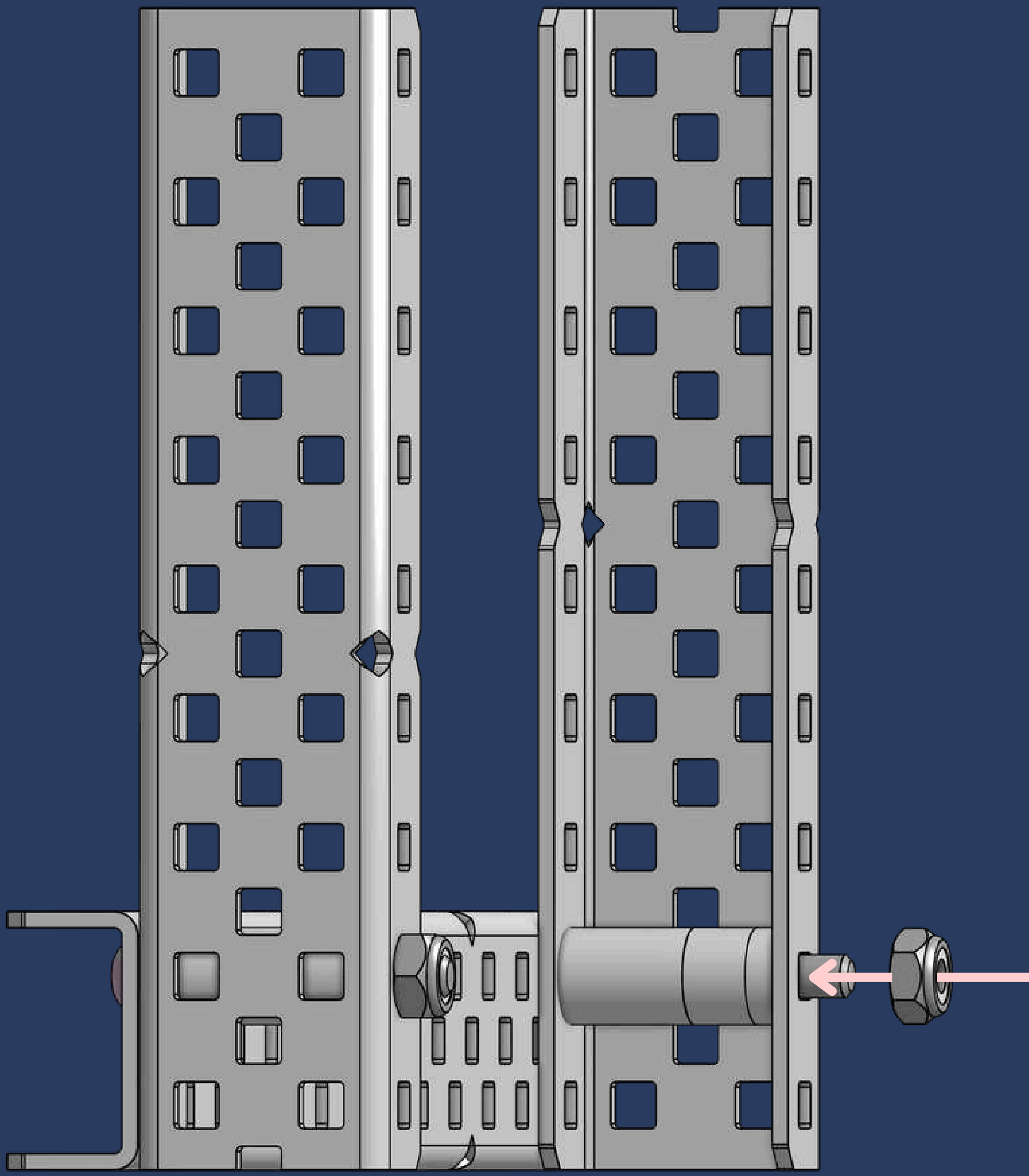
INK-DISPENSIBLE ADVICE

If you don’t have the exact spacers, you can **combine smaller ones** to match the needed size—for example, two 1/4" spacers make a 1/2" spacer!

8



1x
Thin Nylock Nut



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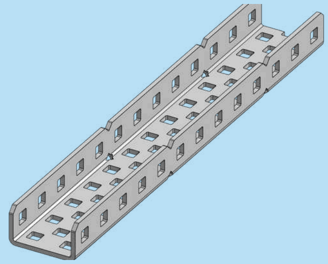
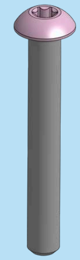
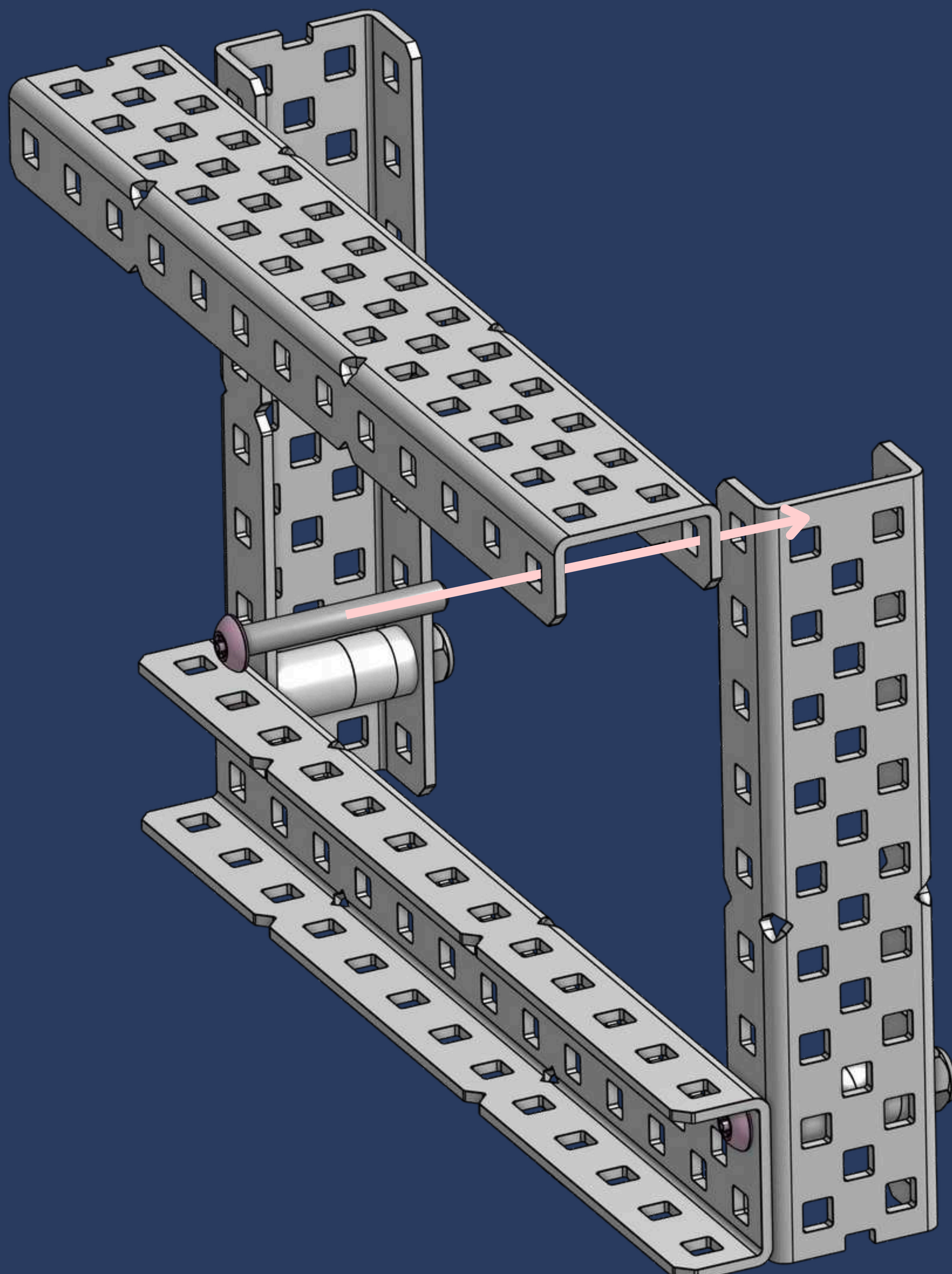


1x
1x2x1x13 Aluminum C-Channel



1x
#8-32 x 1-1/84" Star Drive Screw



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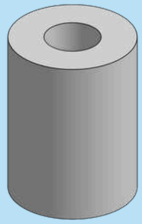

1x
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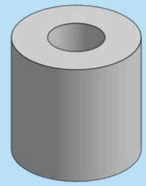

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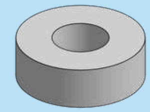

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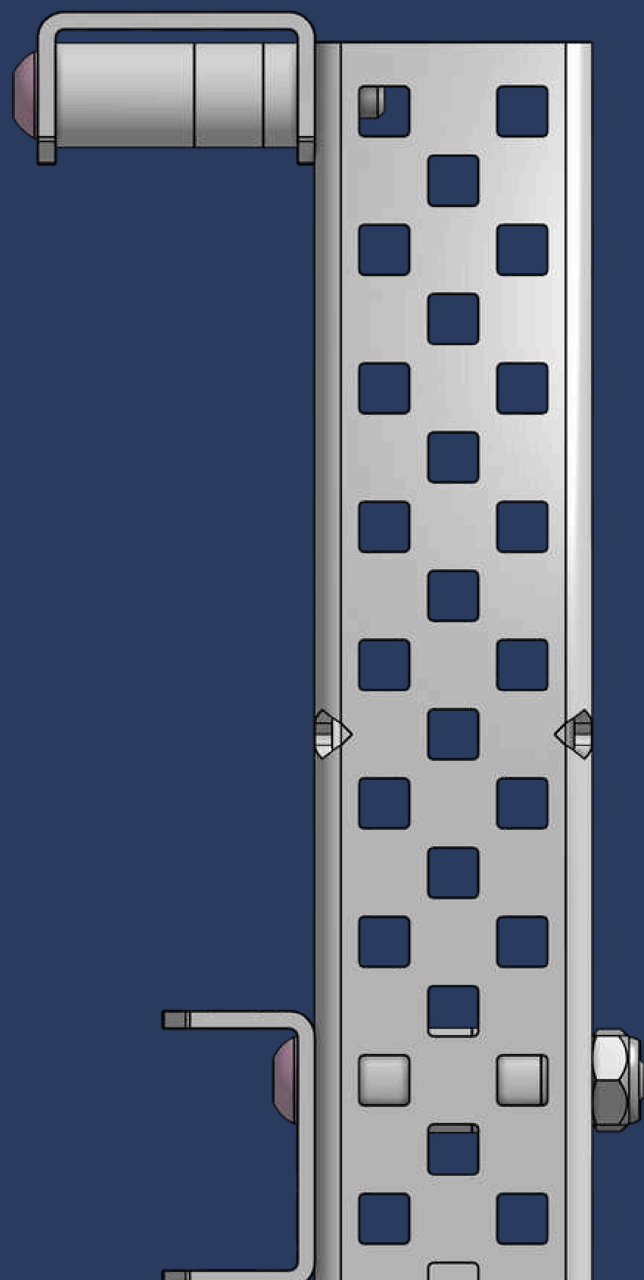

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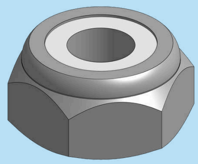
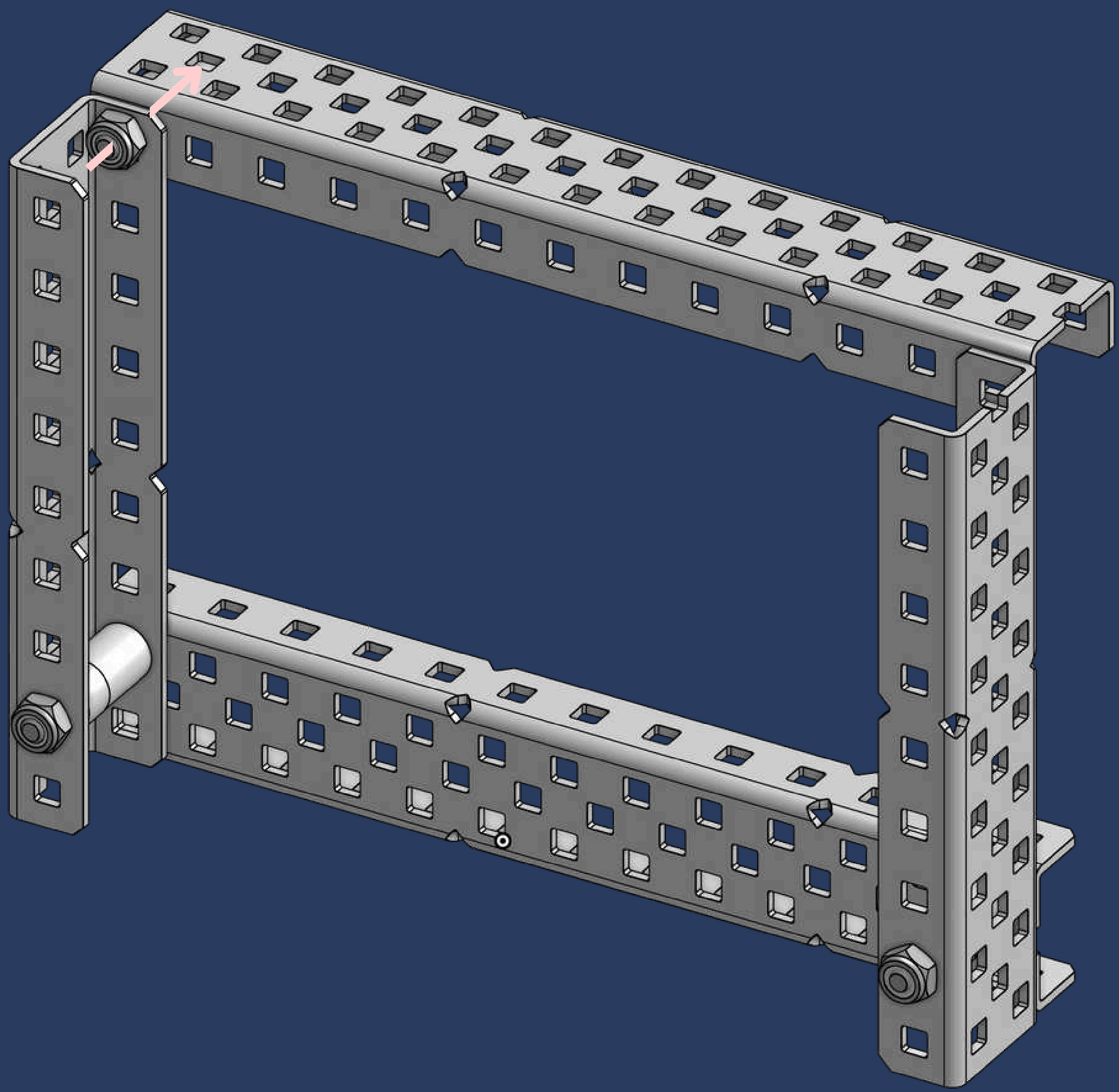


1x
Thin Nylock Nut



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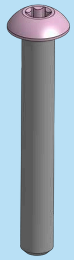
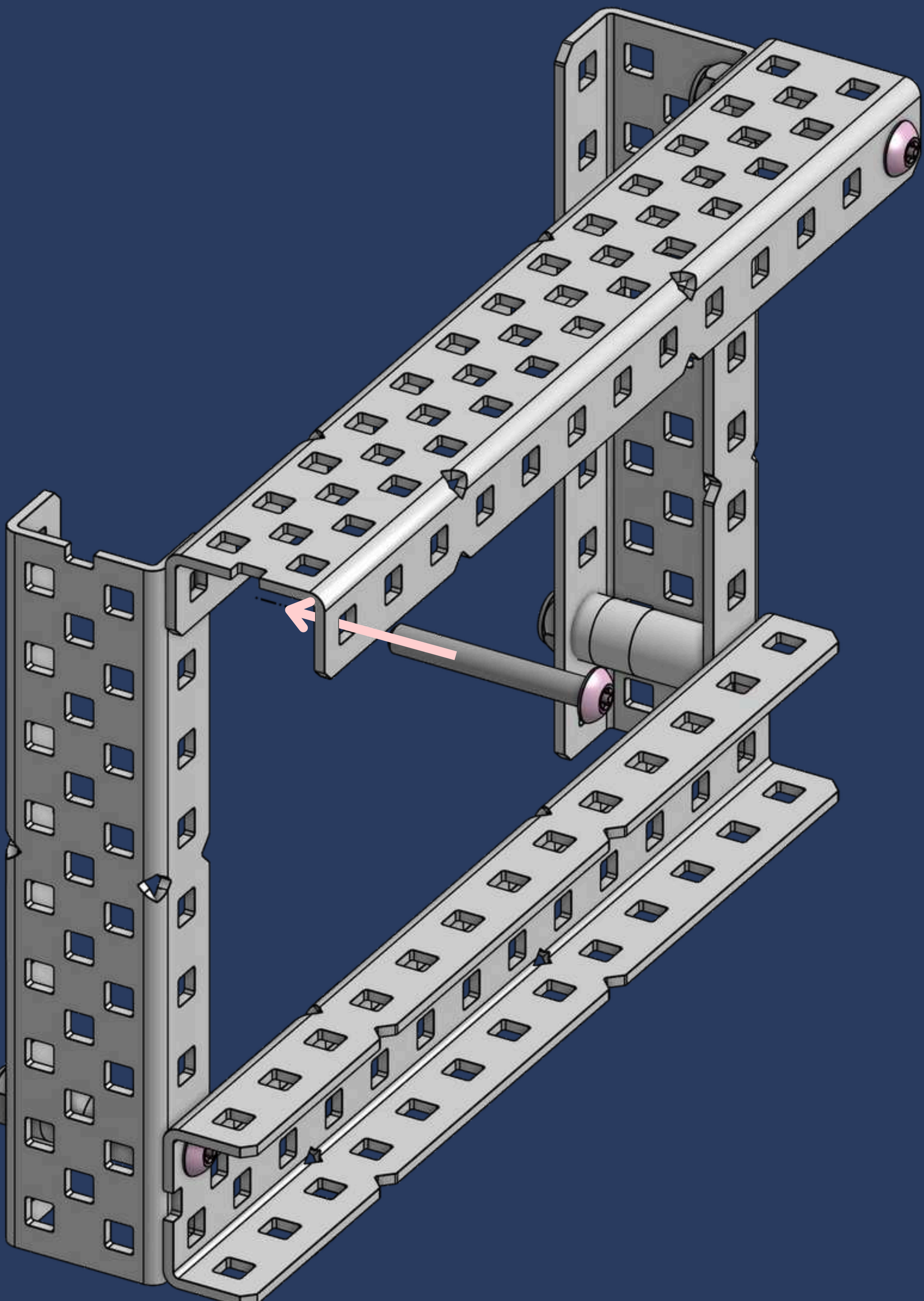

1x
#8-32 x 1-1/84" Star Drive Screw



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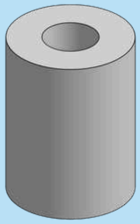
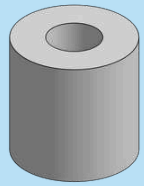


1x
1/2" x 3/8" OD x #8 Nylon Spacer



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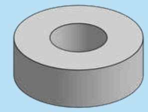
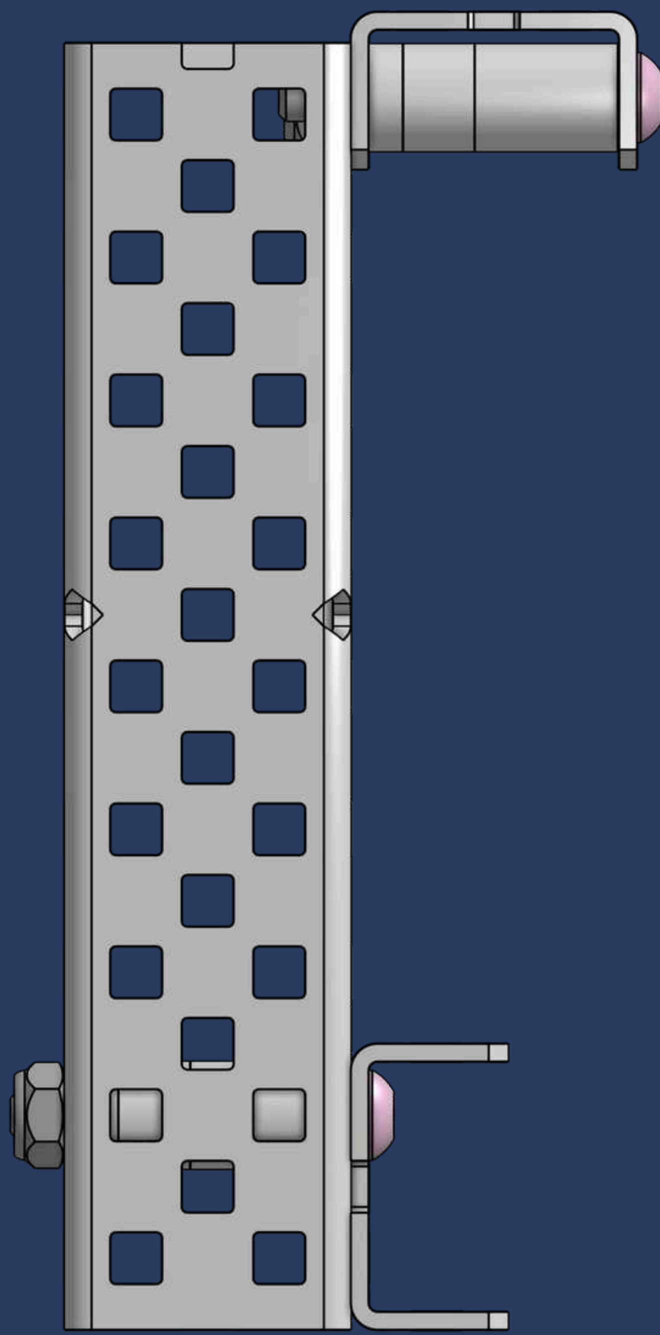


1x
1/8" x 3/8" OD x #8 Nylon Spacer



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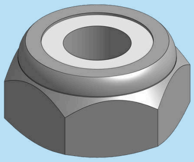
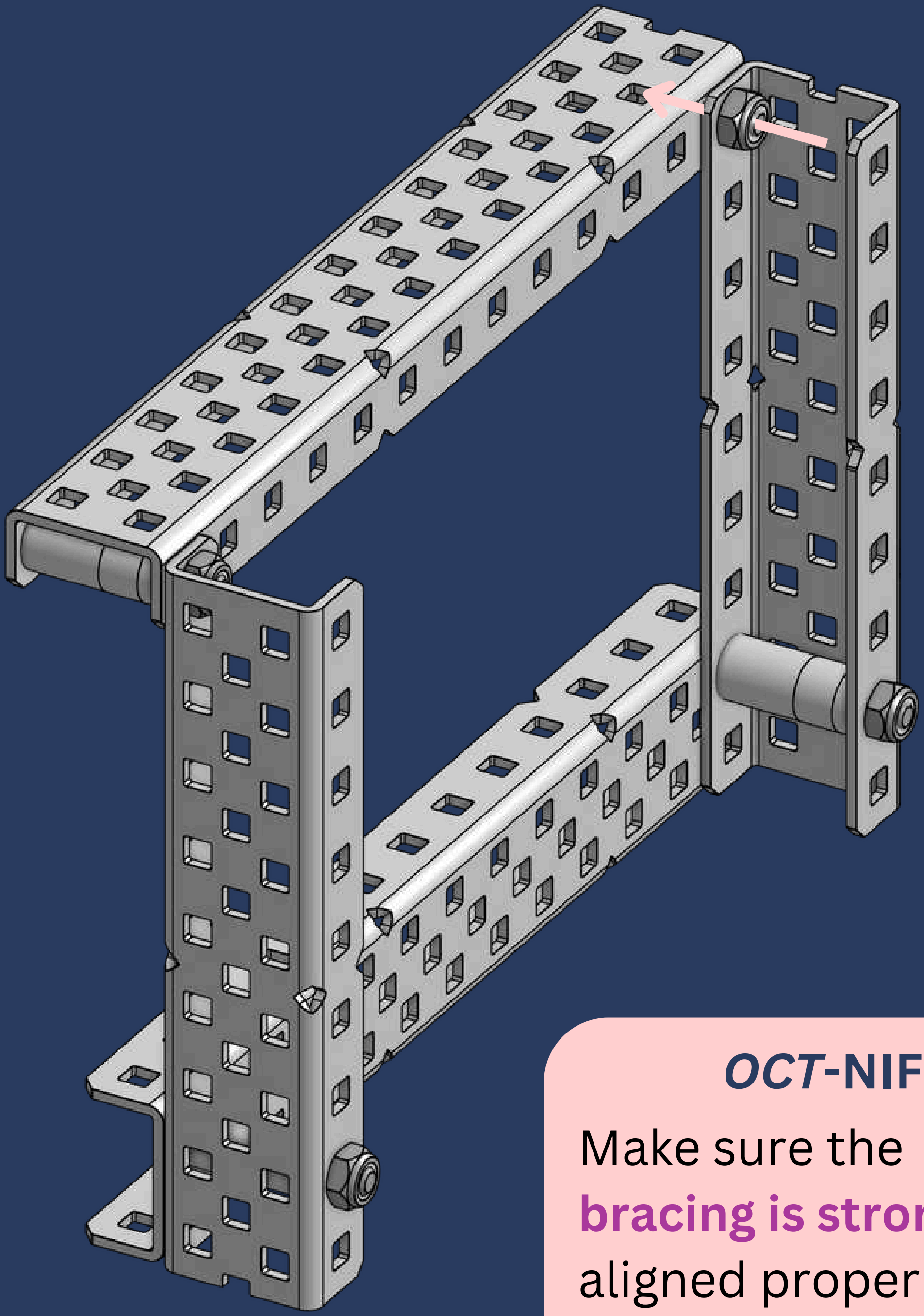


1x
Thin Nylock Nut

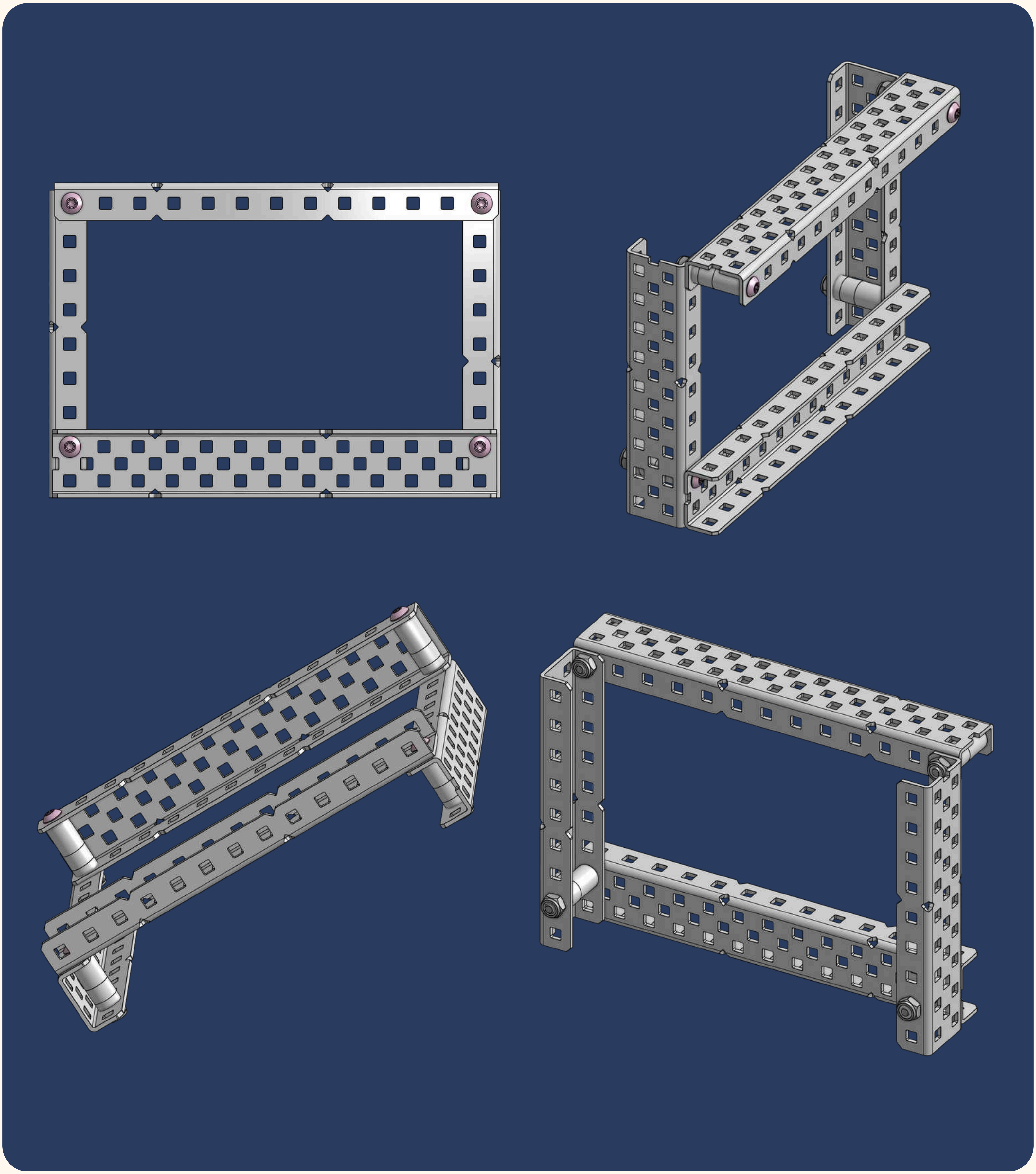


OCT-NIFICIANT INSIGHT

Make sure the **bracing is strong** and aligned properly—building on a weak or uneven structure can cause **issues** later.

CHECKPOINT #1

007



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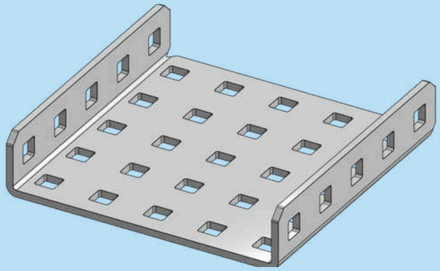
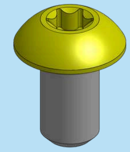
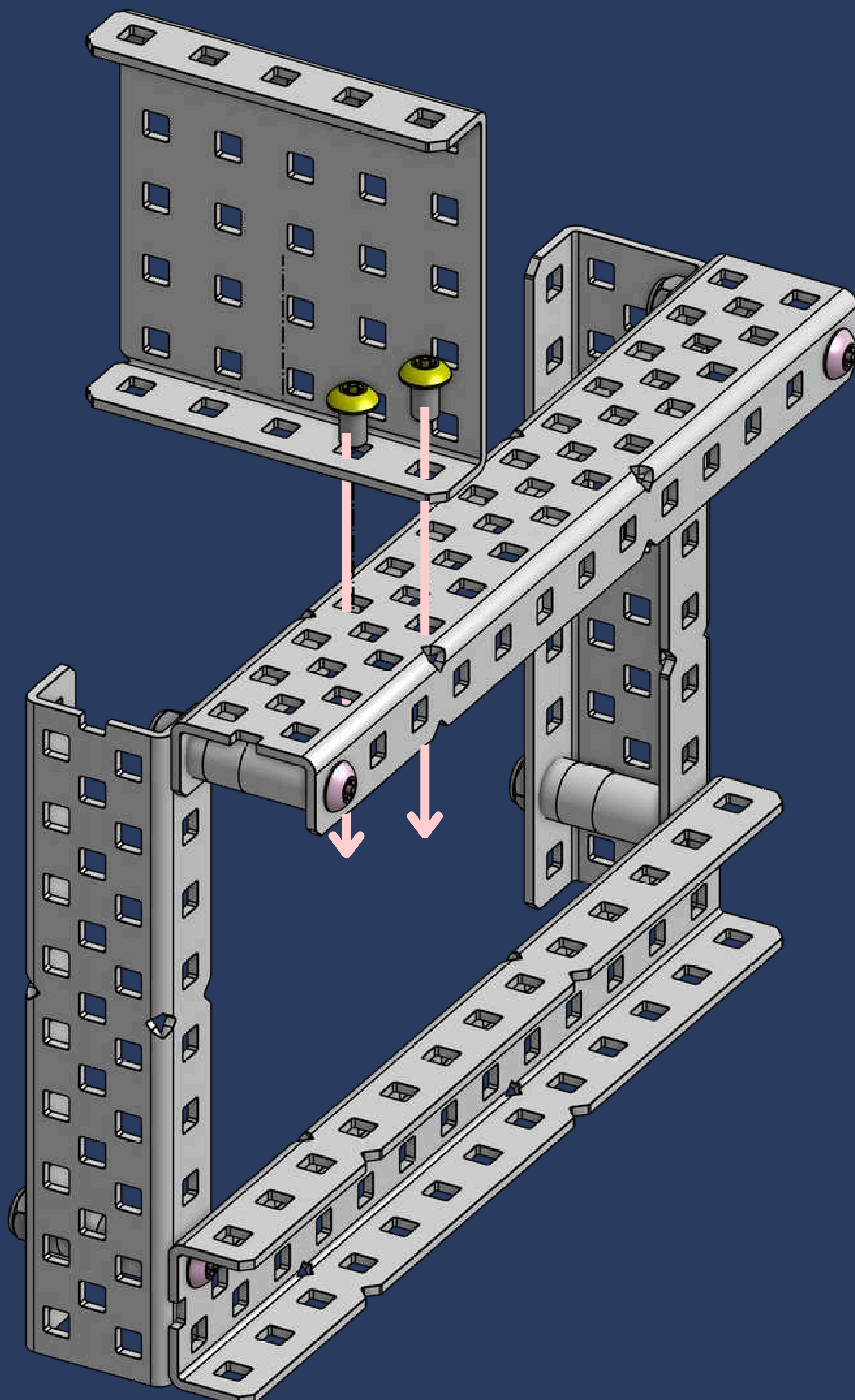


1x
1x5x1x5 Aluminum C-Channel



2x
#8-32 x 1/4" Star Drive Screw



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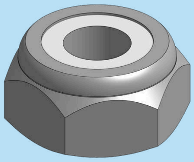
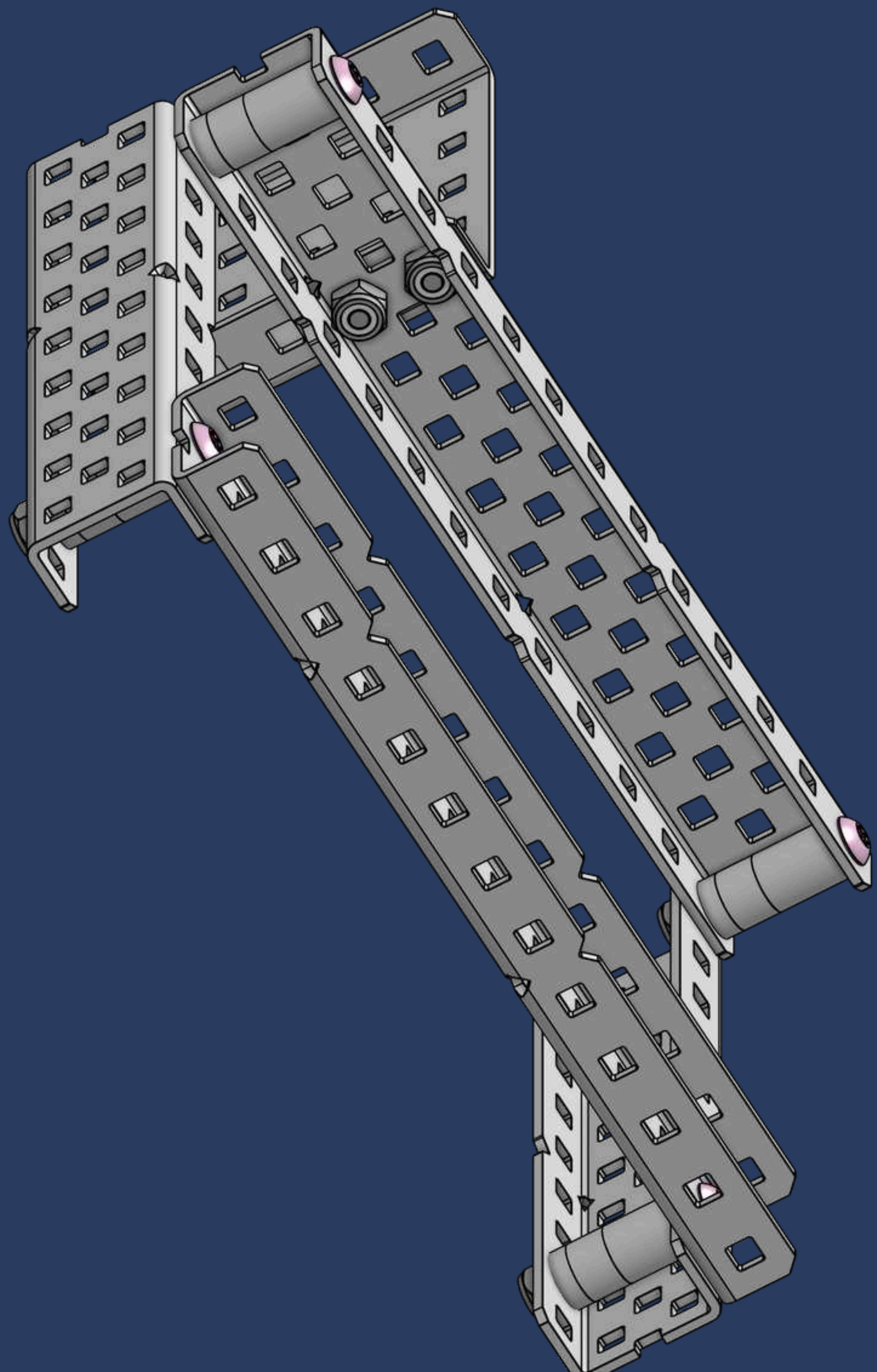


2x
Thin Nylock Nut



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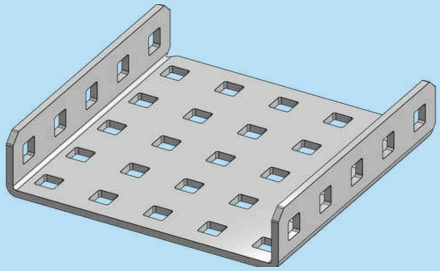
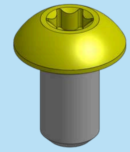
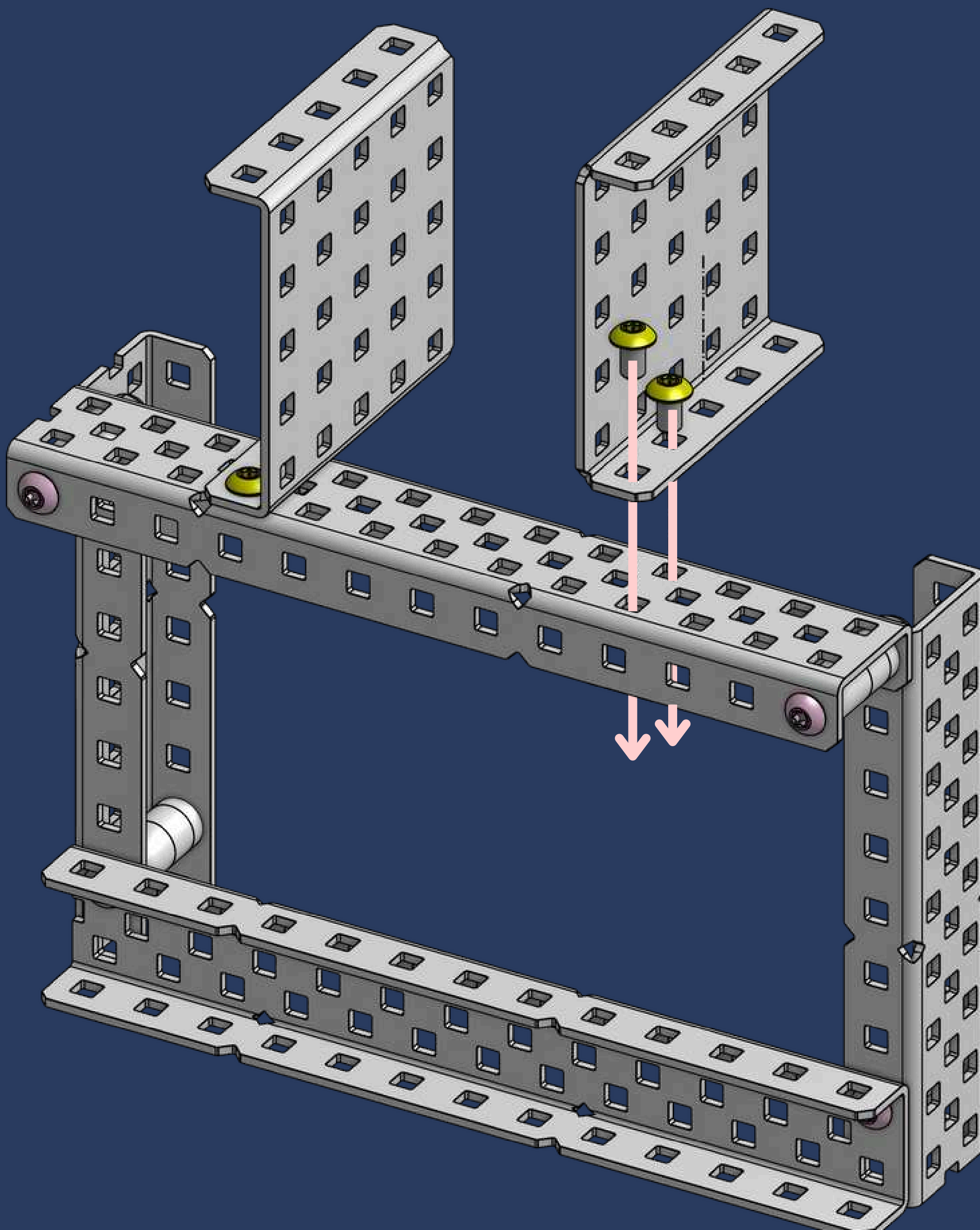


1x
1x5x1x5 Aluminum C-Channel



2x
#8-32 x 1/4" Star Drive Screw



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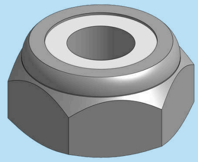
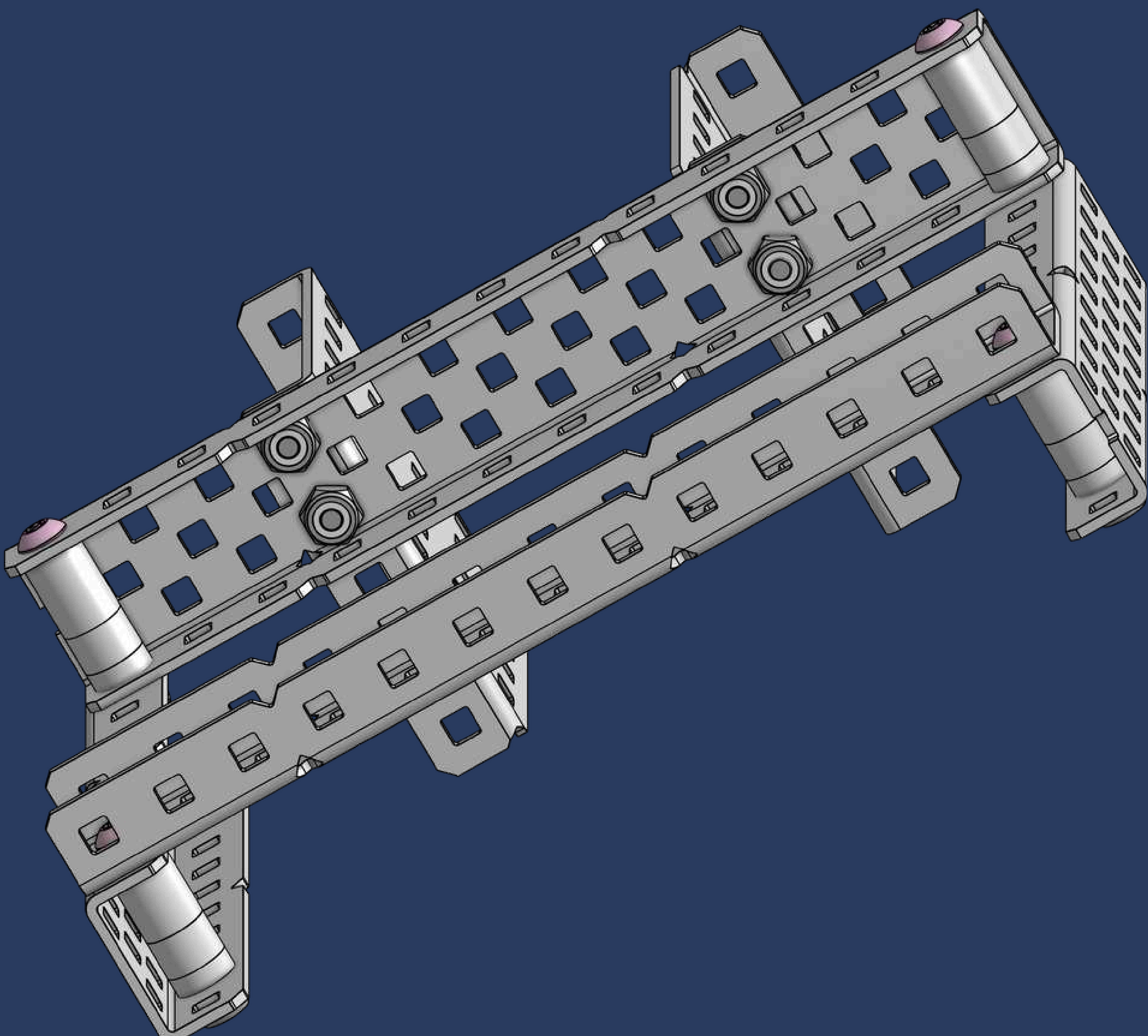


2x
Thin Nylock Nut



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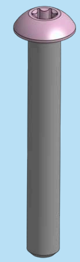
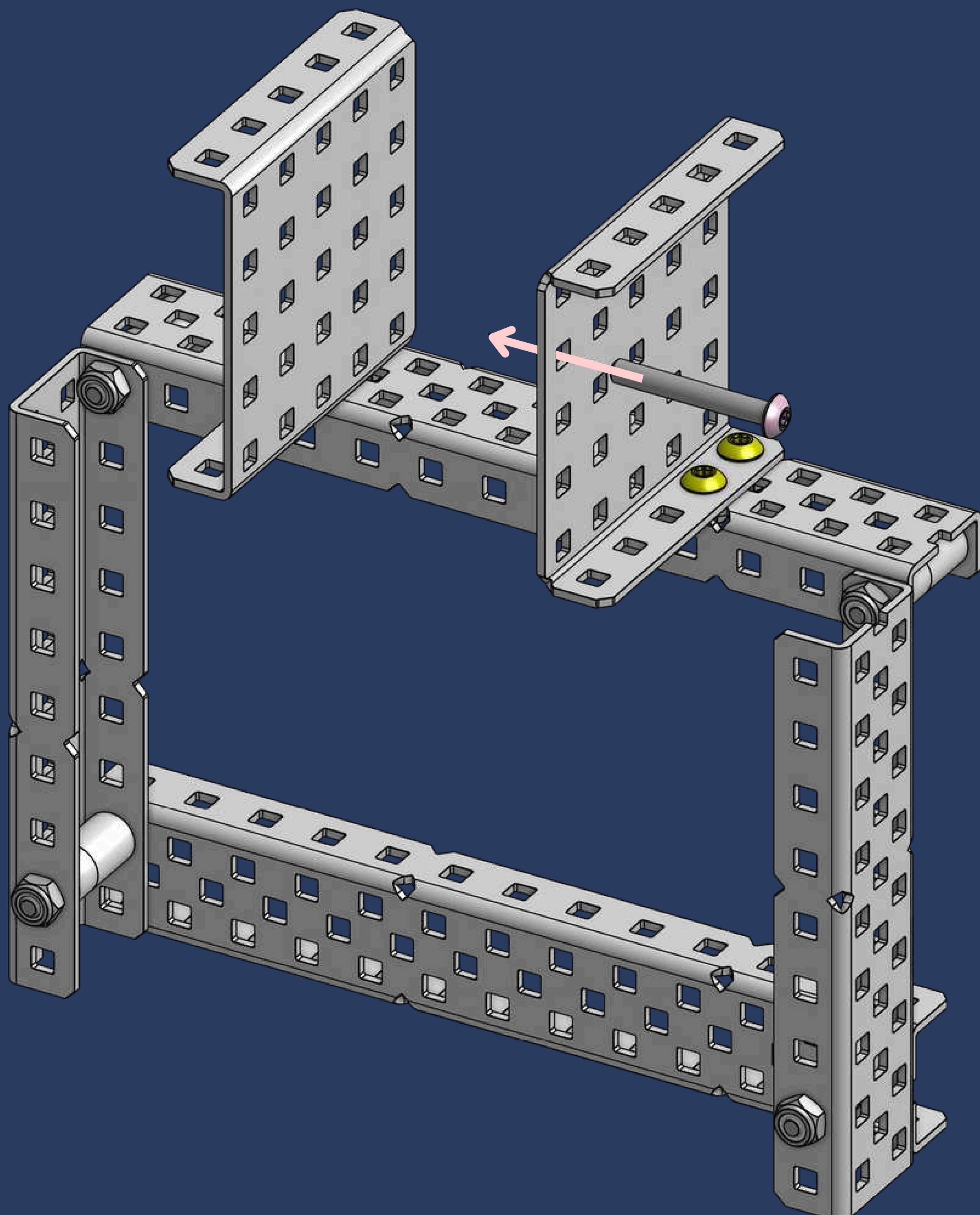


1x
#8-32 x 1-1/84" Star Drive Screw



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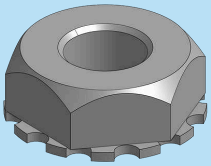
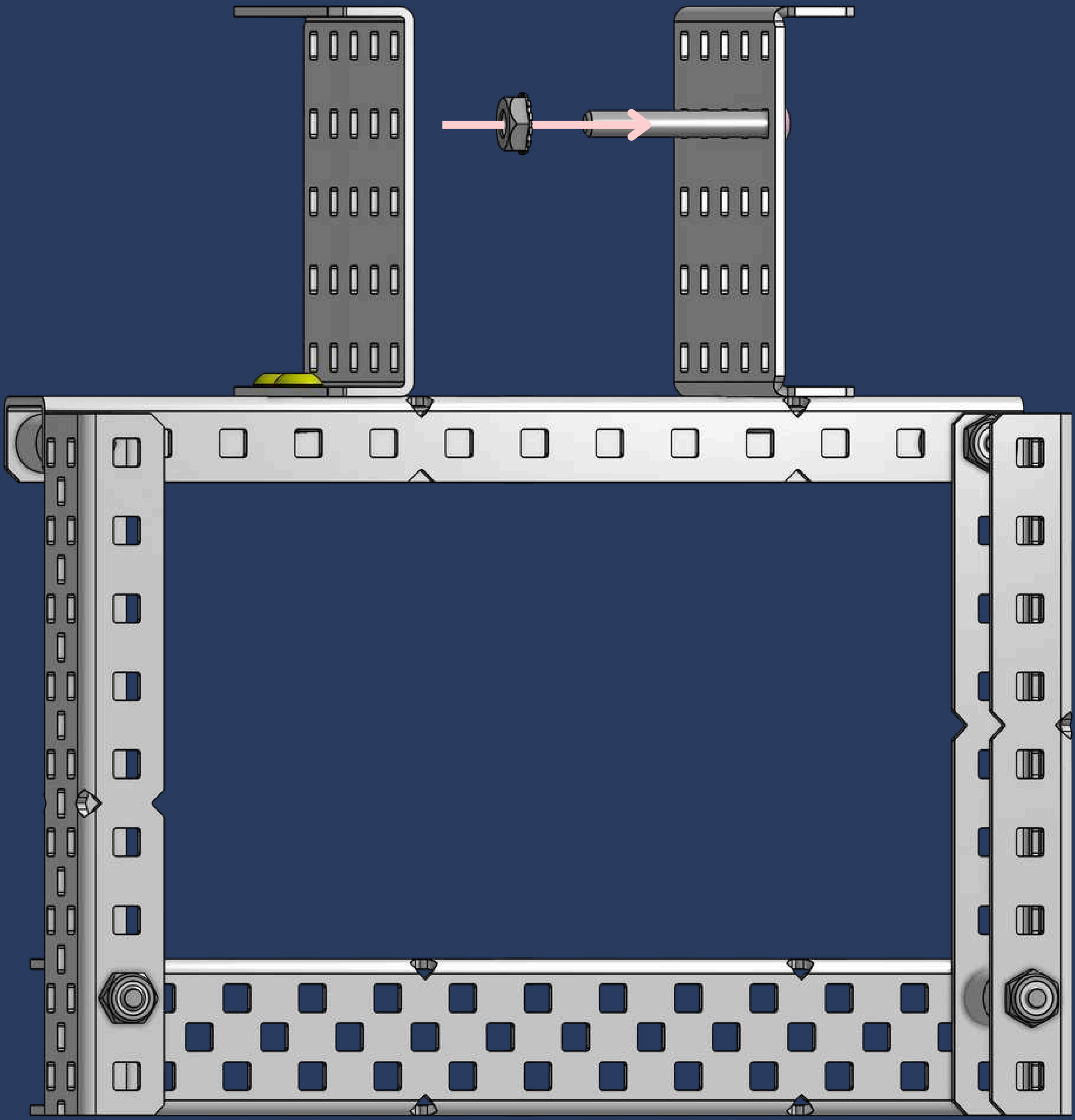


1x
Keps Nut



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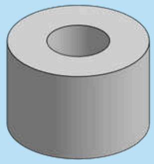
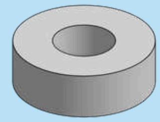


1x
3/8" x 3/8" OD x #8 Nylon Spacer



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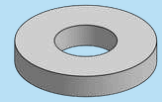
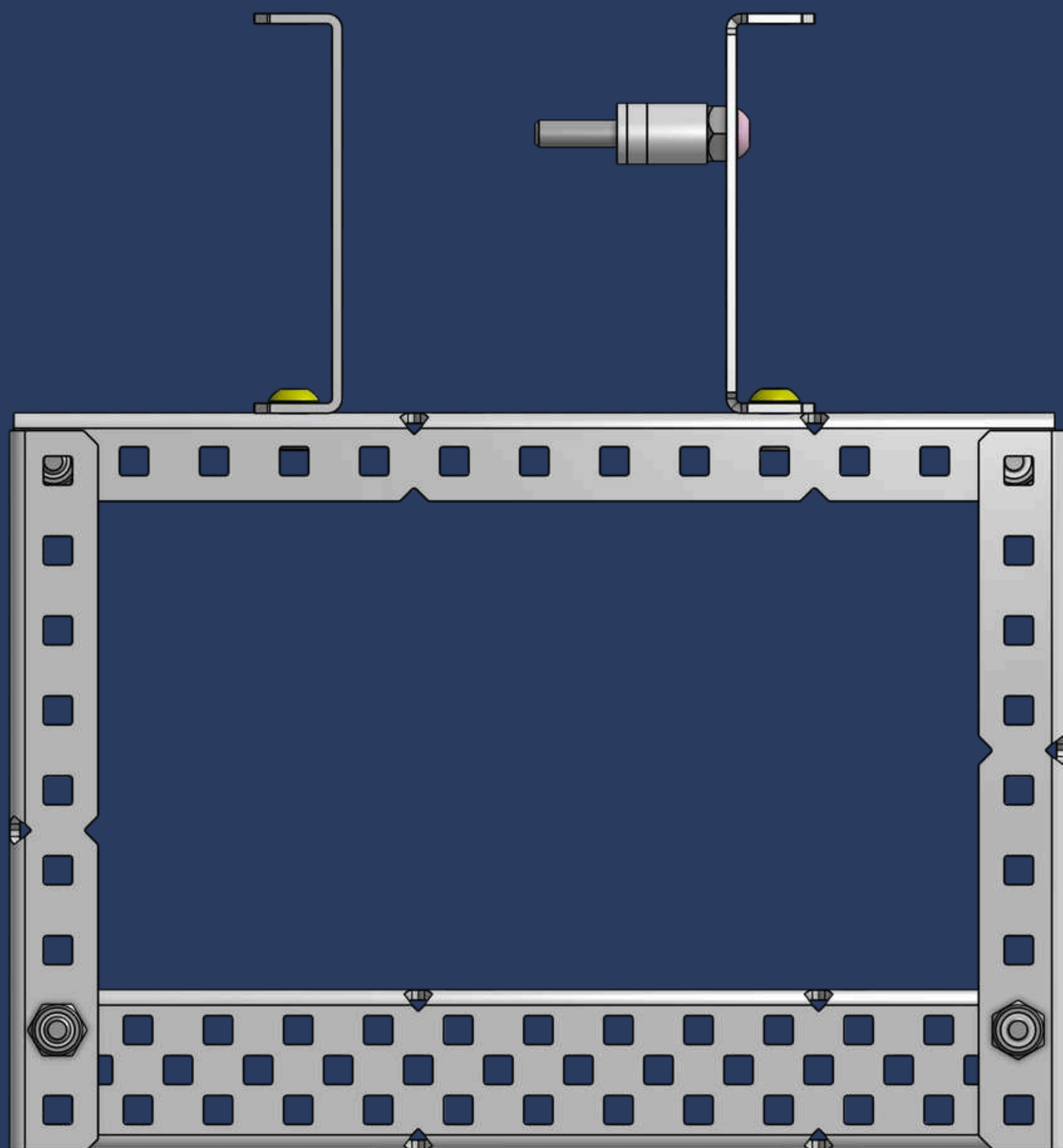
1/8" x 3/8" OD x #8 Nylon Spacer



1x
1/16" x 3/8" OD x #8 Nylon Spacer



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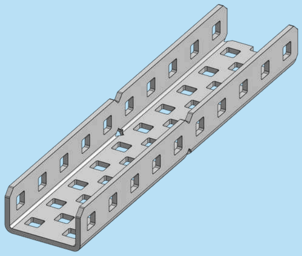
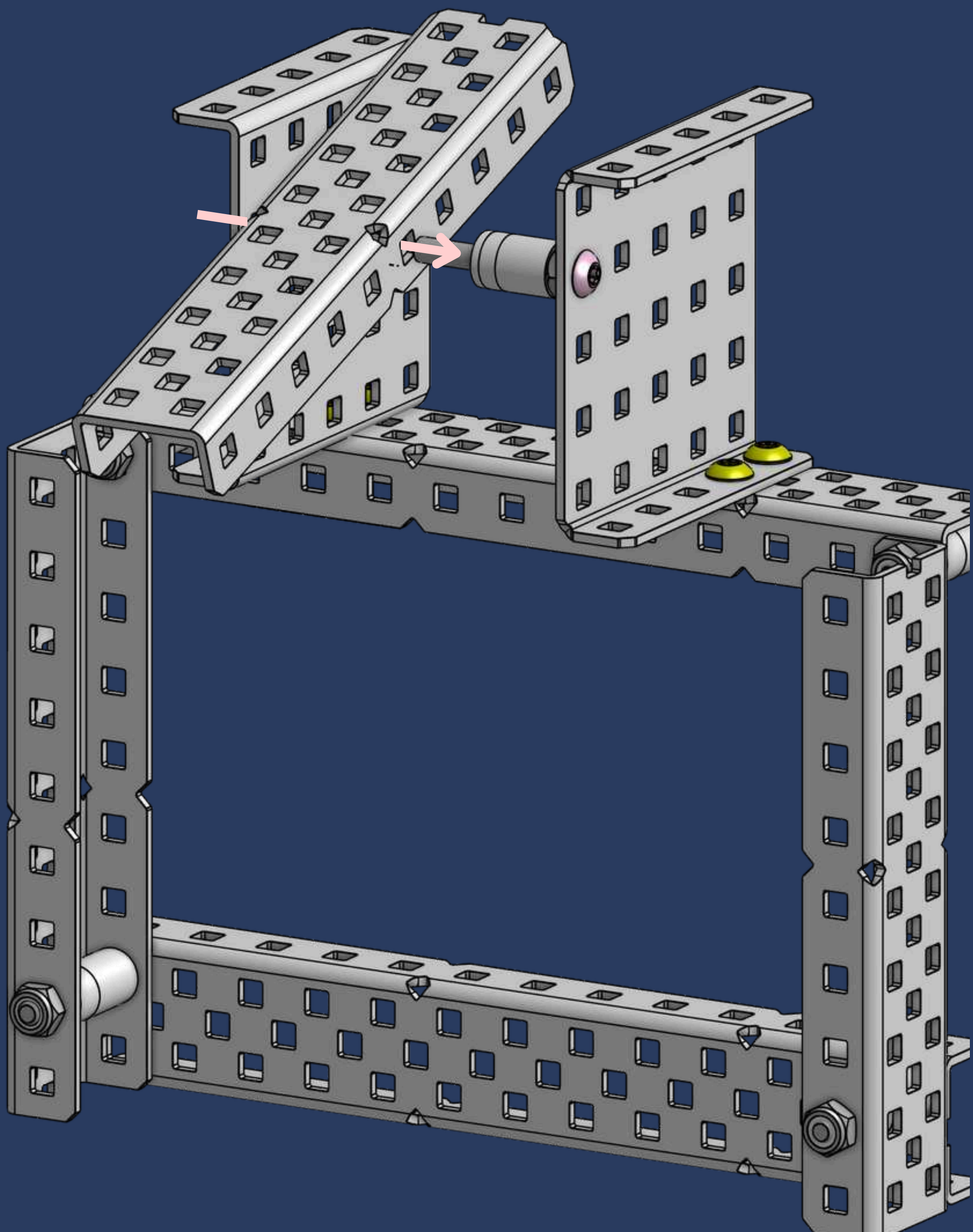


1x
1x2x1x10 Aluminum C-Channel



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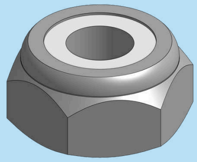
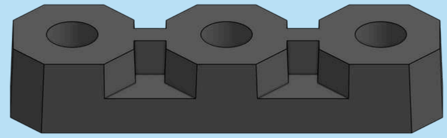
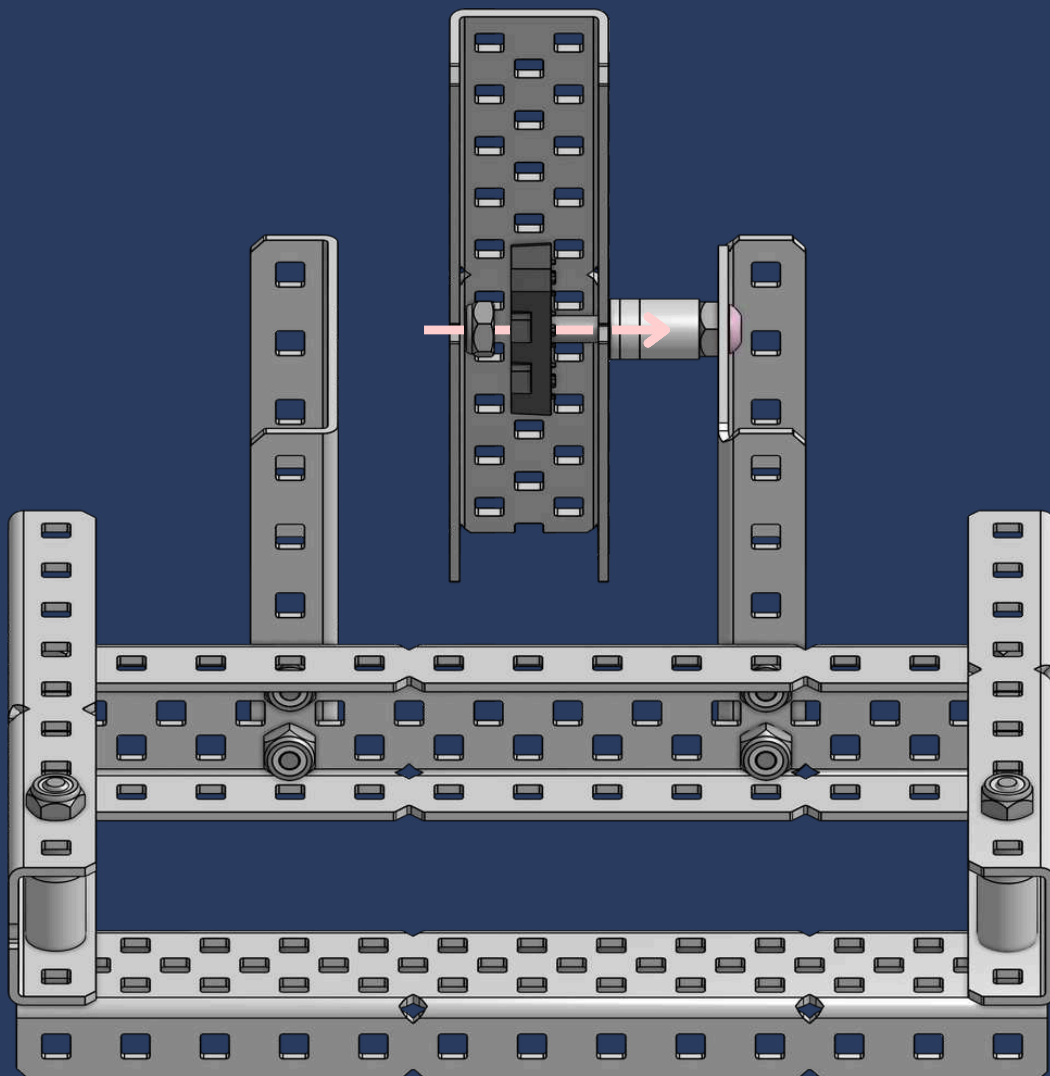


1x
Thin Nylock Nut



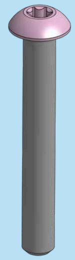
1x
Flat Bearing



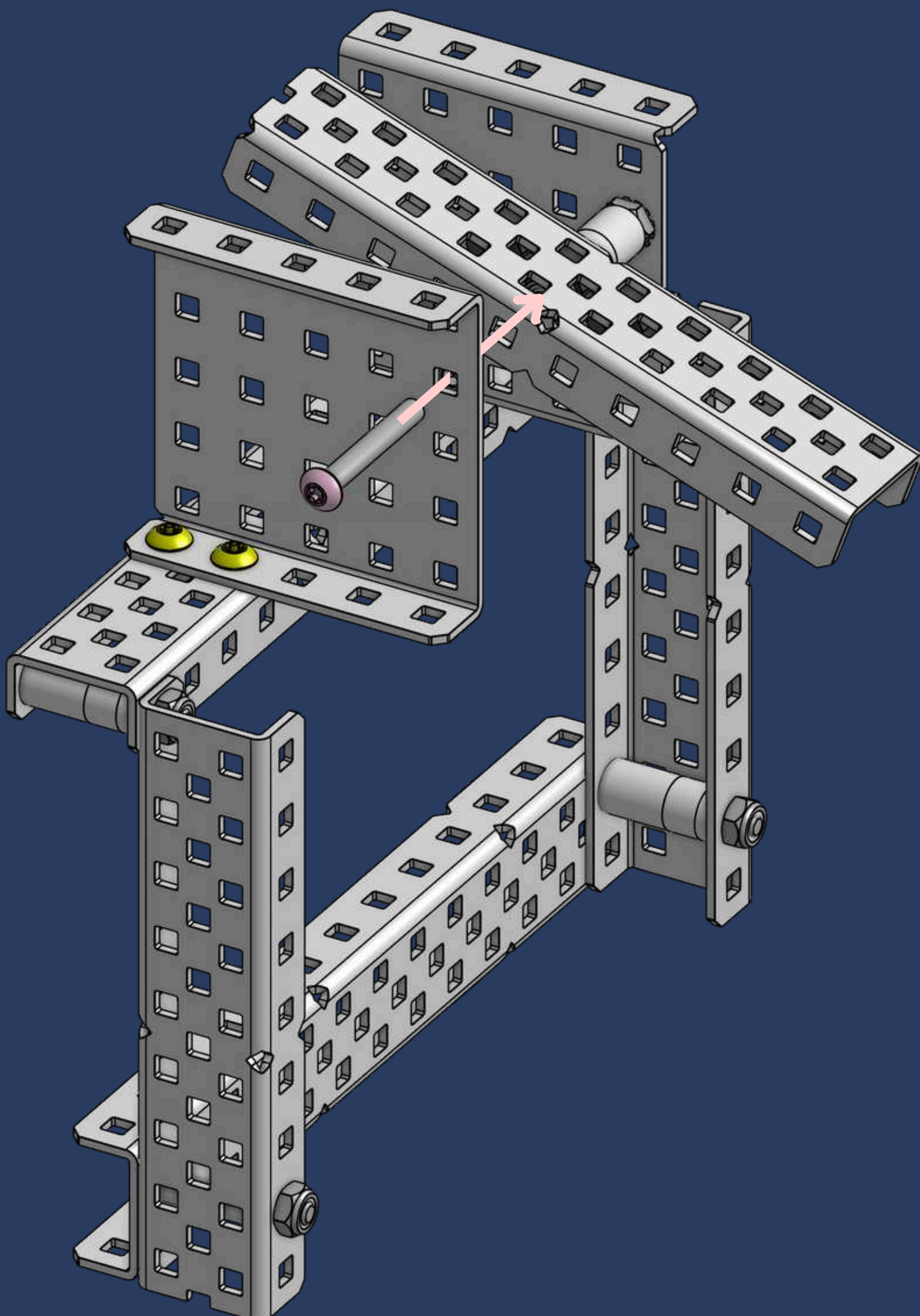
INK-STANT ADVICE

This is our first **screw joint**! A screw joint is a connection where a screw serves as a **pivot**, allowing parts to rotate. In VEX, it typically includes a screw, kep nut, spacers, and a nylock nut. The **nylock nut** should be **slightly loosened** to let the c-channel pivot freely.

24



1x
#8-32 x 1-1/84" Star Drive Screw



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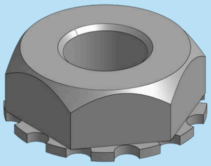
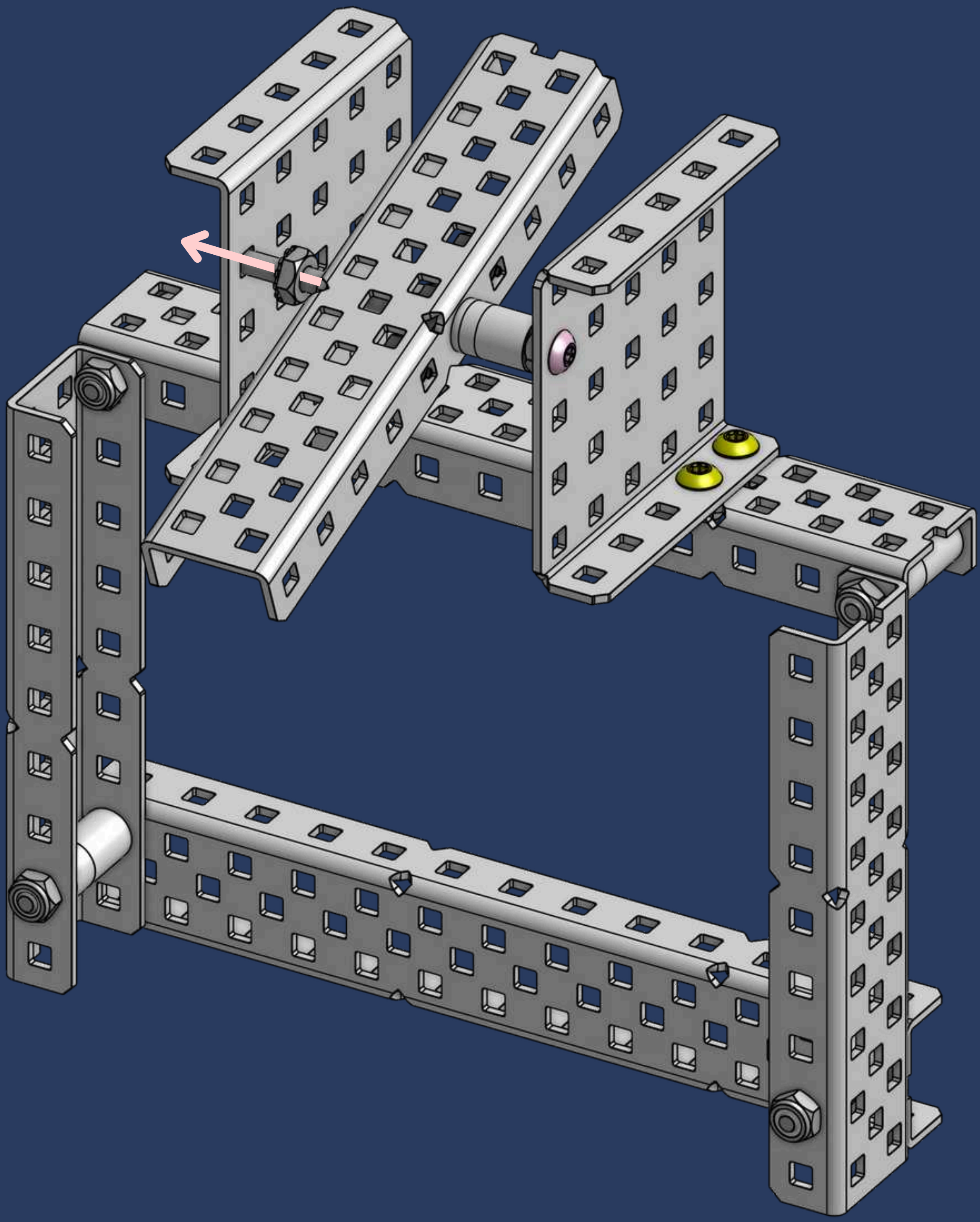


1x
Keps Nut



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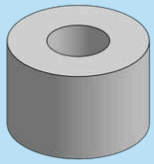
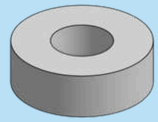
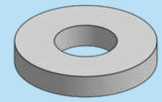


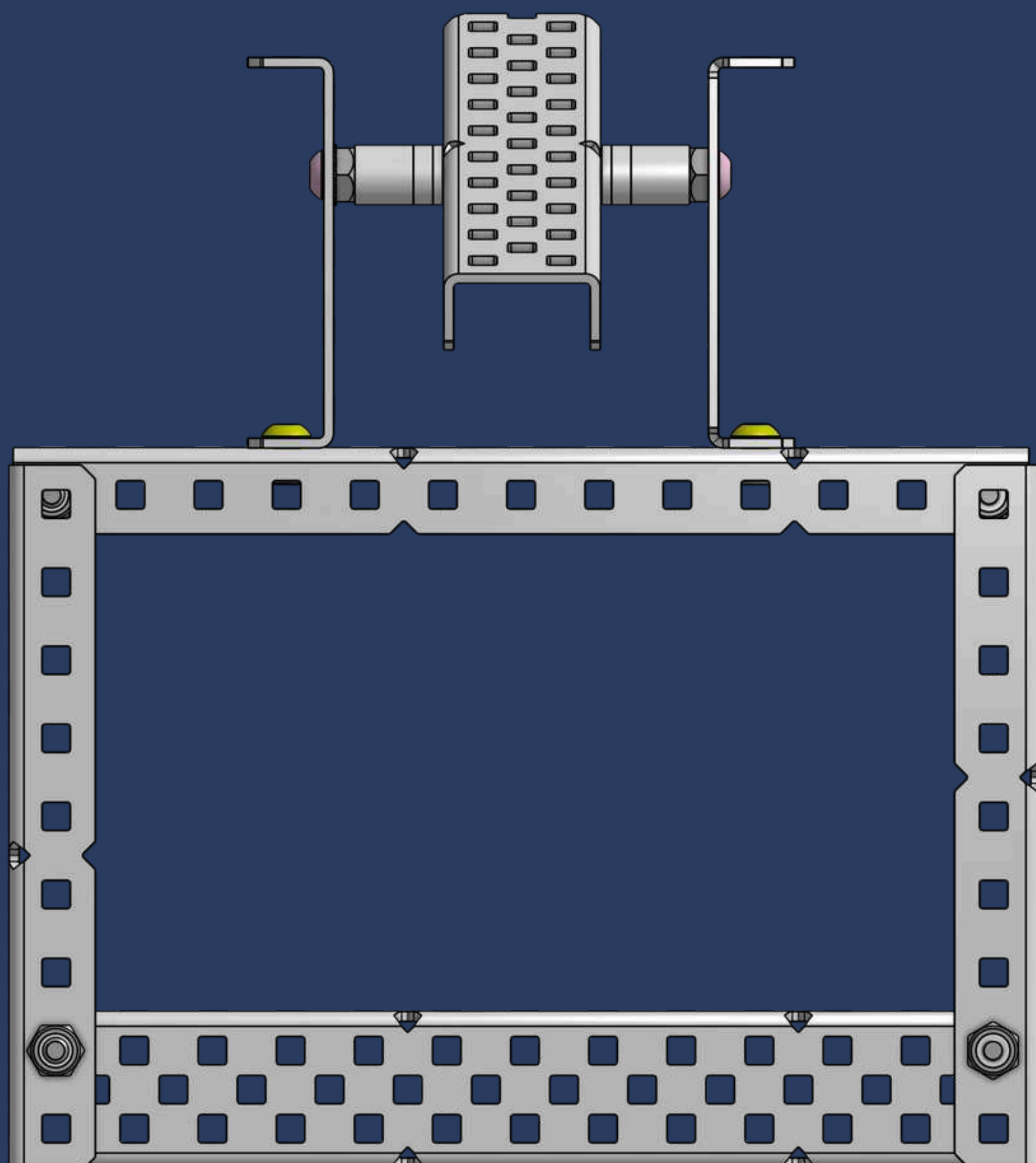
1x
3/8" x 3/8" OD x #8 Nylon Spacer



1x
1/8" x 3/8" OD x #8 Nylon Spacer



1x
1/16" x 3/8" OD x #8 Nylon Spacer



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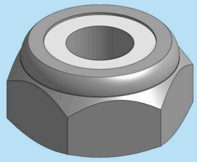
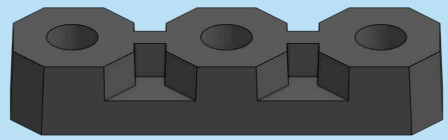
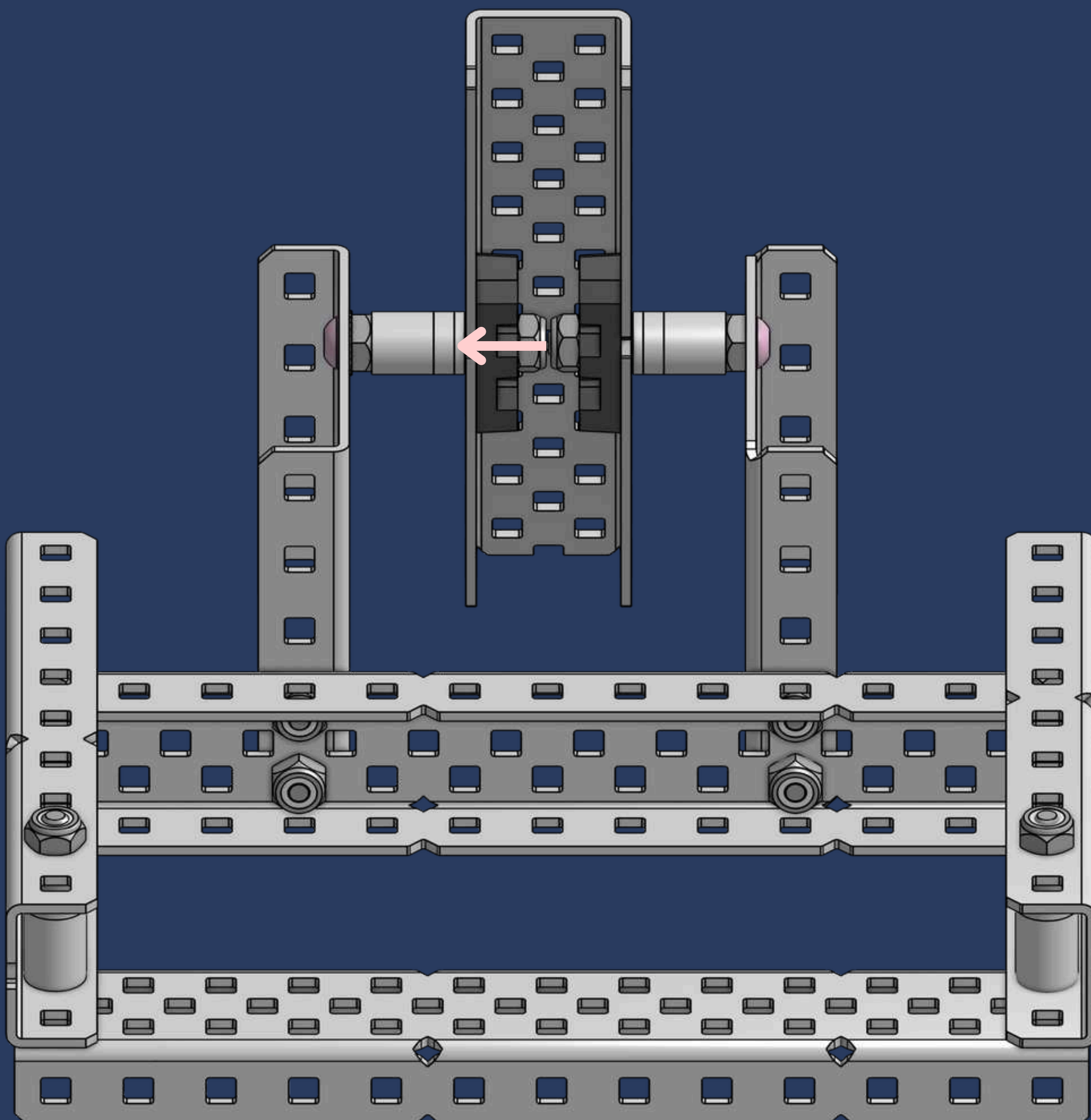


1x
Thin Nylock Nut



1x
Flat Bearing

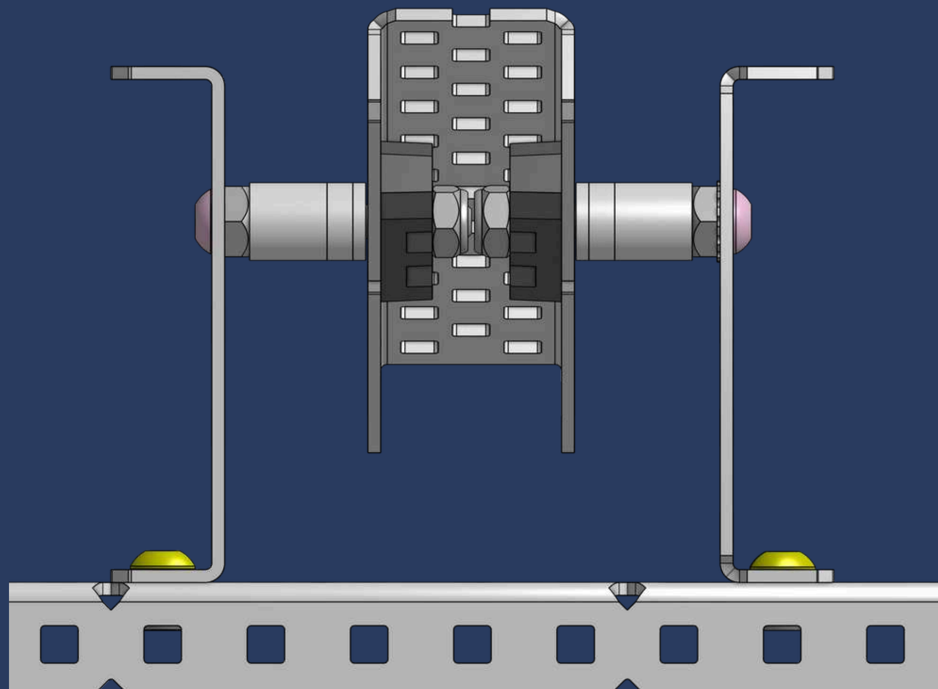


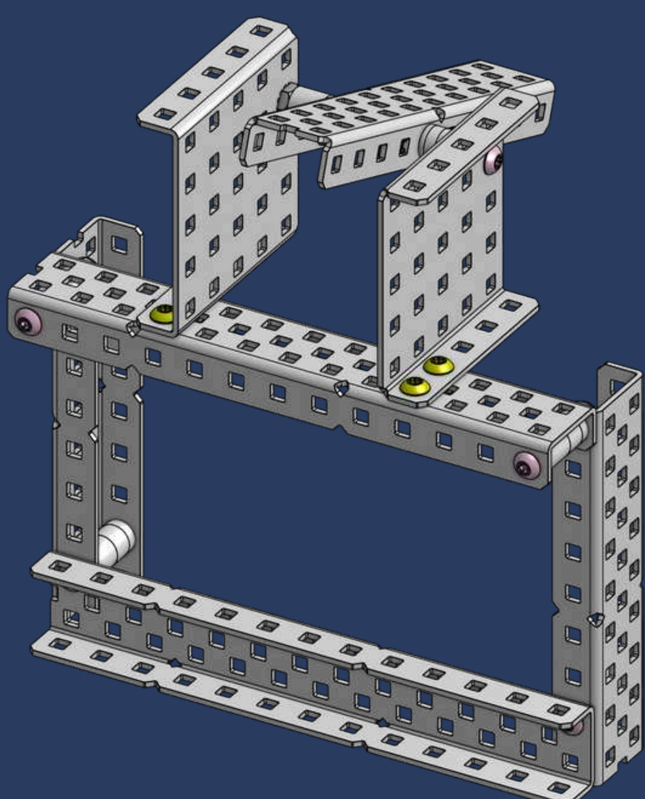
OCTO-CREDIBLE HACK

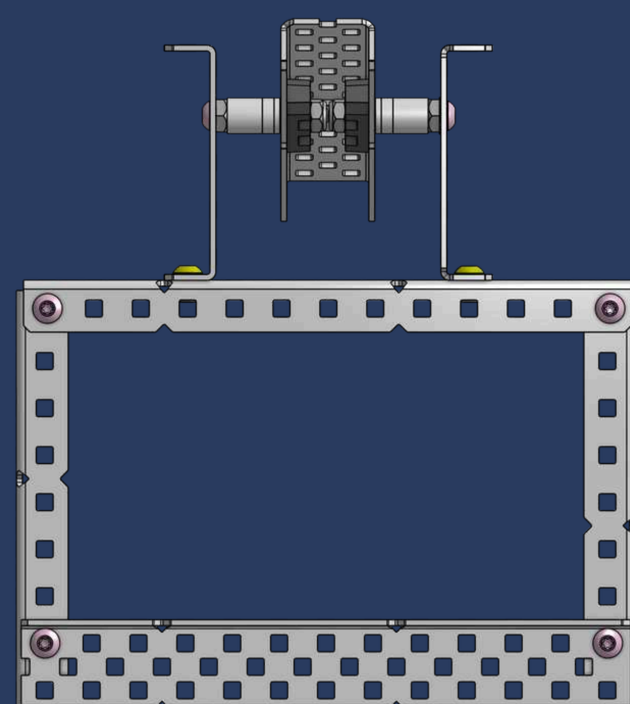
It’s a tight fit, but doable! Follow step 23, then on the second screw, tighten the kep nut halfway, add spacers one at a time, adjust the kep nut after each spacer, and fully tighten it after all spacers are in place!

CHECKPOINT #2

008







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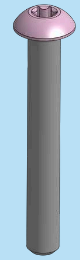
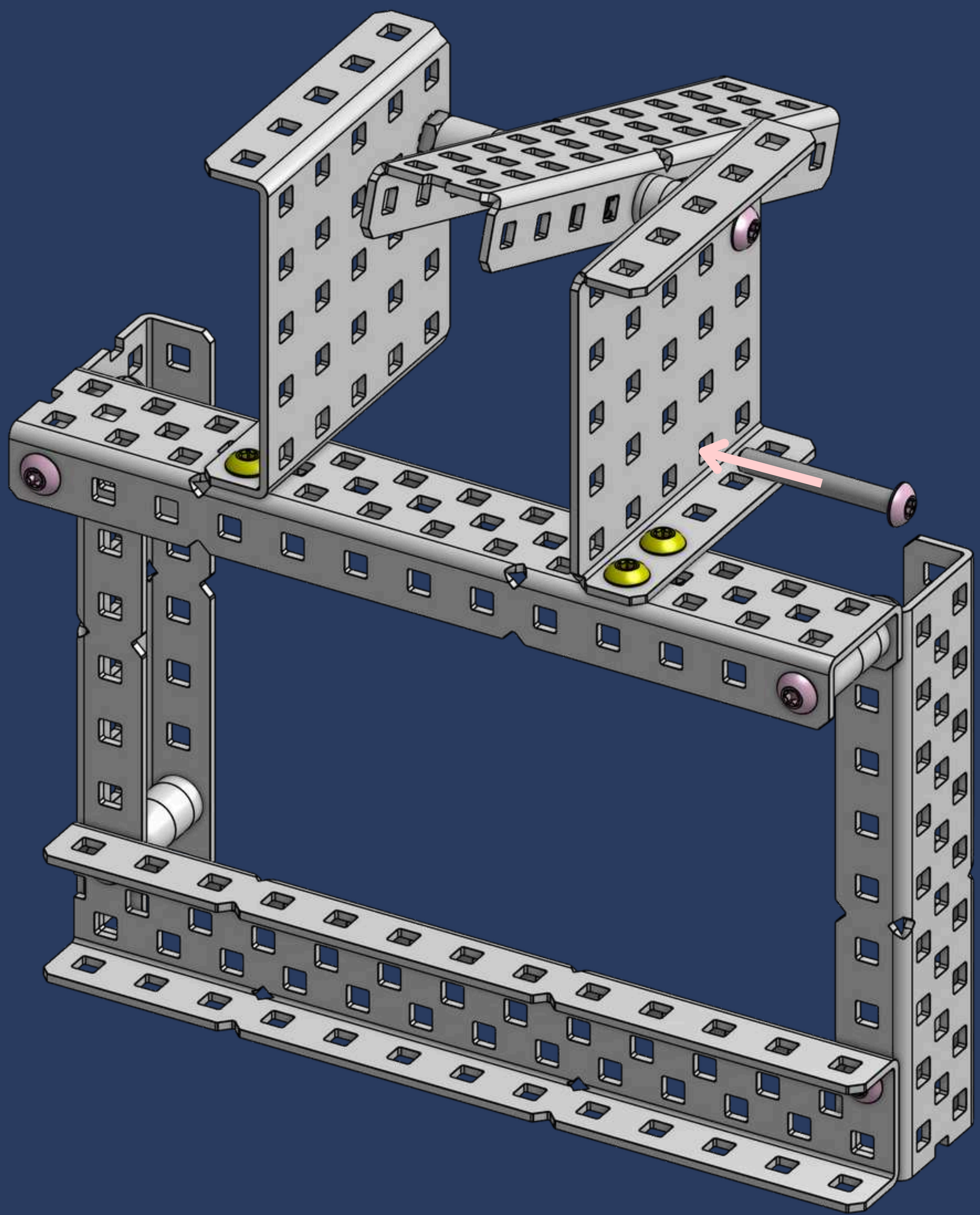


1x
#8-32 x 1-1/84" Star Drive Screw



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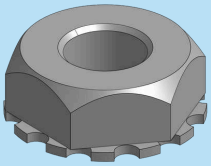
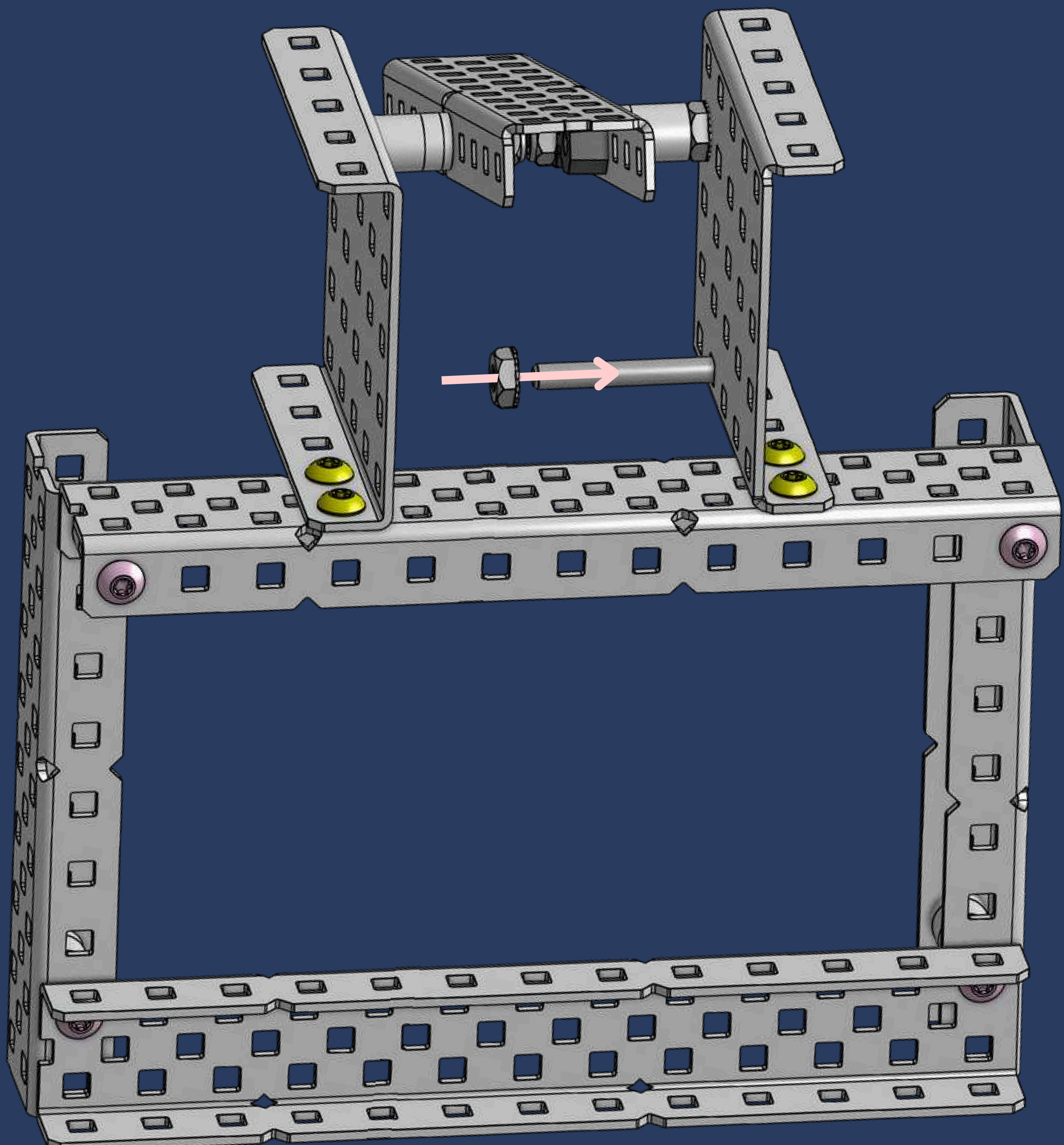


1x
Keps Nut



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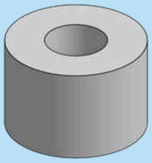
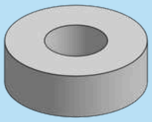
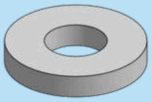


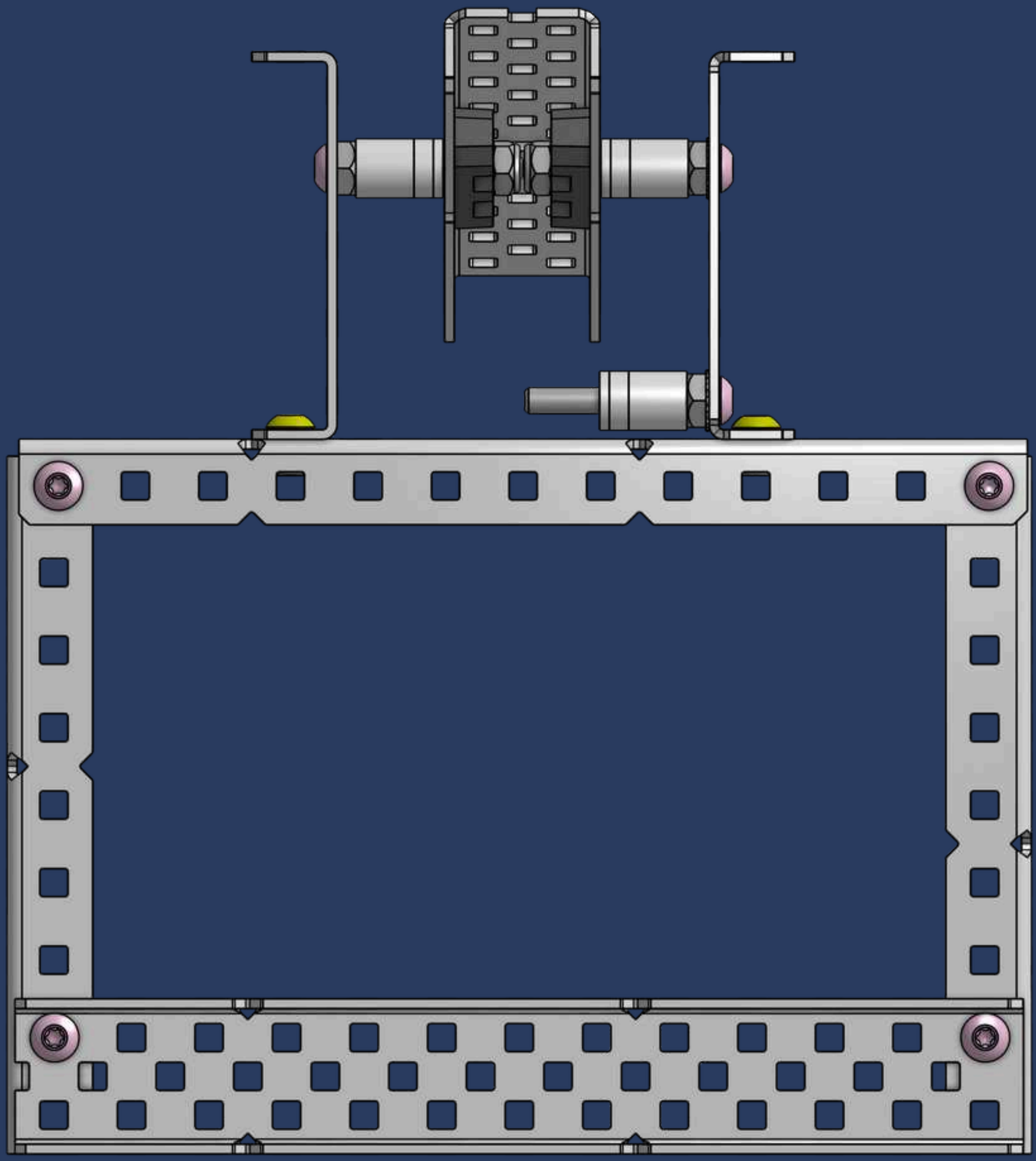
1x
3/8" x 3/8" OD x #8 Nylon Spacer



1x
1/8" x 3/8" OD x #8 Nylon Spacer



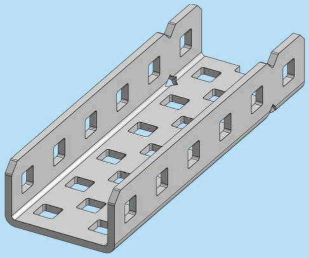
1x
1/16" x 3/8" OD x #8 Nylon Spacer



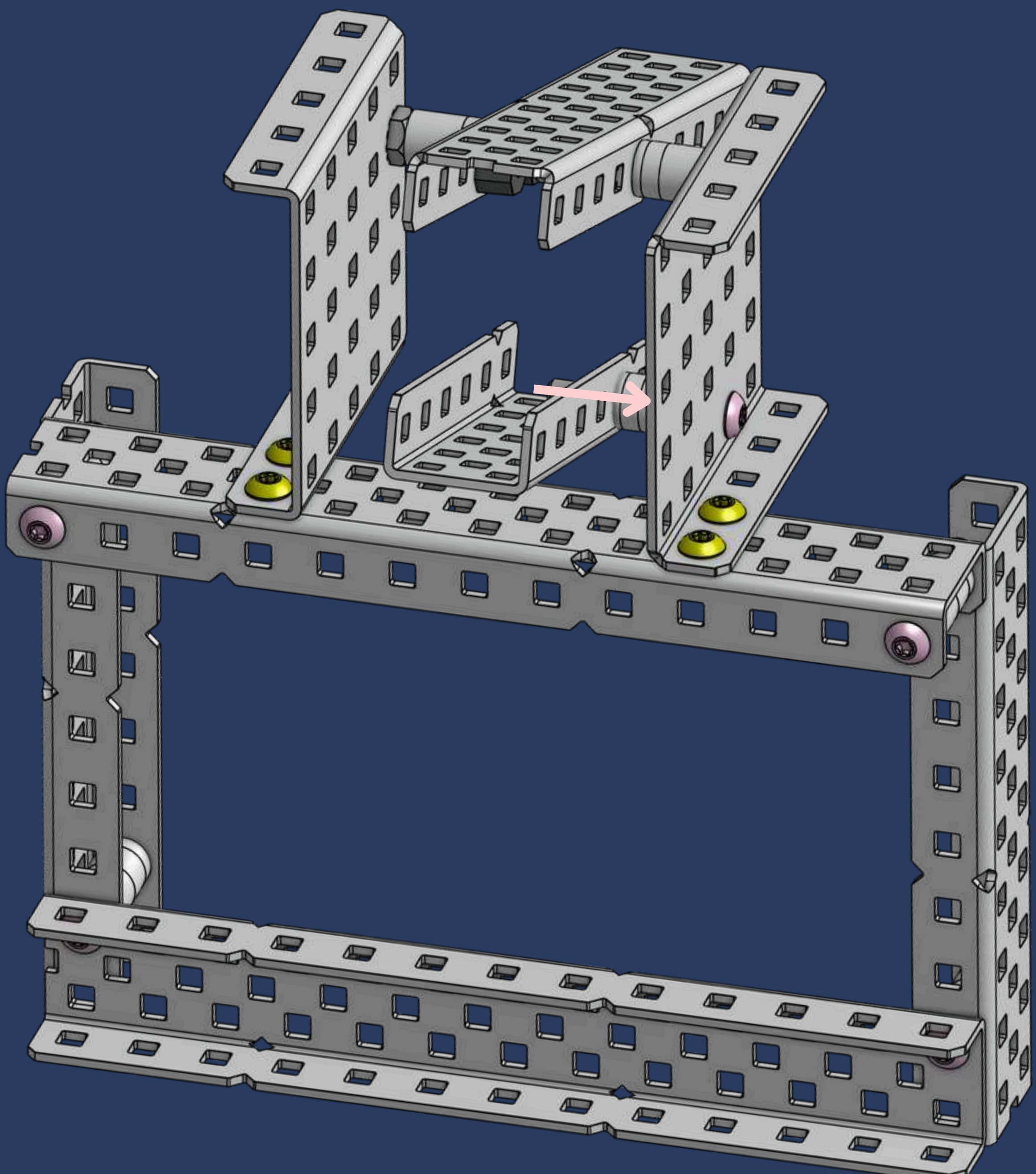
INK-SPIRING IDEA

If needed, **loosen nearby parts** for better access to your current step, like adjusting a c-channel to add spacers, then **shift everything back** once you're done.

31



1x
1x2x1x6 Aluminum C-Channel



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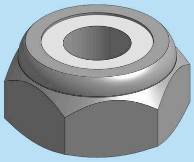
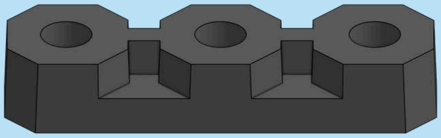
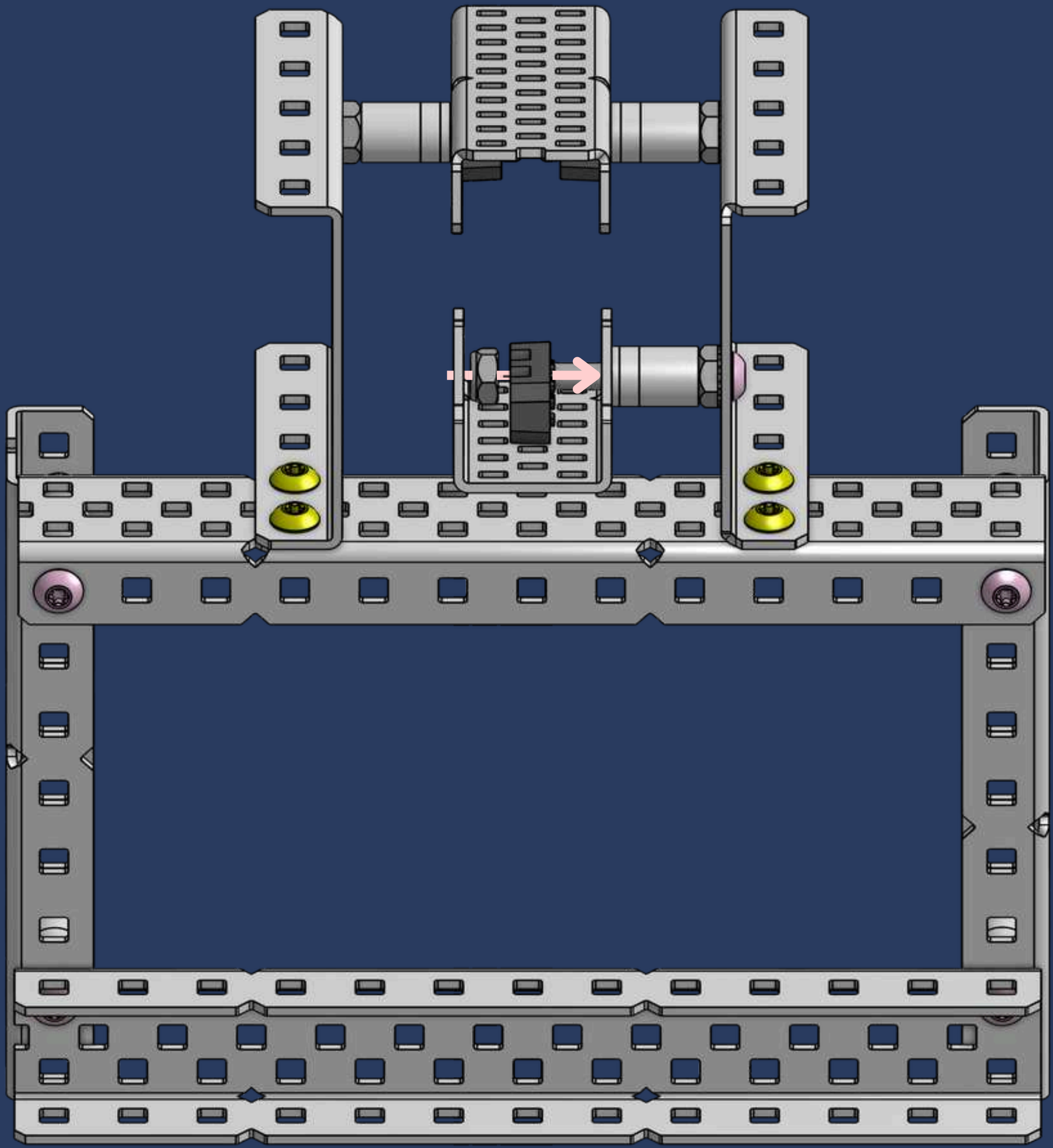


1x
Thin Nylock Nut



1x
Flat Bearing



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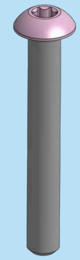
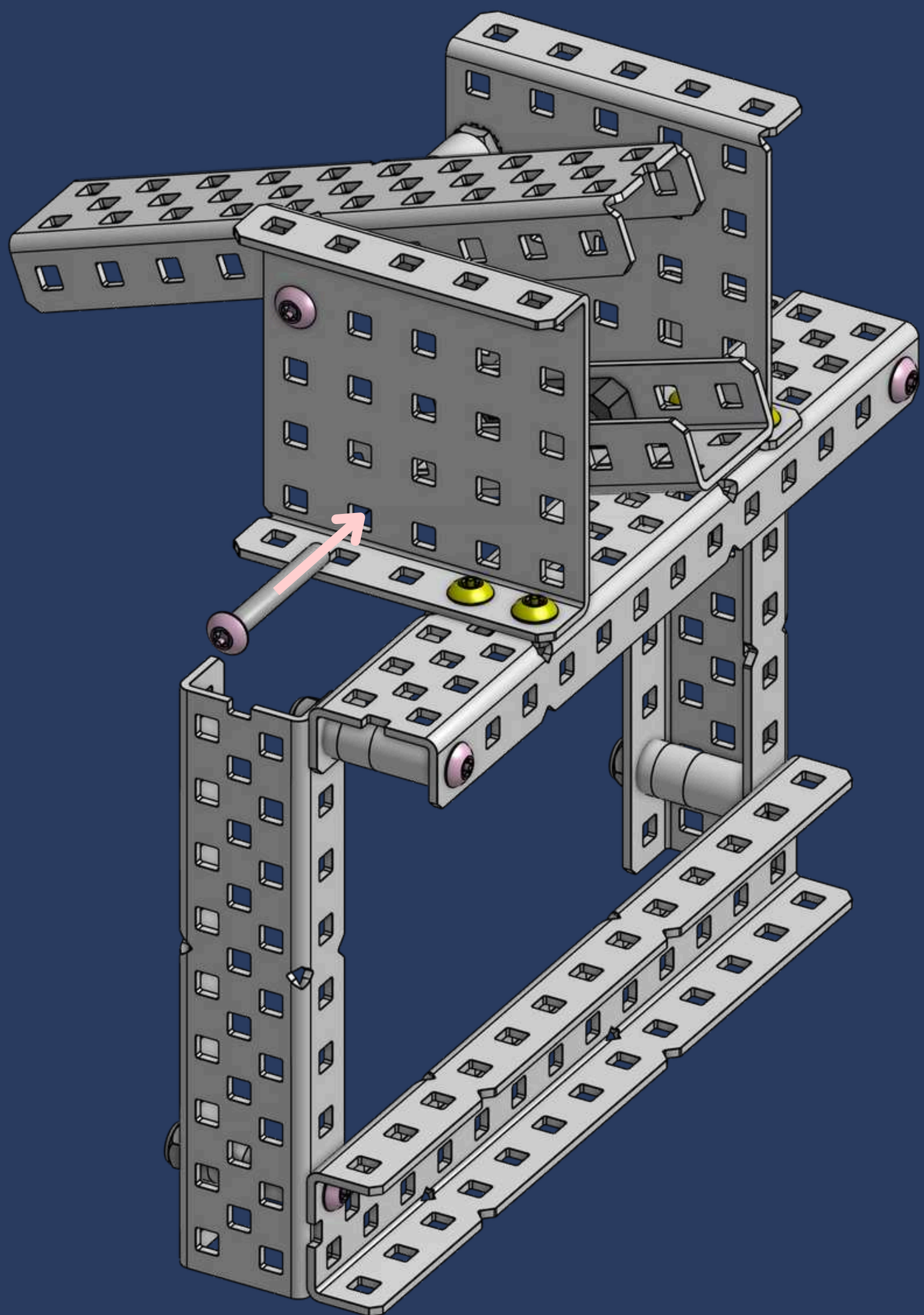


1x
#8-32 x 1-1/84" Star Drive Screw



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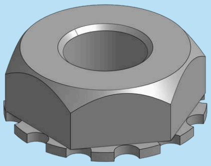
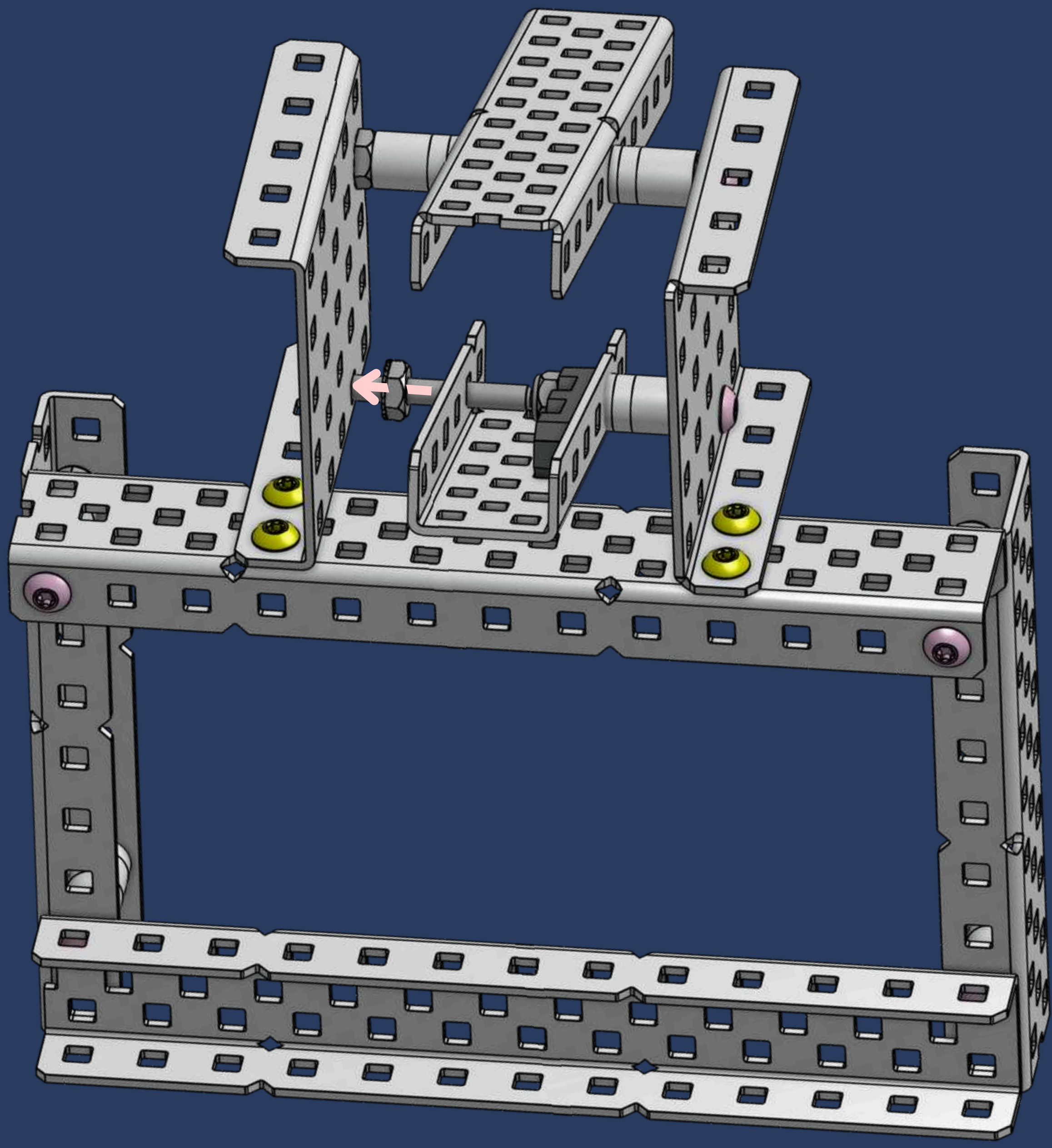


1x
Keps Nut



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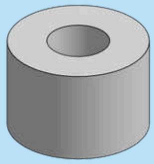
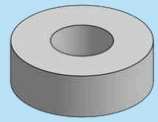
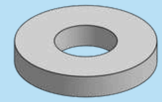


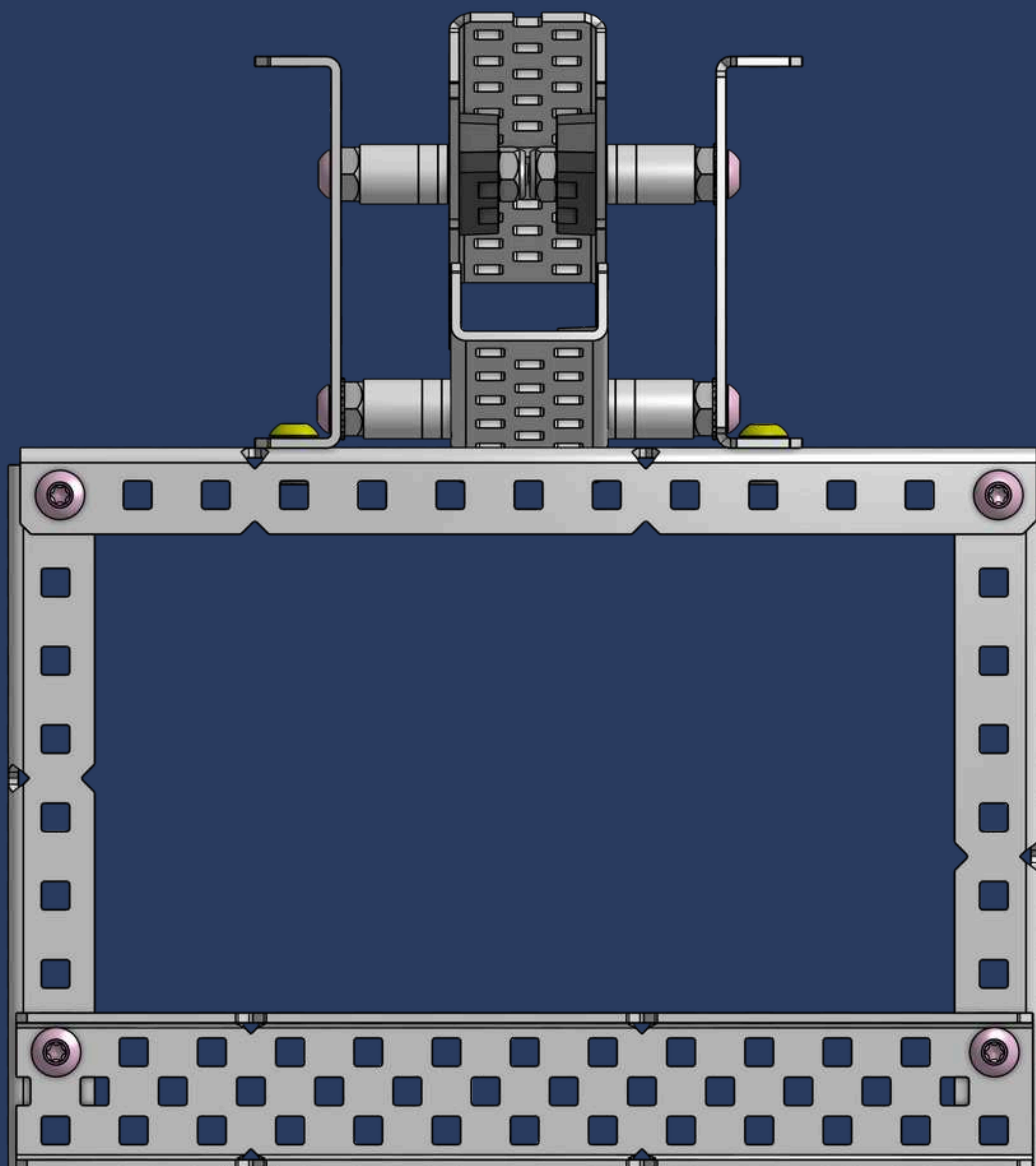
1x
3/8" x 3/8" OD x #8 Nylon Spacer



1x
1/8" x 3/8" OD x #8 Nylon Spacer



1x
1/16" x 3/8" OD x #8 Nylon Spacer



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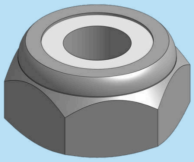
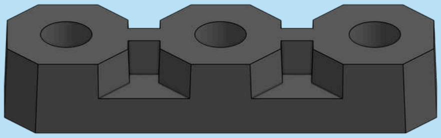
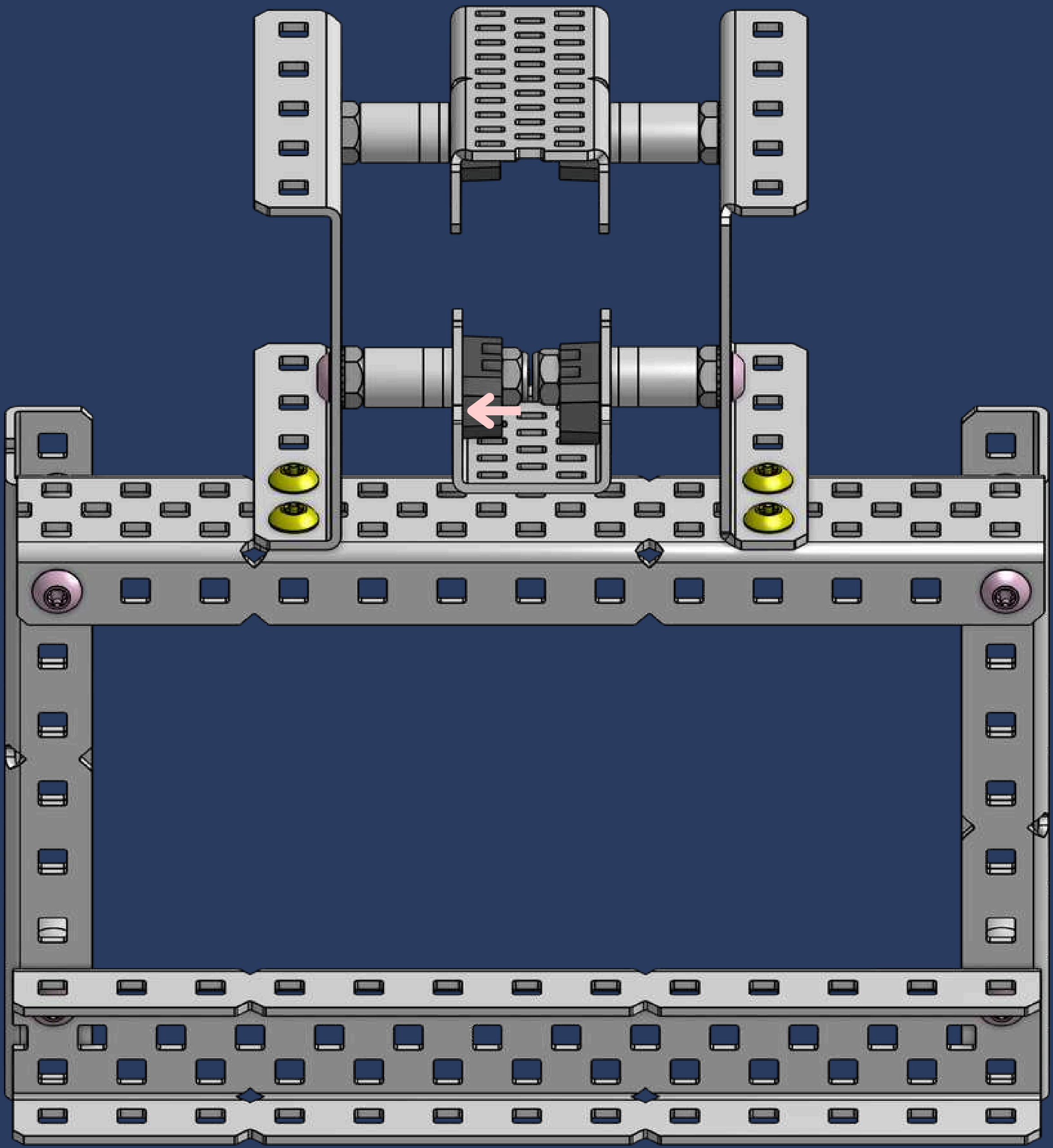


1x
Thin Nylock Nut

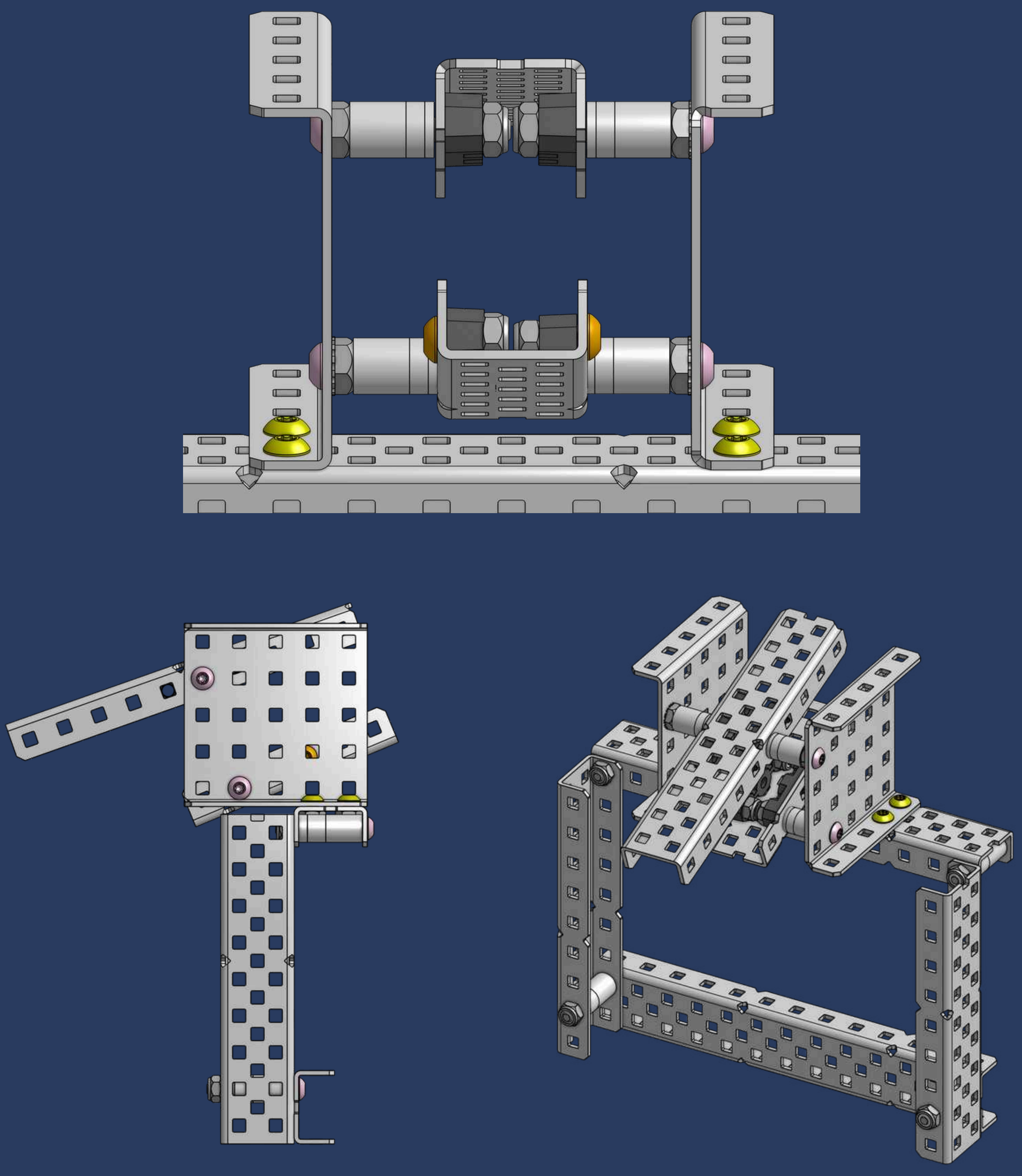


1x
Flat Bearing



CHECKPOINT #3

009



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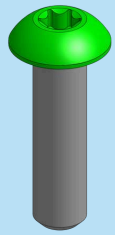
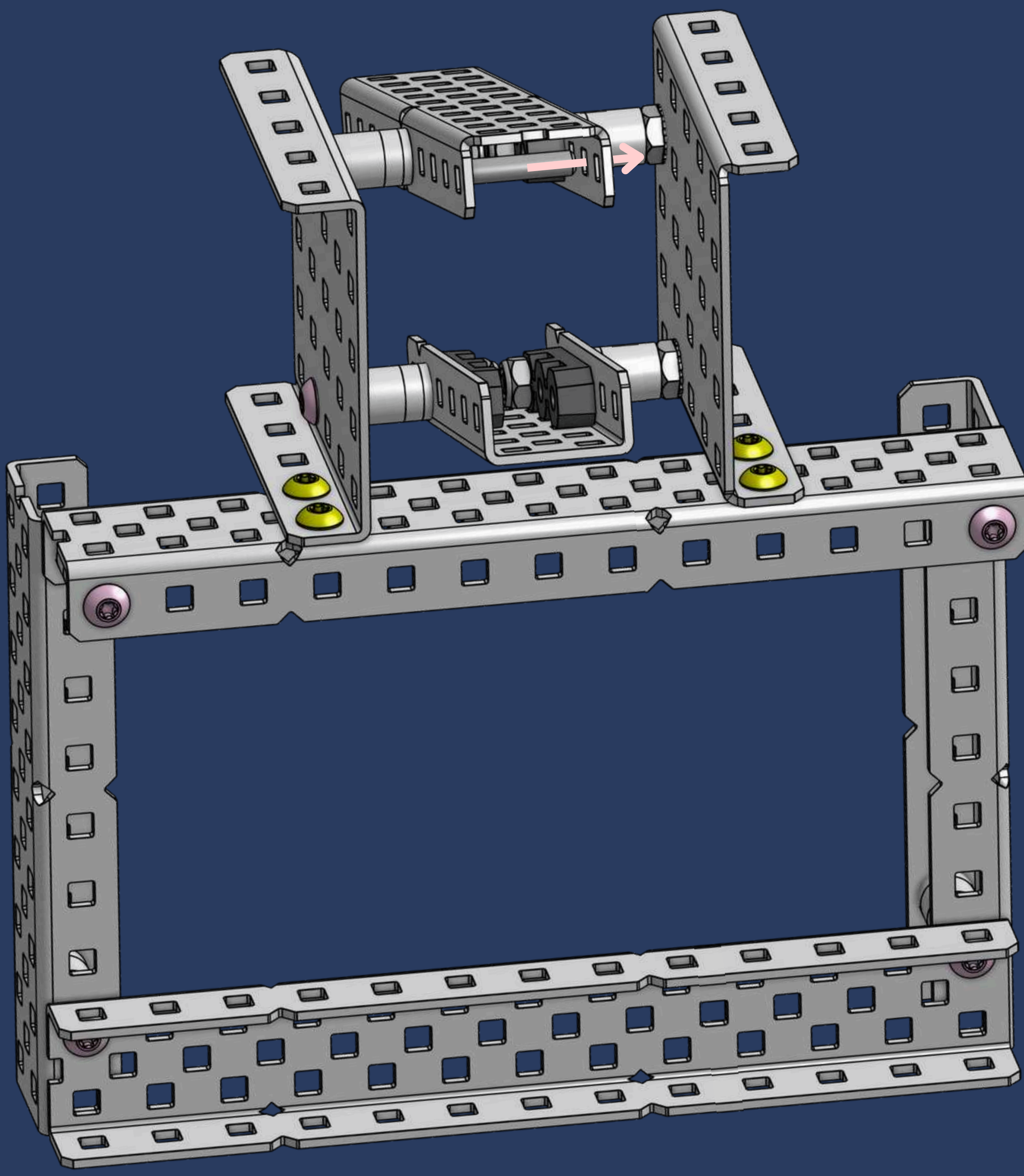


1x
#8-32 x 5/8" Star Drive Screw



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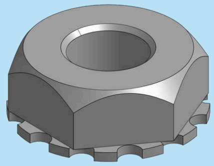
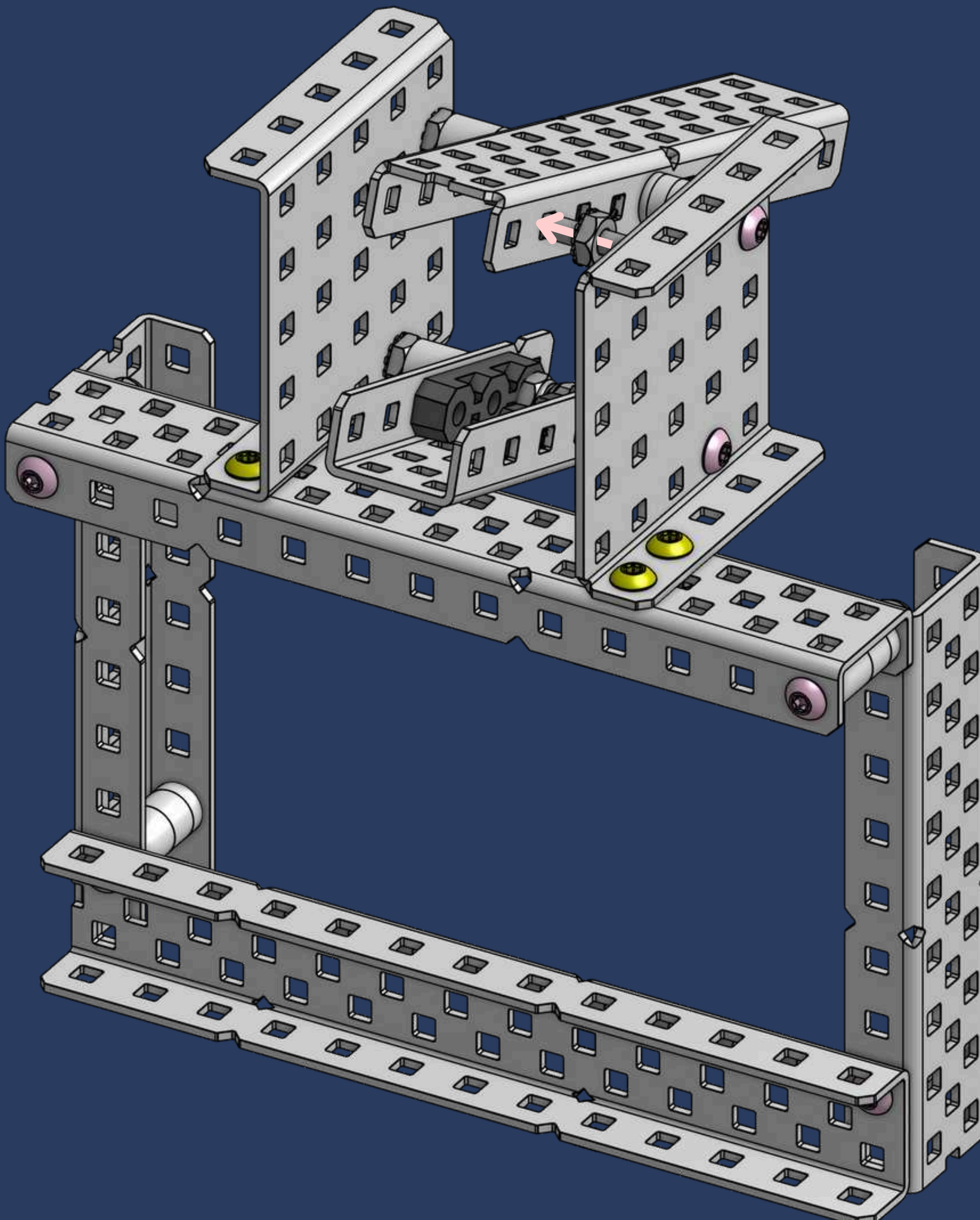


1x
Keps Nut



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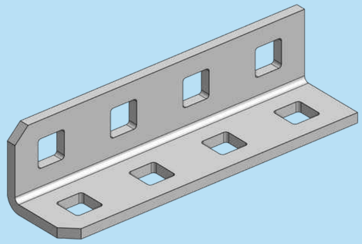
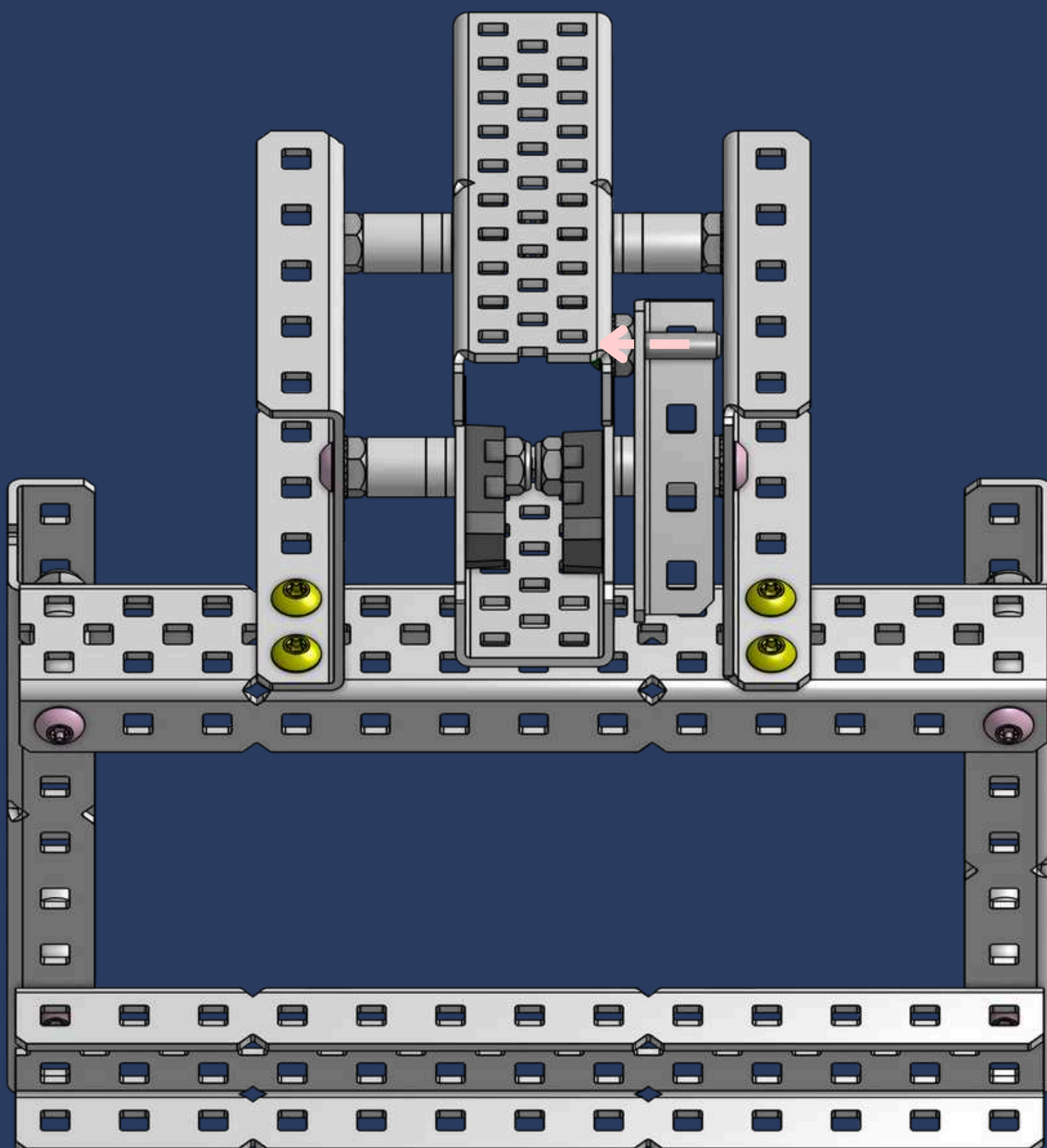


1x
1x1x4 Aluminum Angle



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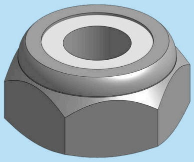
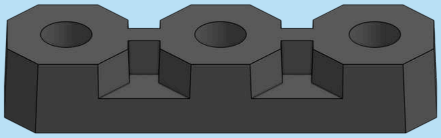
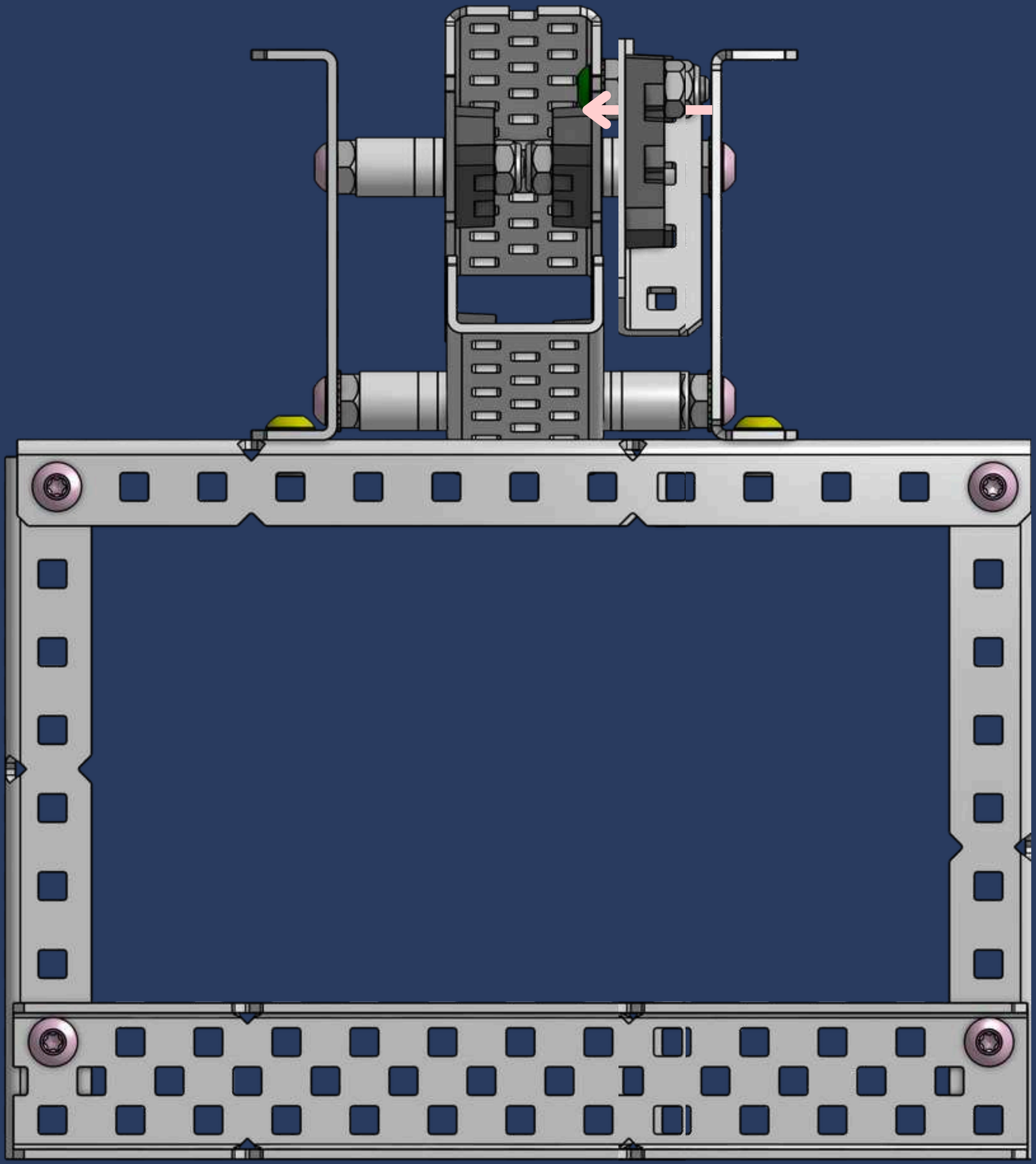


1x
Thin Nylock Nut



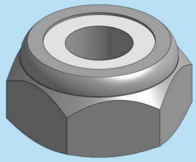
1x
Flat Bearing



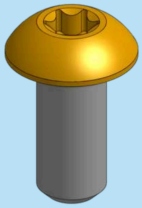
TENTACULAR SIGHT

The part you just built is the **lever** that **locks** the clamp! When the piston activates, the lever locks at an angle, making it **difficult to unlock**.

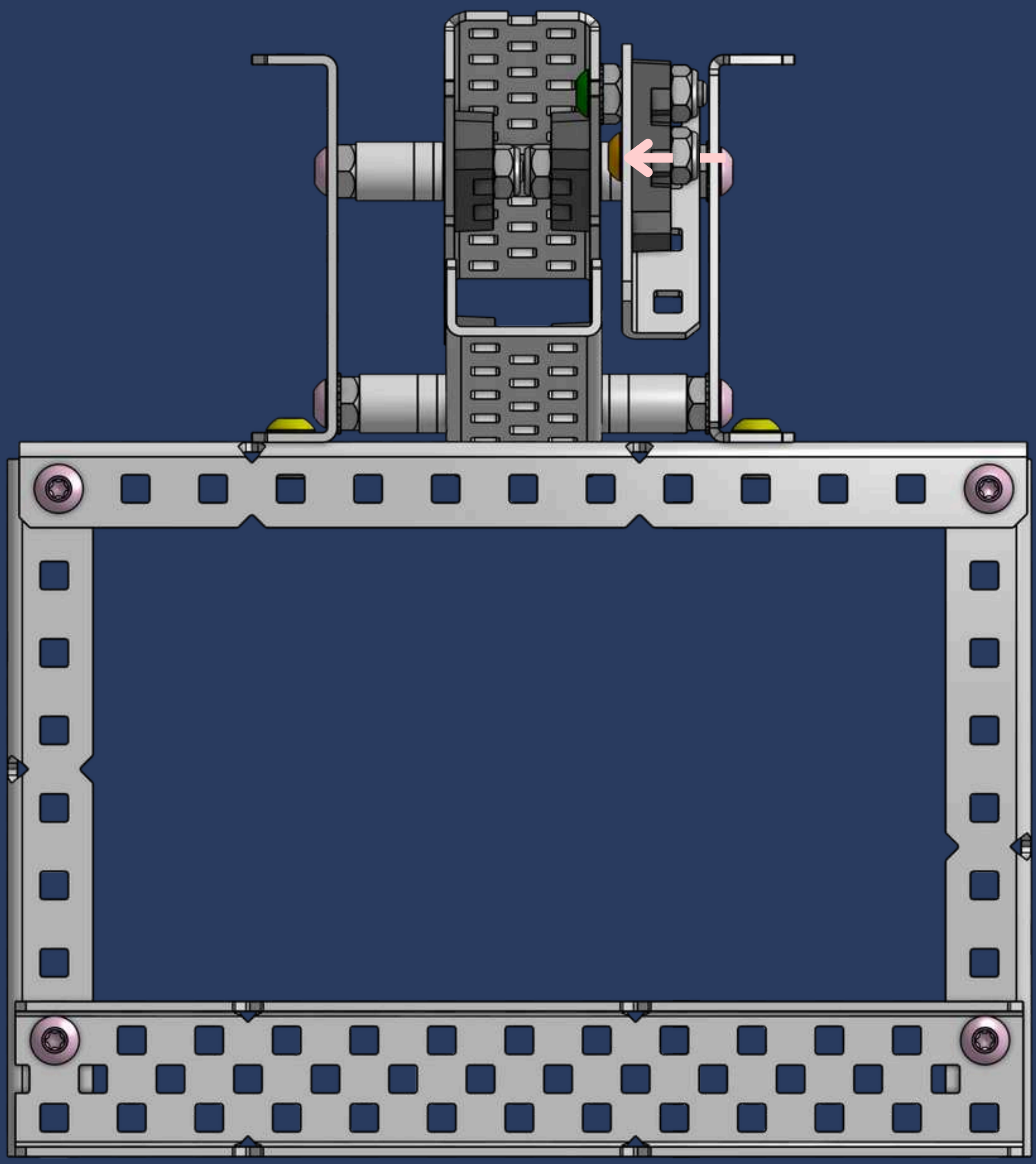
41



1x
Thin Nylock Nut



1x
#8-32 x 3/8" Star Drive Screw



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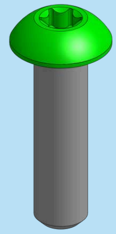
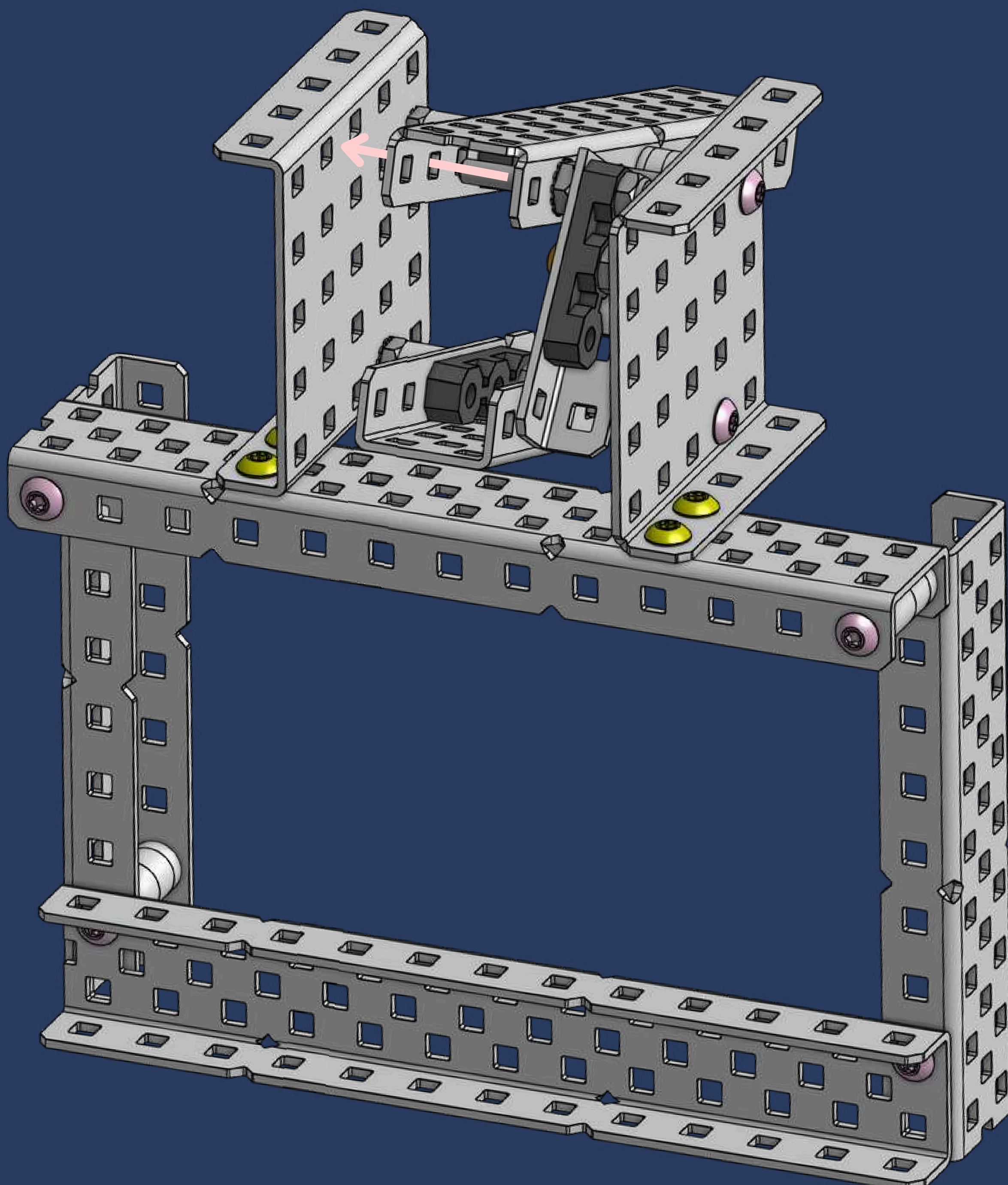


1x
#8-32 x 5/8" Star Drive Screw



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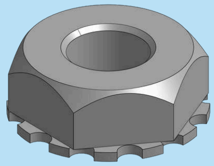
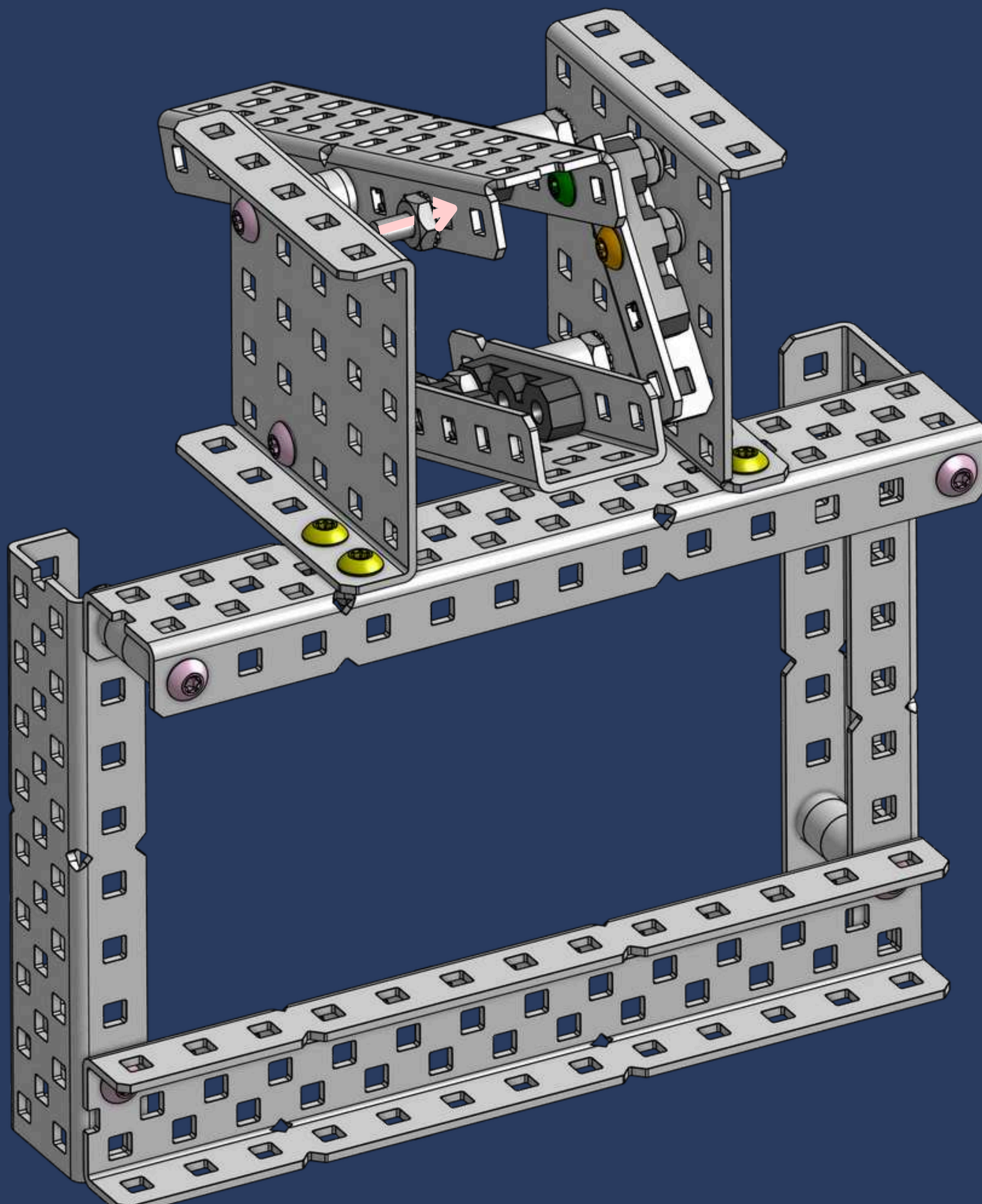


1x
Keps Nut



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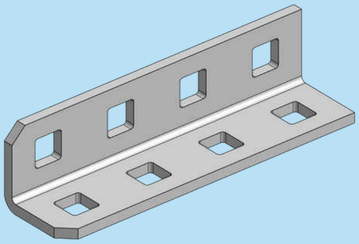
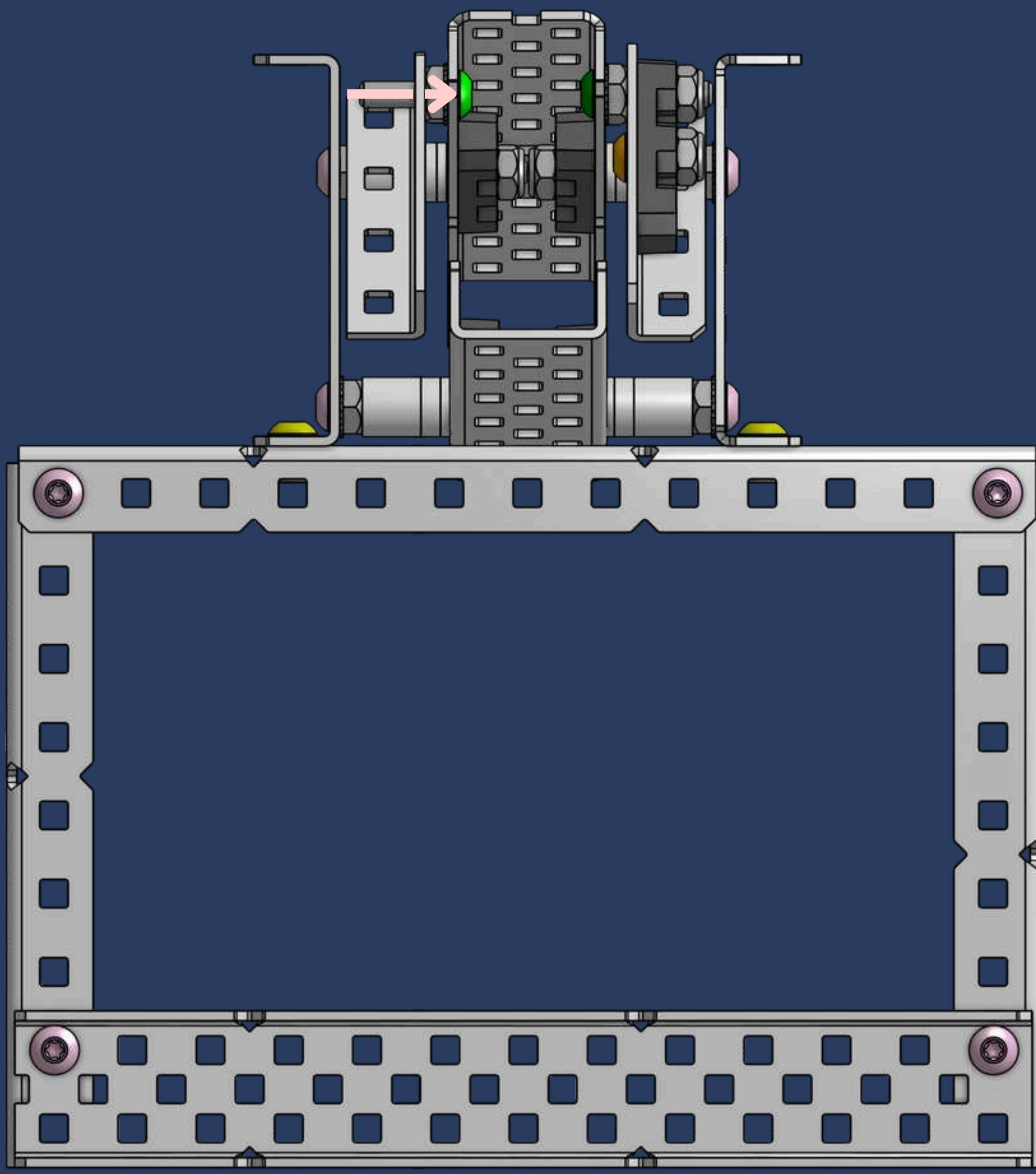


1x
1x1x4 Aluminum Angle



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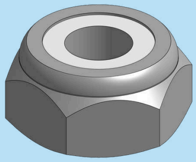
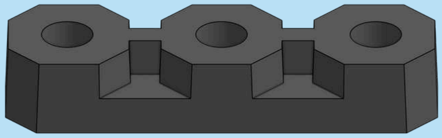
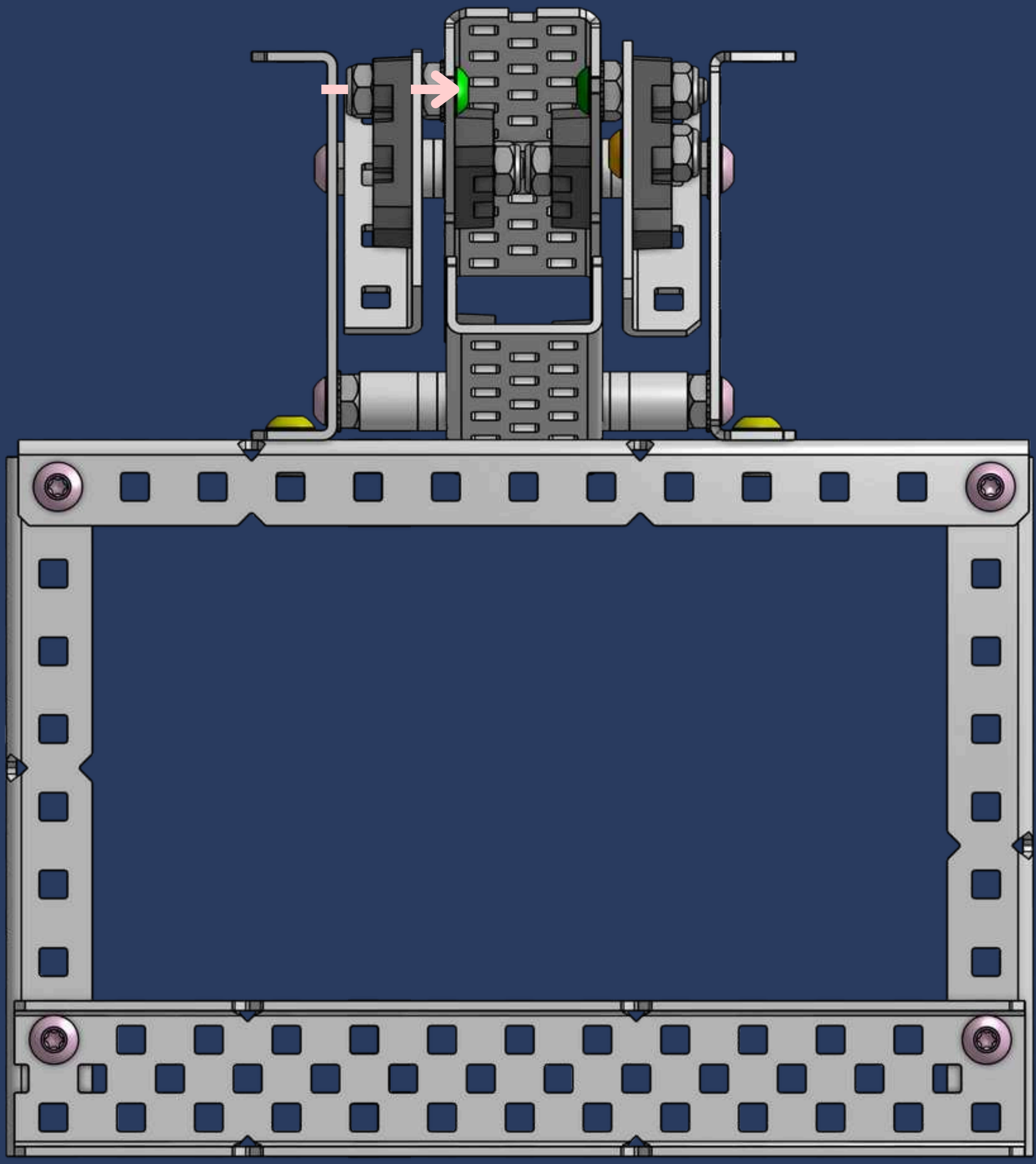


1x
Thin Nylock Nut



1x
Flat Bearing



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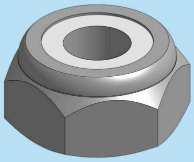
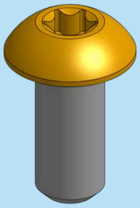
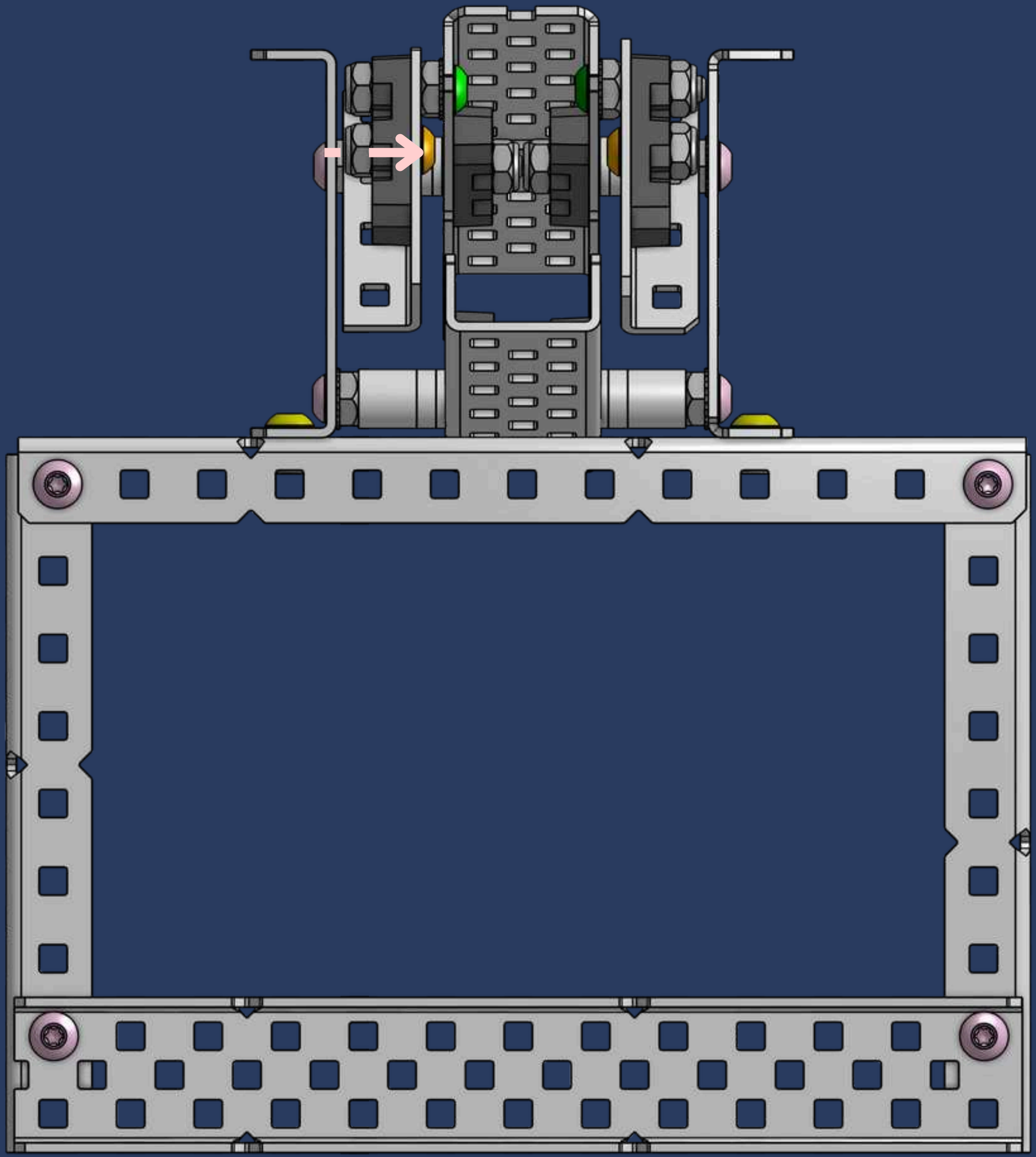


1x
Thin Nylock Nut

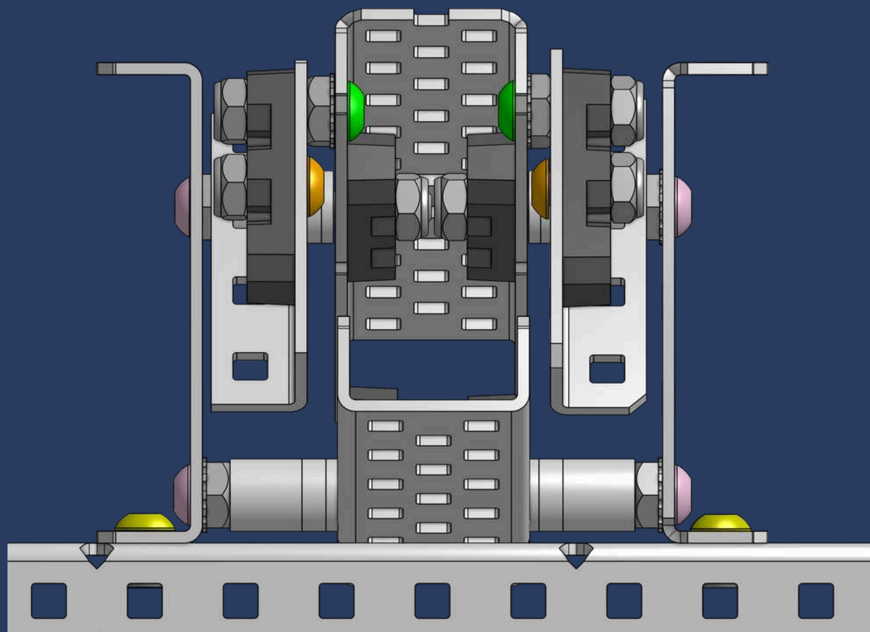


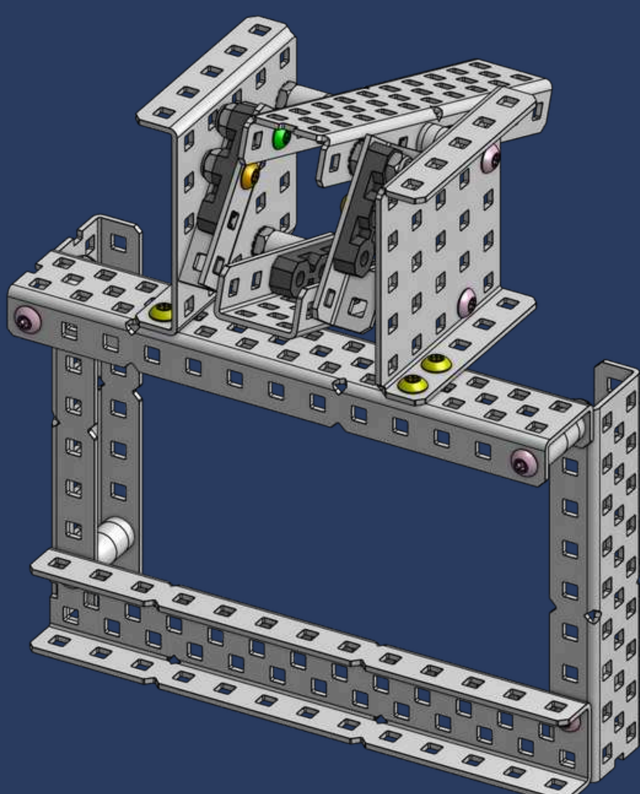
1x
#8-32 x 3/8" Star Drive Screw

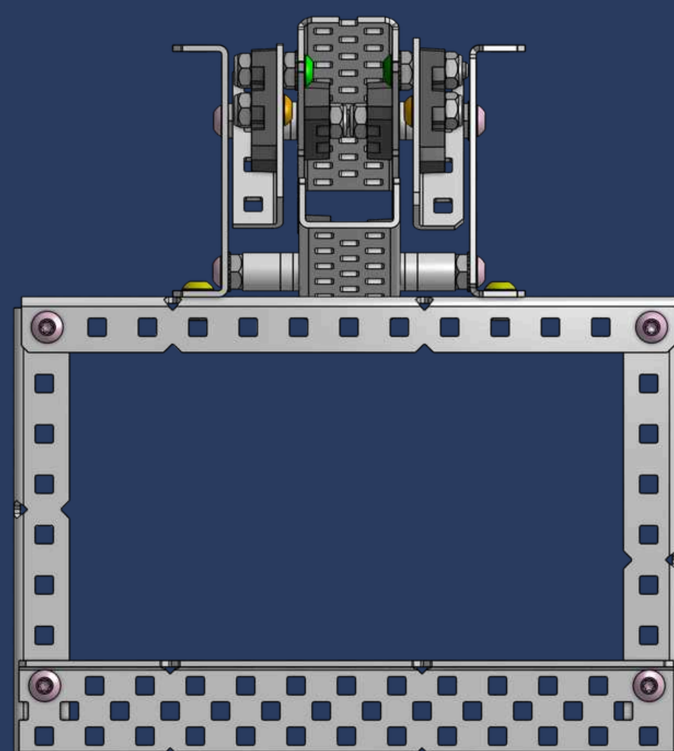


CHECKPOINT #4

010







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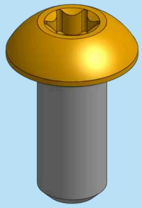
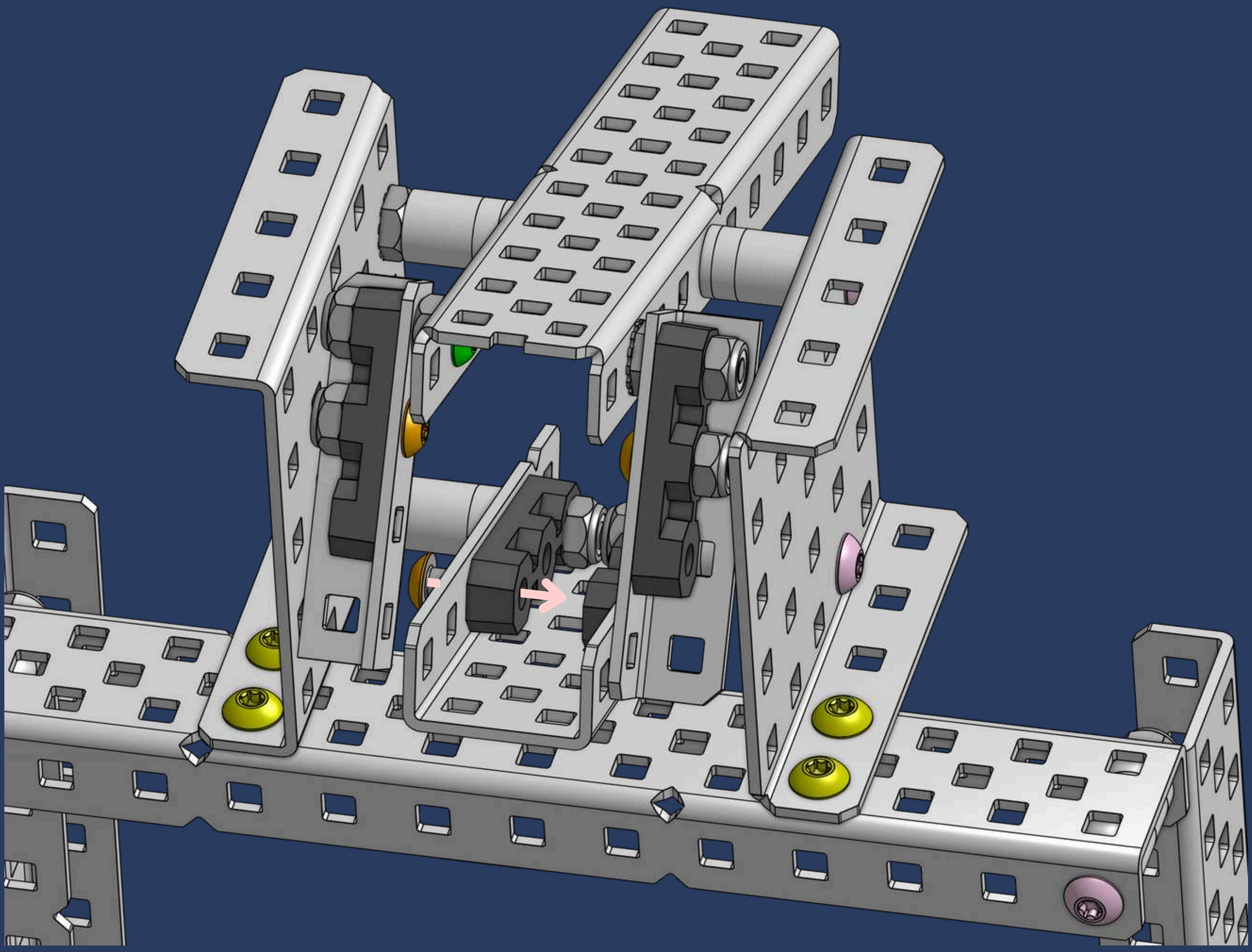


1x
#8-32 x 3/8" Star Drive Screw



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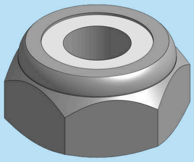
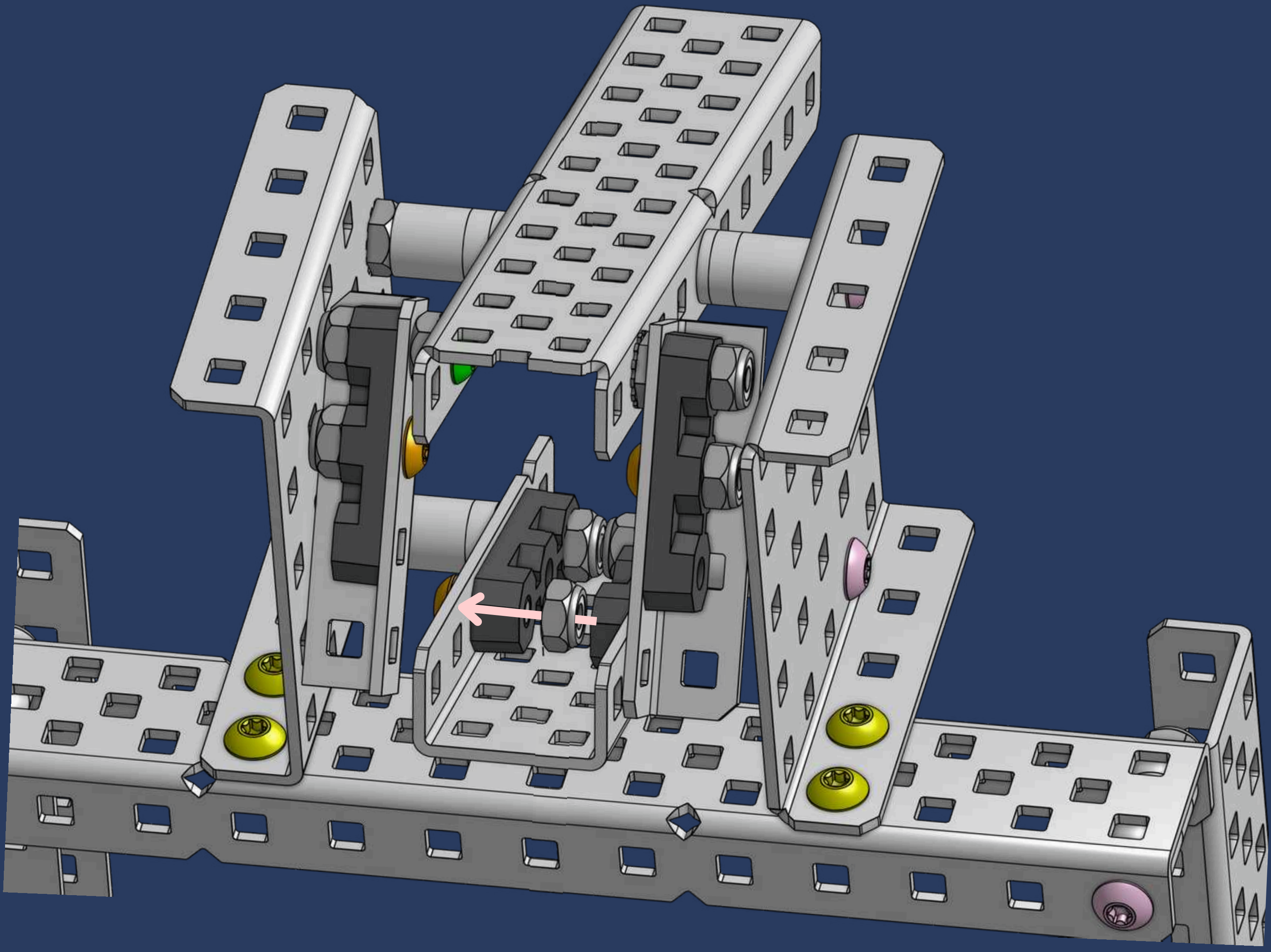


1x
Thin Nylock Nut



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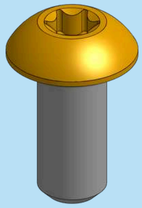
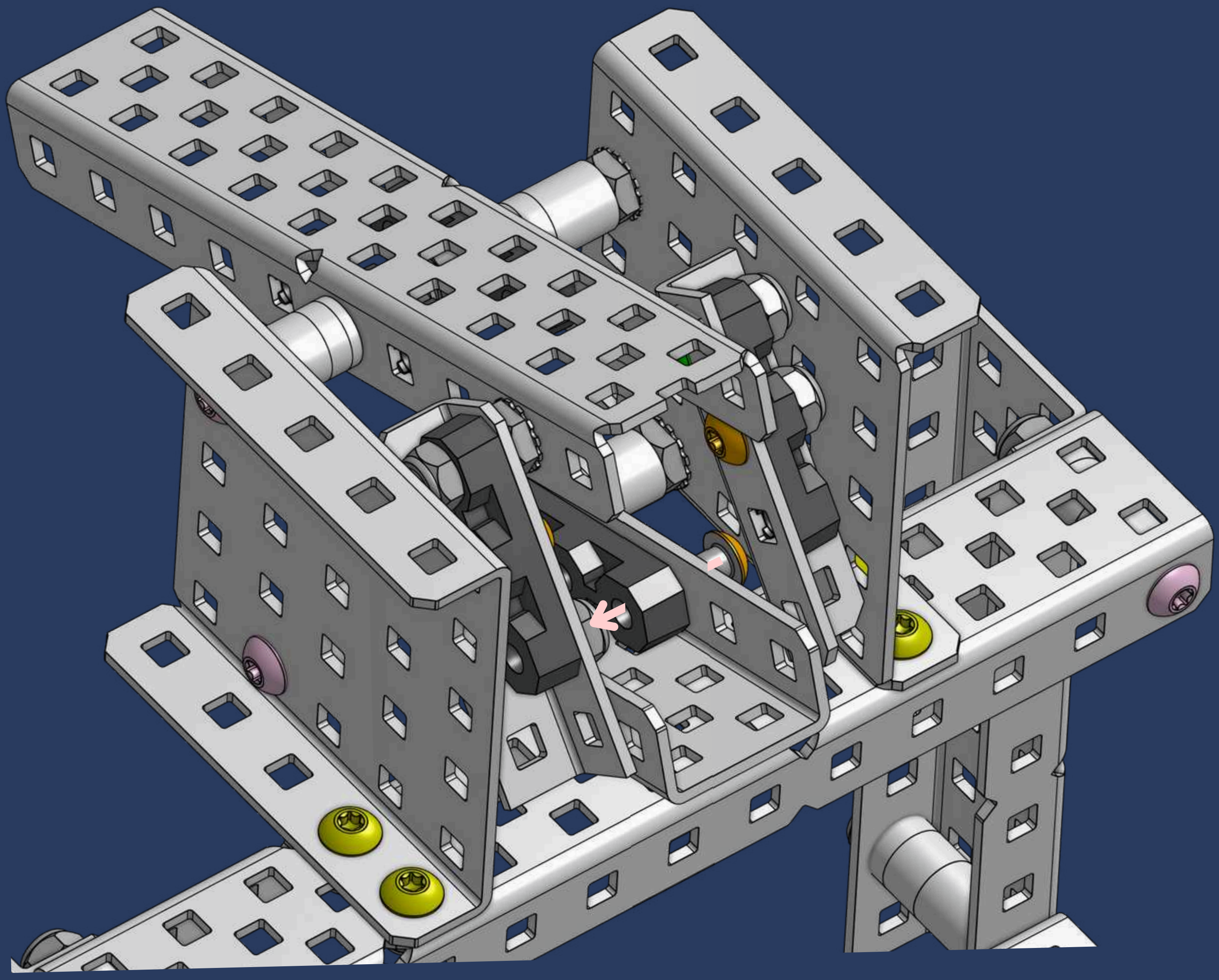


1x
#8-32 x 3/8" Star Drive Screw



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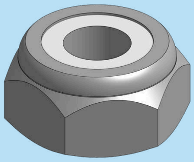
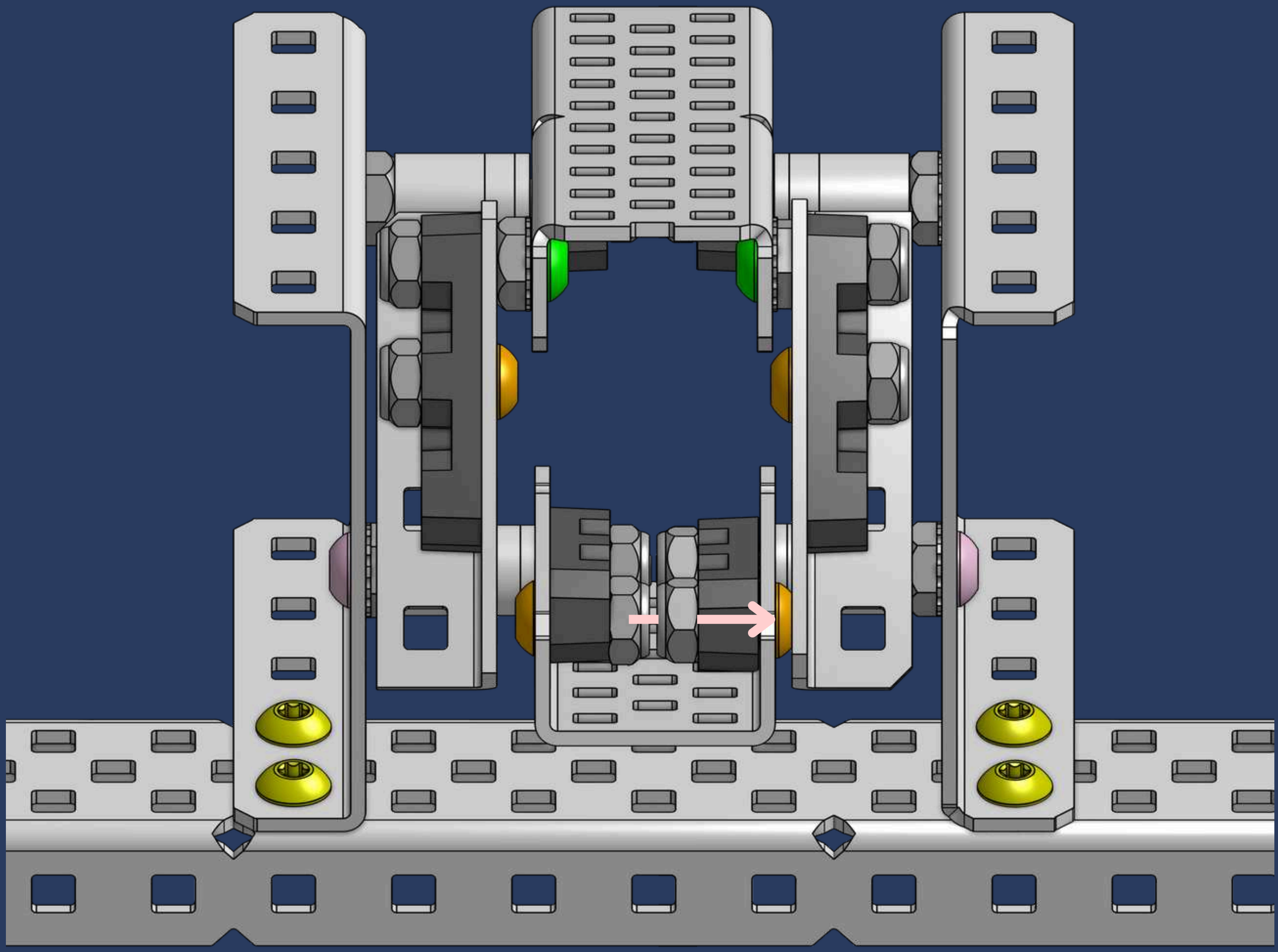


1x
Thin Nylock Nut



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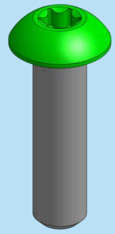
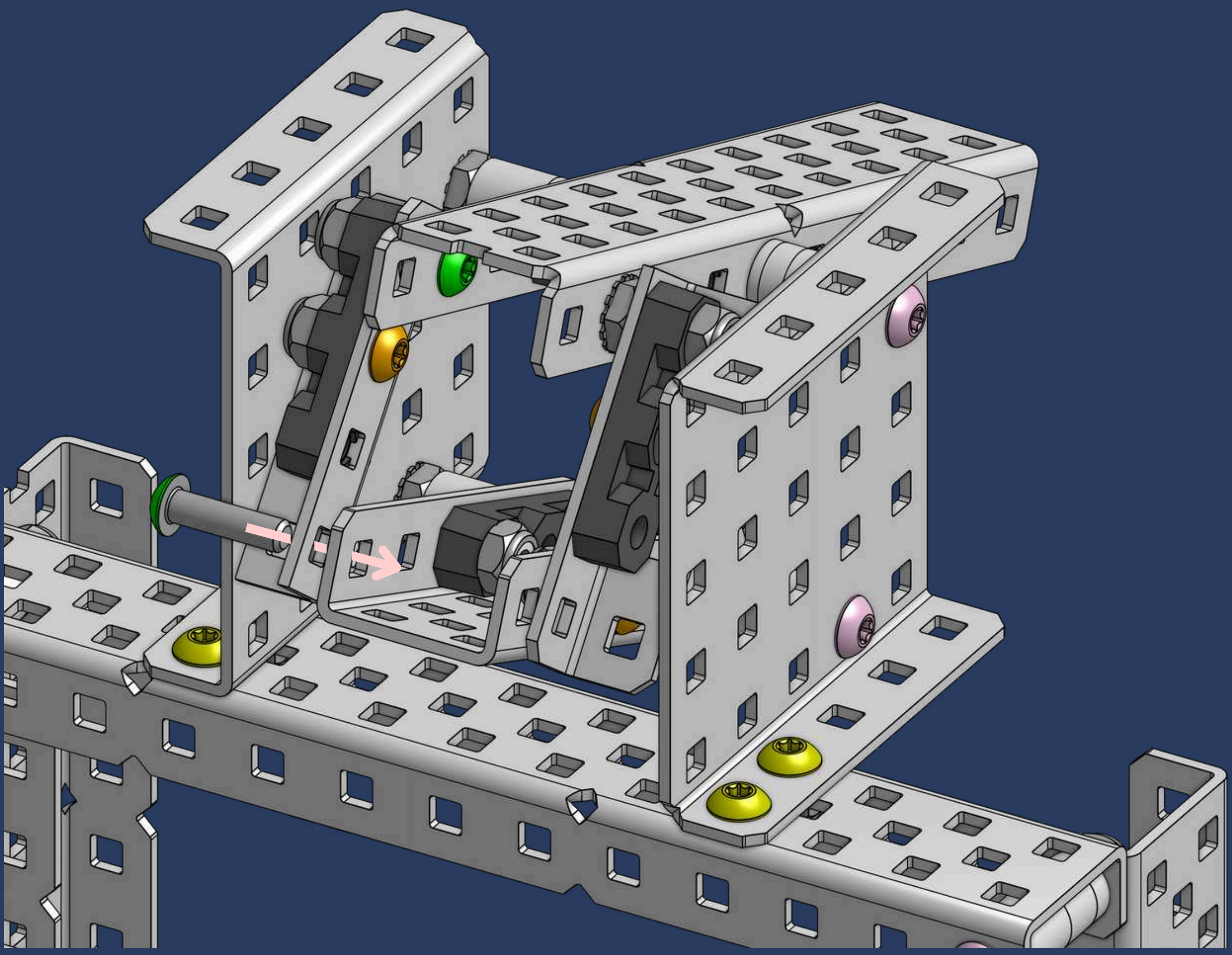


1x
#8-32 x 5/8" Star Drive Screw



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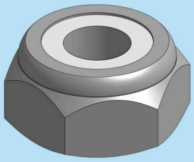
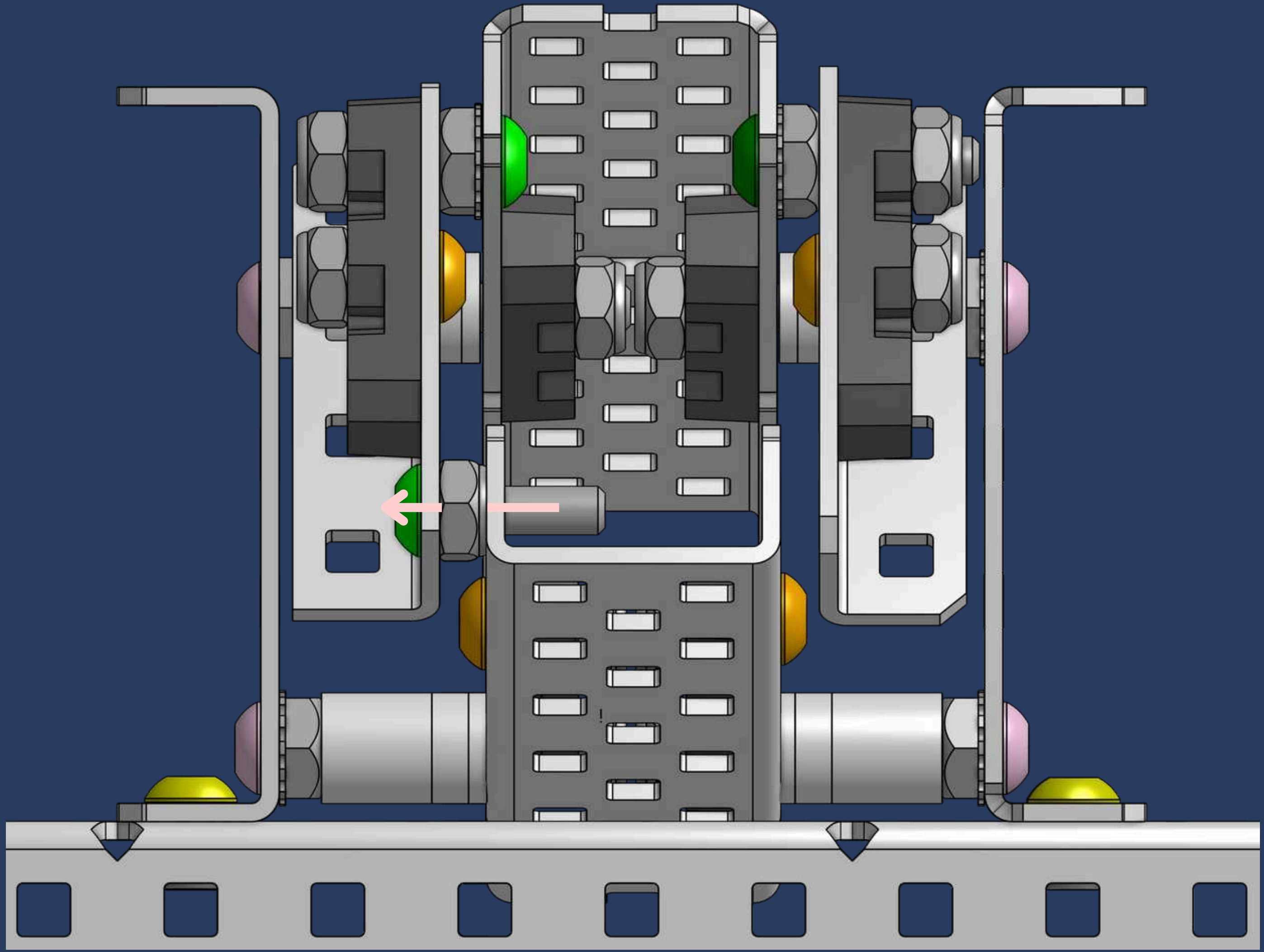


1x
Thin Nylock Nut



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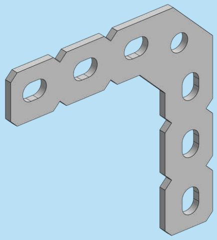
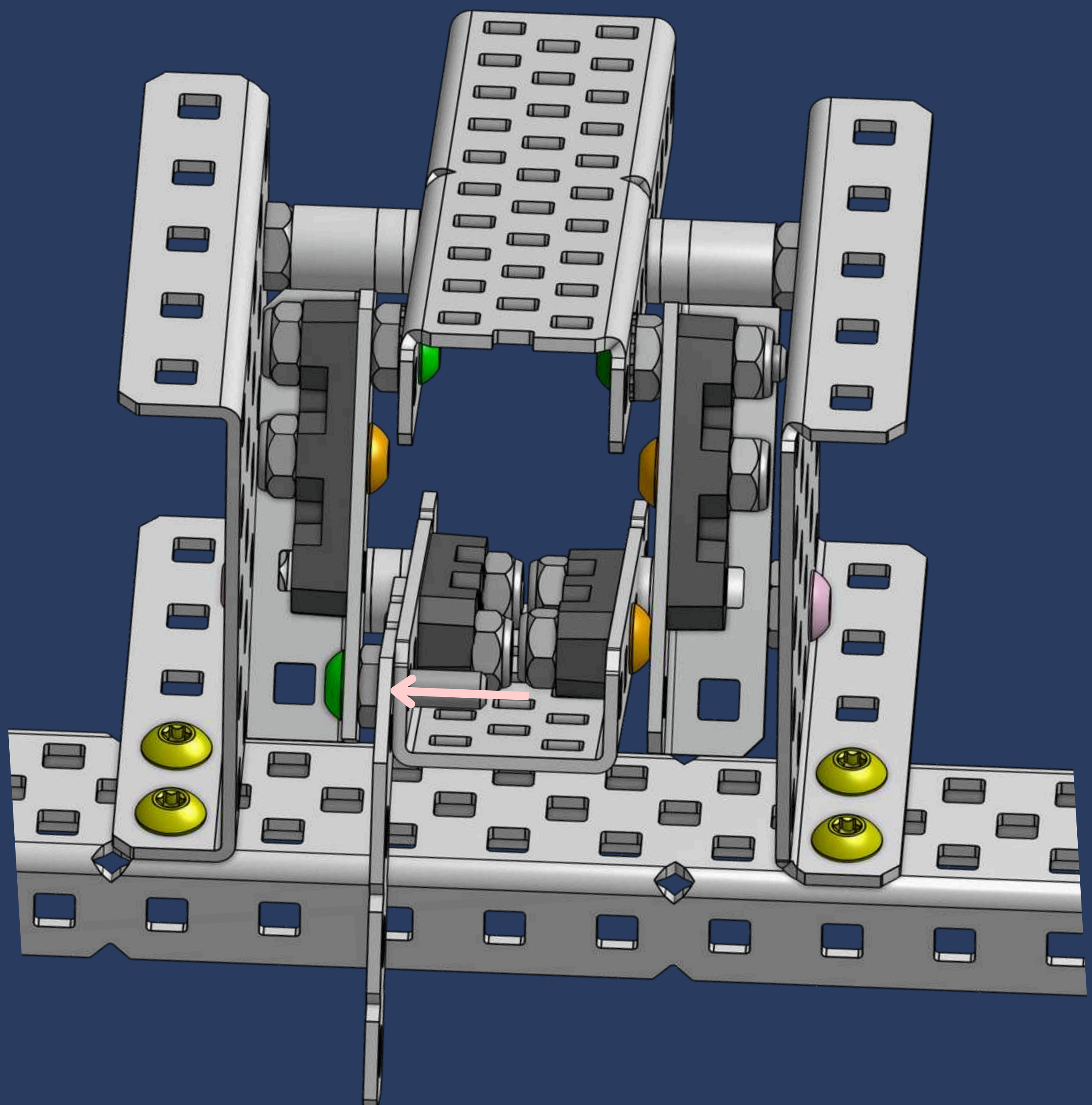


1x
90 Degree Large Flat Gusset



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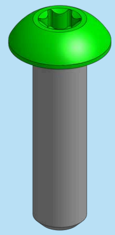
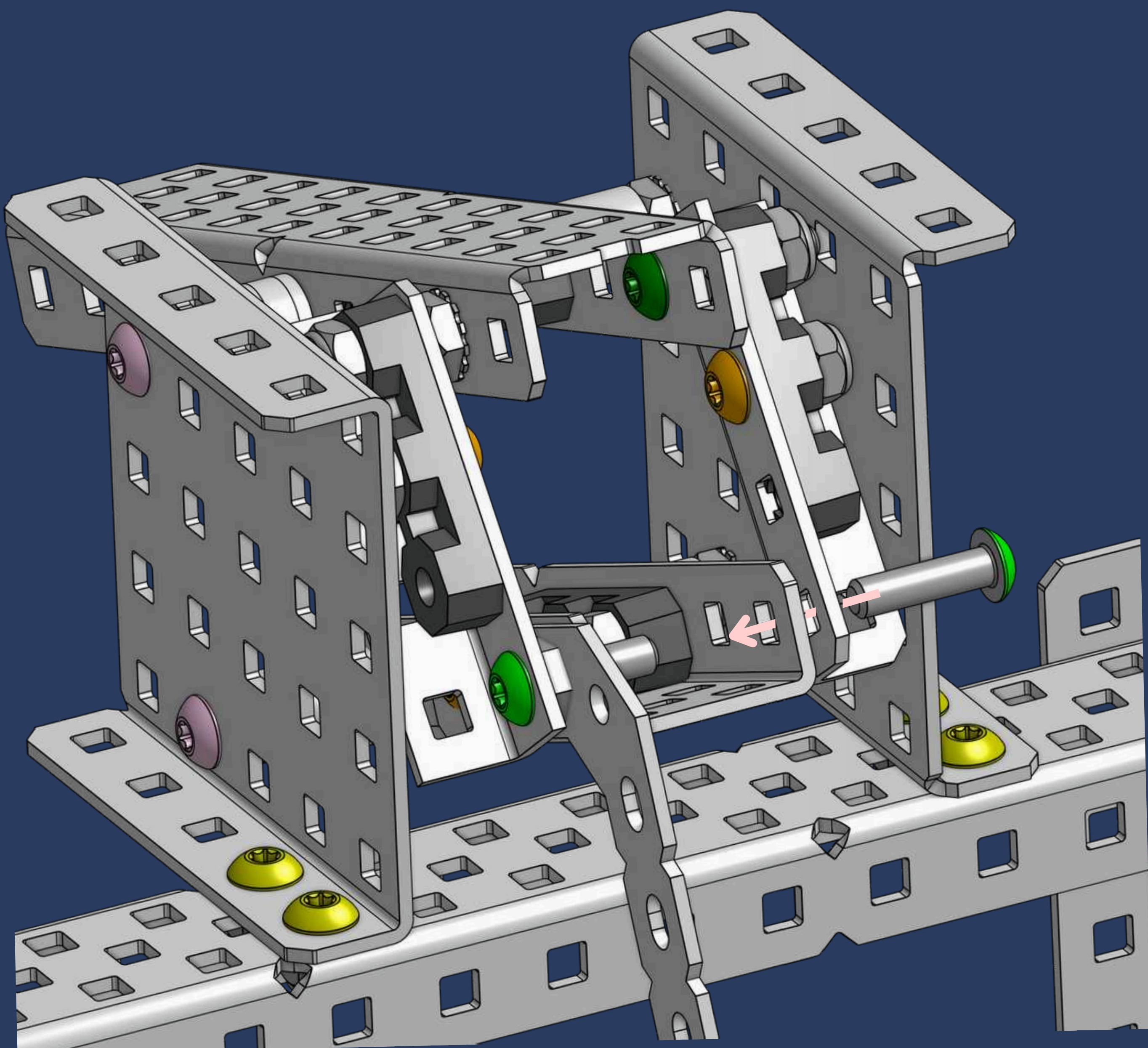


1x
#8-32 x 5/8" Star Drive Screw



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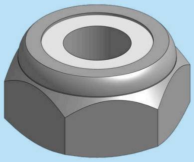
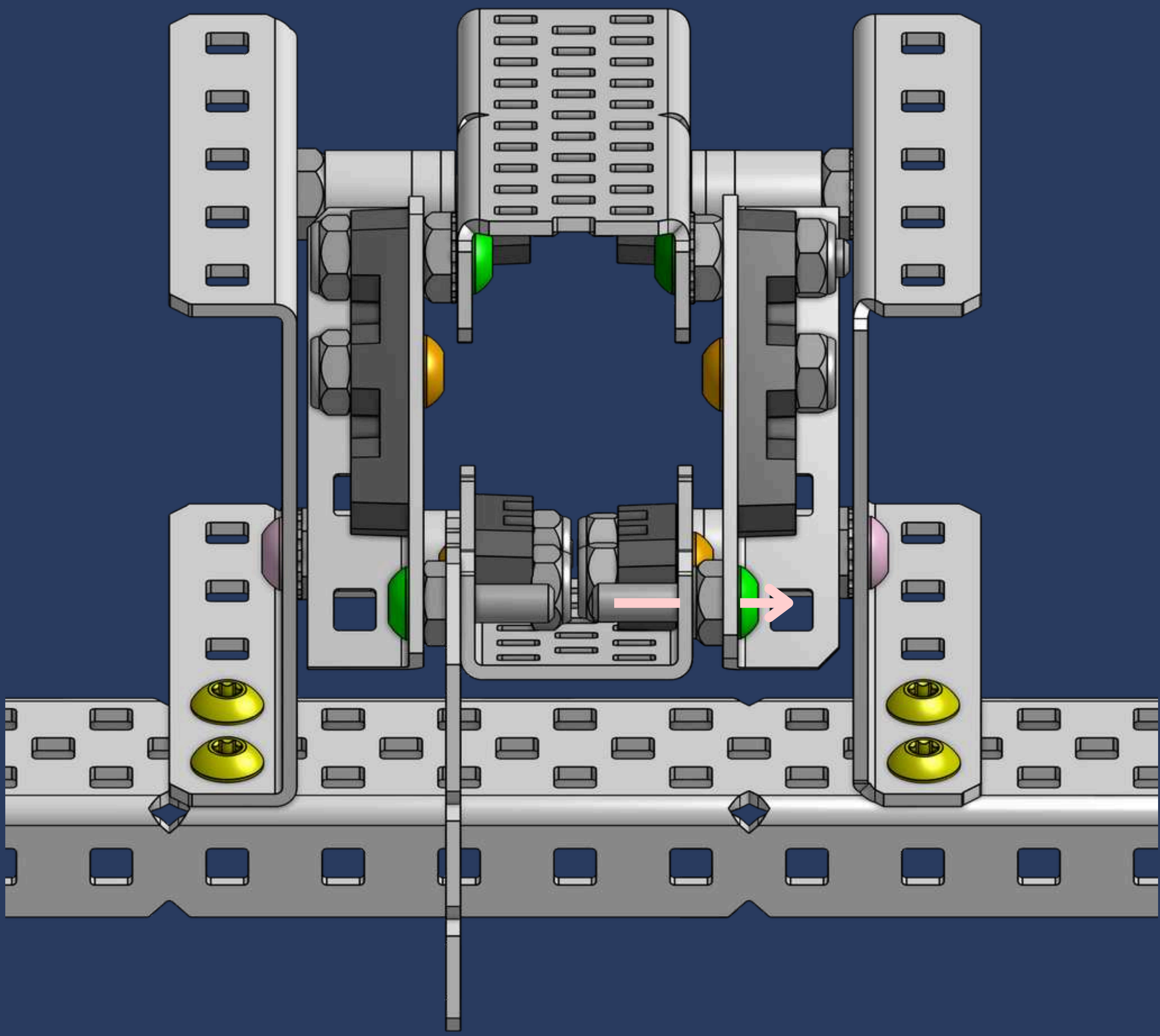


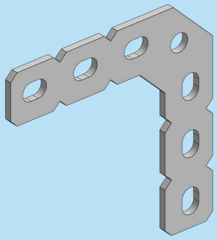
1x
Thin Nylock Nut



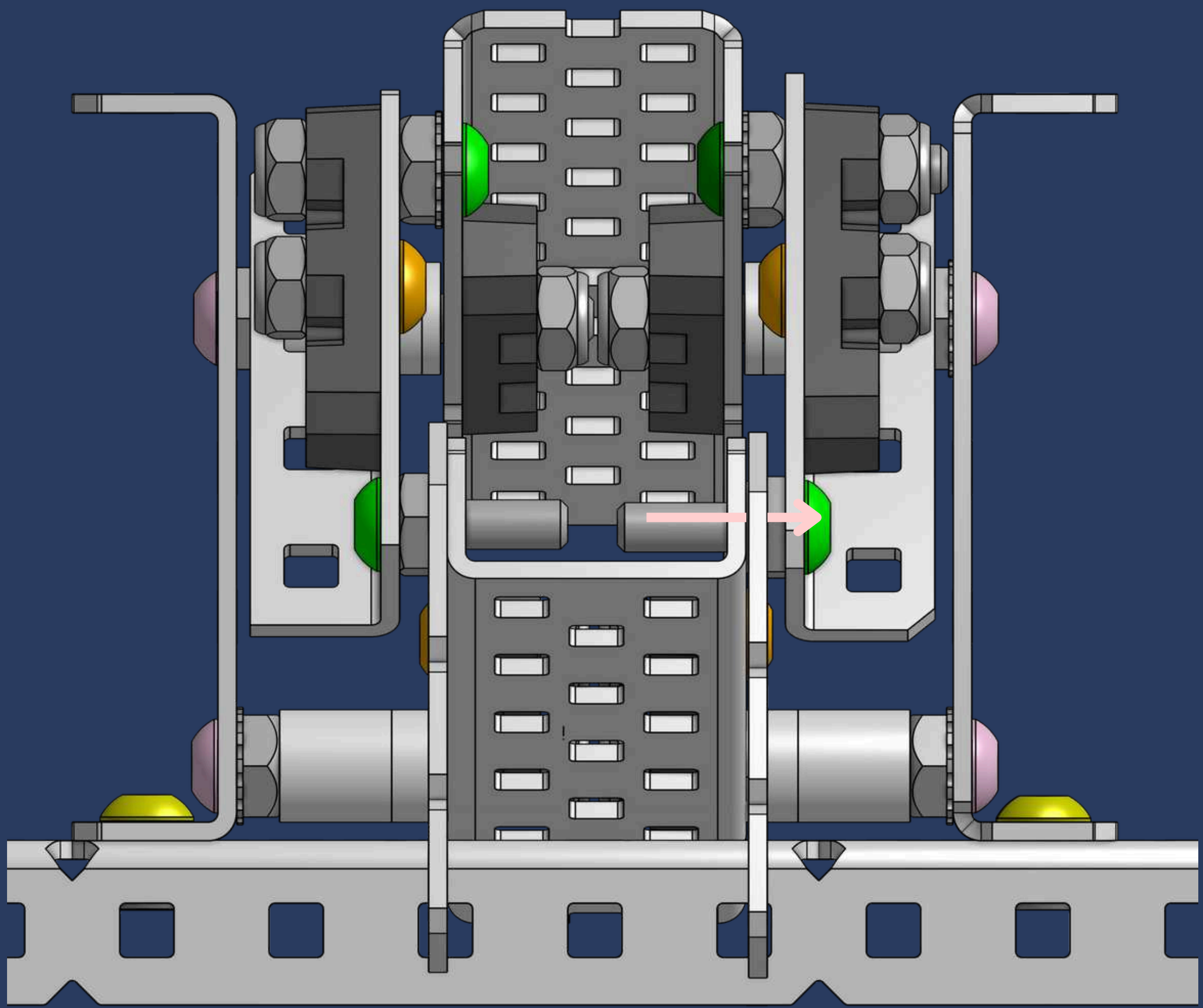
OCT-SOME VIEW

For more **stability**, consider adding **extra screws** onto the gusset and secure it with a nylock at the end!

56



1x
90 Degree Large Flat Gusset



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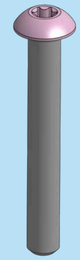
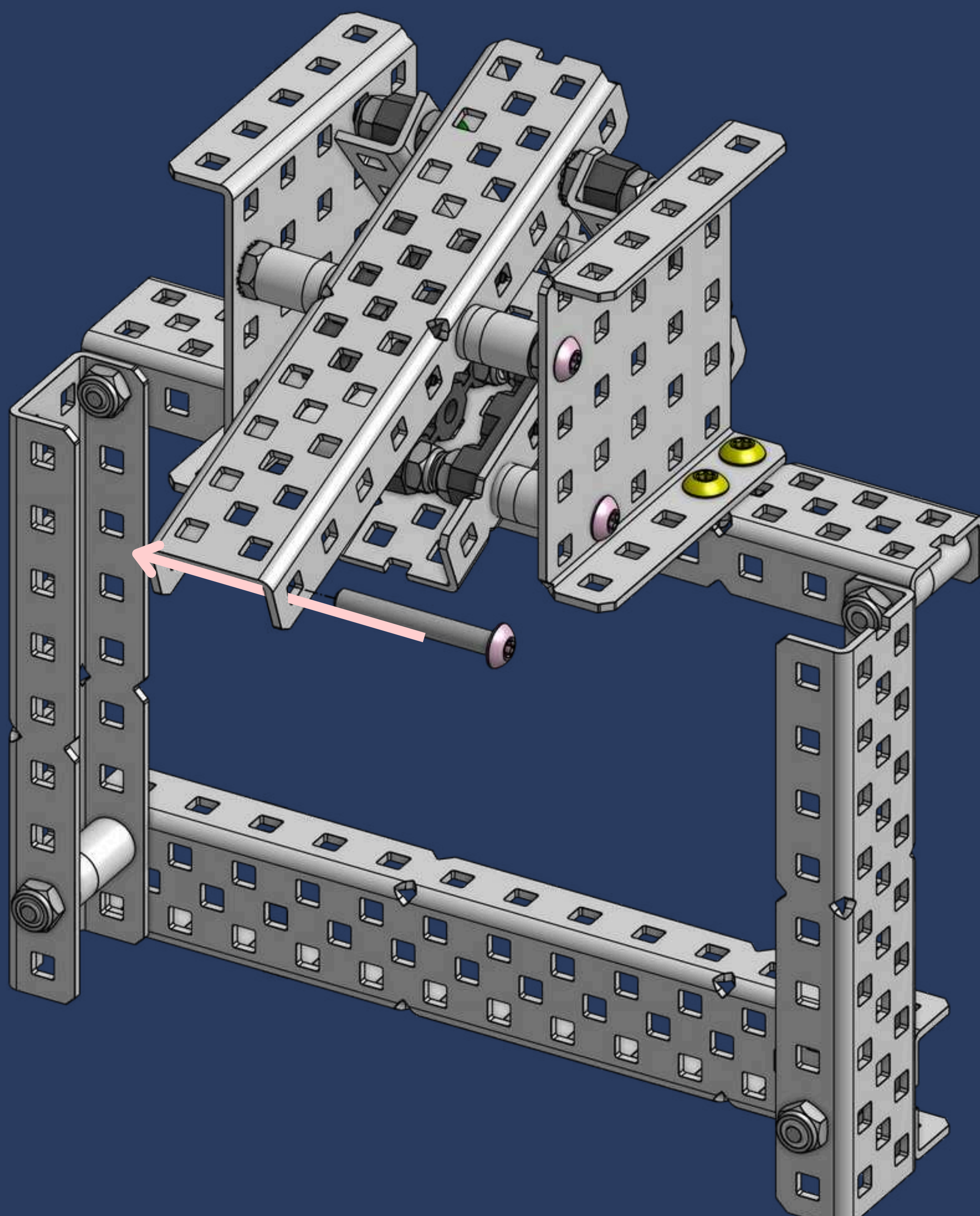


1x
#8-32 x 1-1/84" Star Drive Screw



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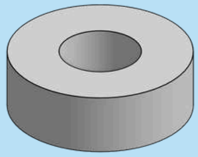
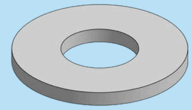
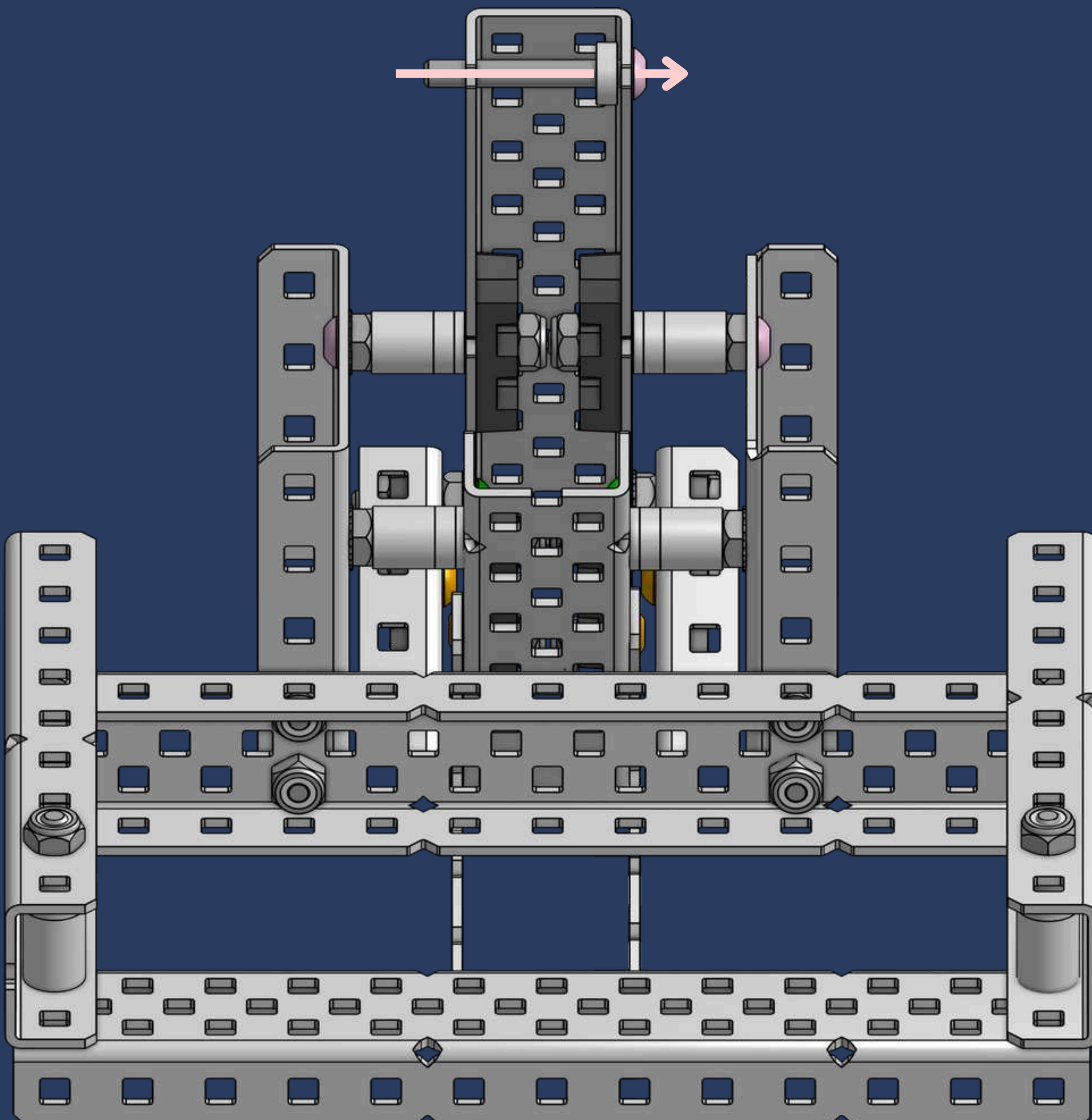


1x
1/8" x 3/8" OD x #8 Nylon Spacer



1x
1/32" x 3/8" OD x #8 Nylon Spacer



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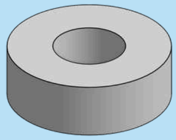
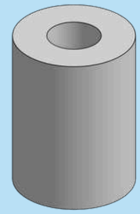
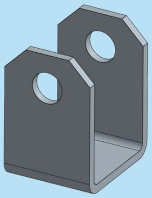


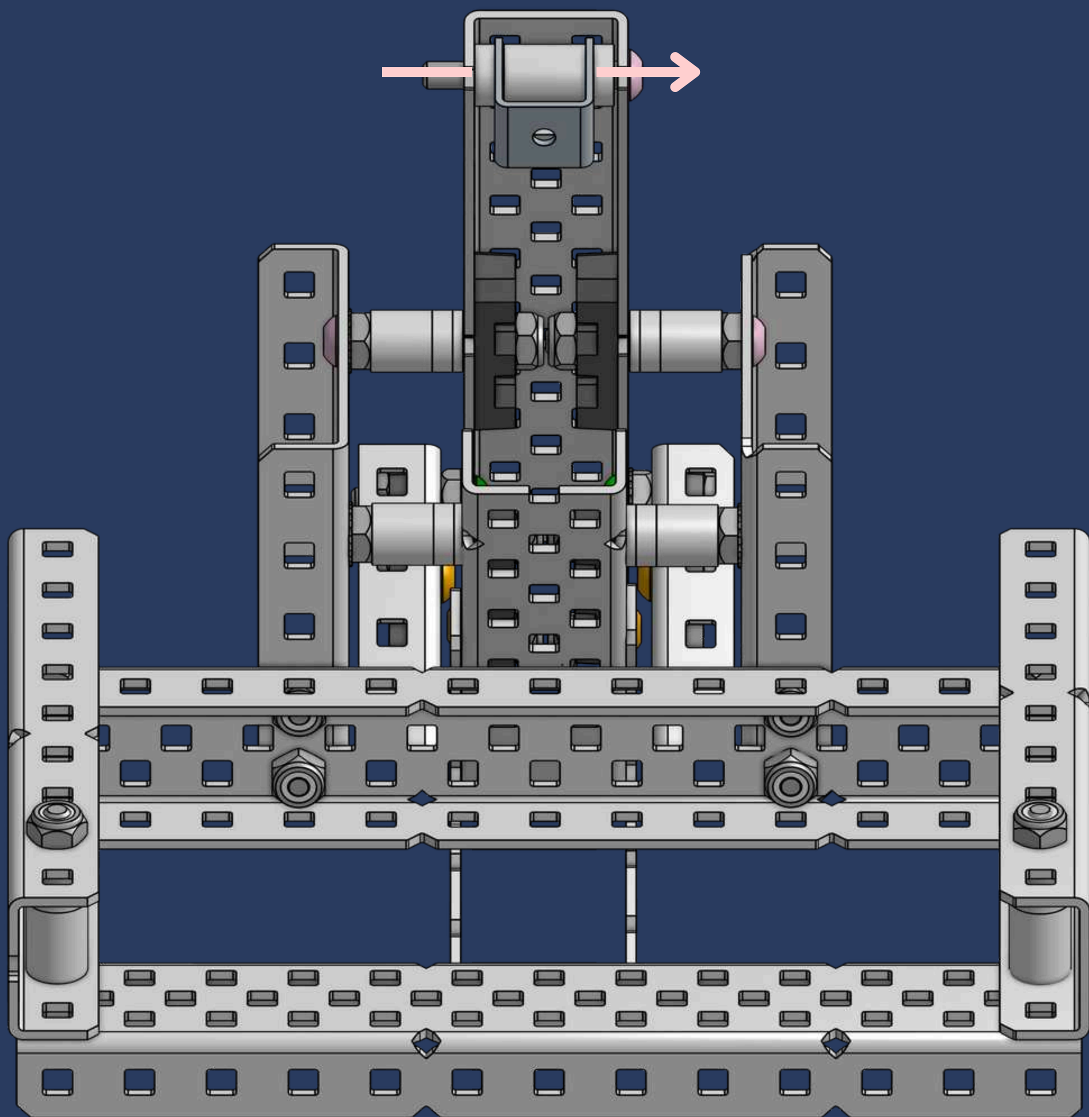
1x
1/8" x 3/8" OD x #8 Nylon Spacer



1x
1/2" x 3/8" OD x #8 Nylon Spacer



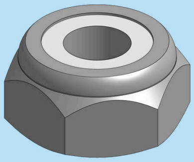
1x
Cylinder Mount



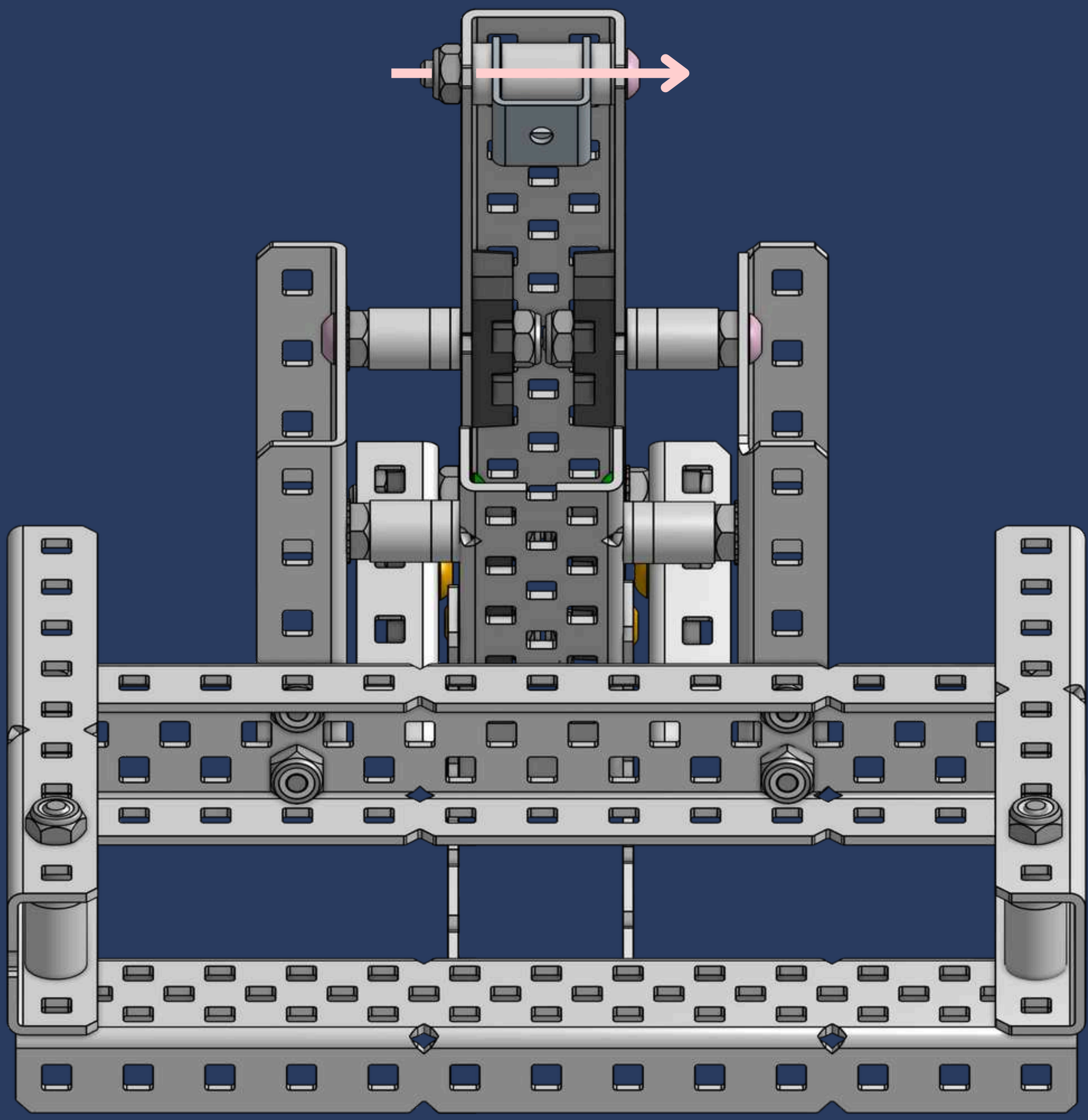
EIGHT-POINT ADVICE

For the **new pneumatic piston** with threaded ends that fit a shaft collar, use a 1/4" spacer + 1/8" spacer, then a shaft collar, followed by a 1/4" spacer again. This method will be shown in the **video**, while the current page method is for **older pistons**.

60



1x
Thin Nylock Nut



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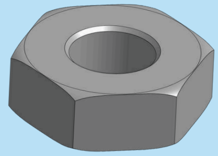

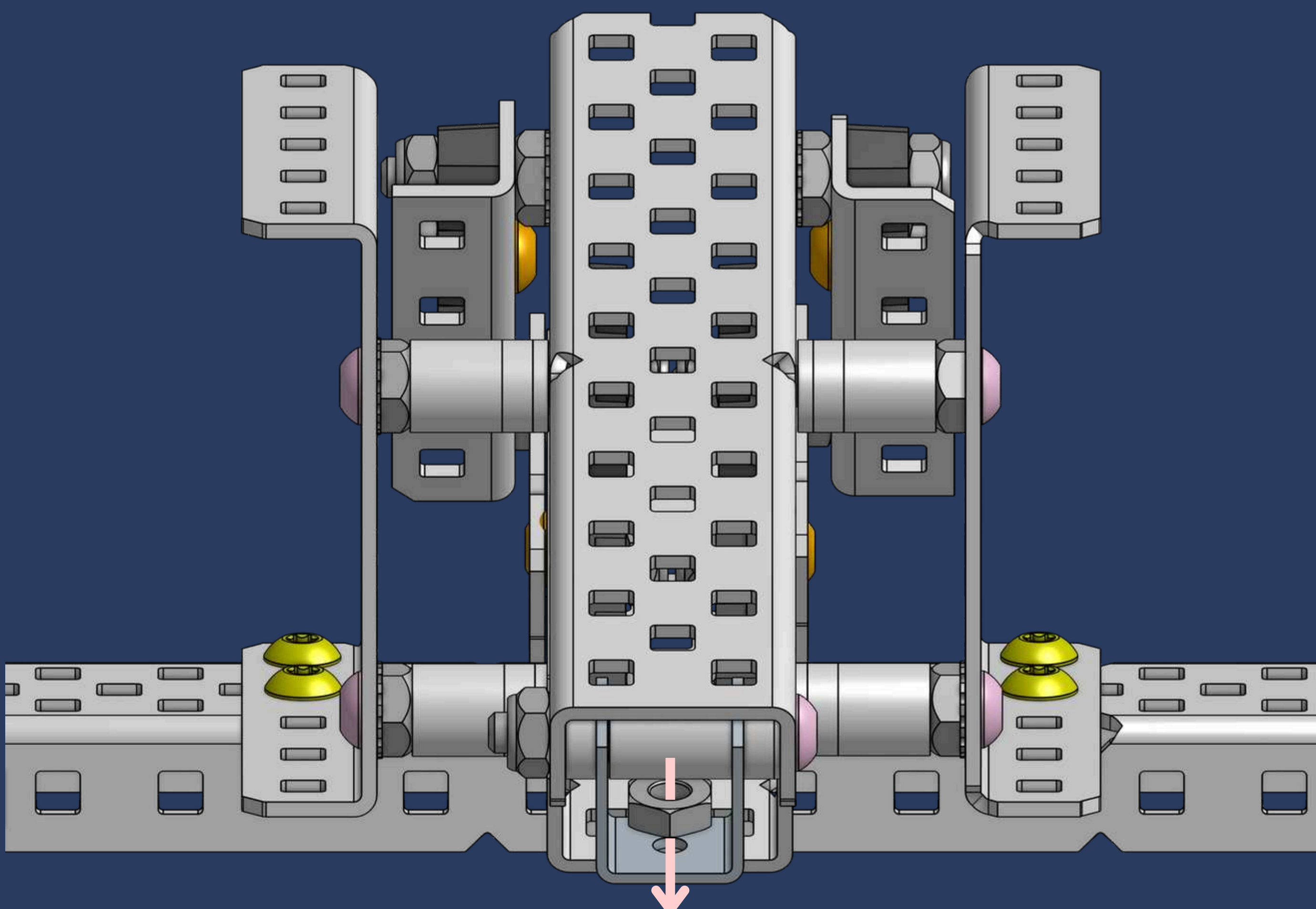

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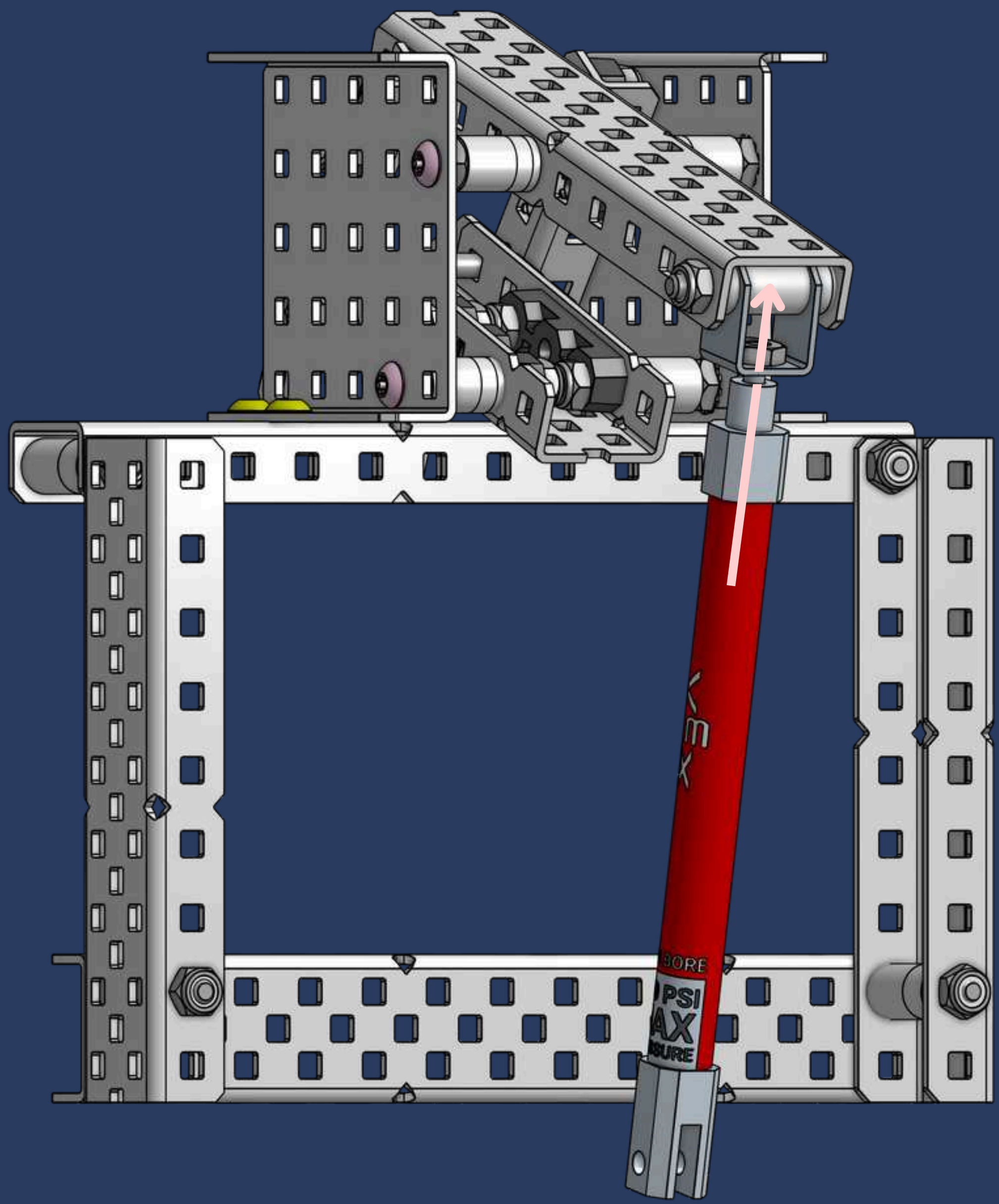
1x
75mm Stroke Pneumatic Cylinder



TENTACLE-TASTIC TIP

For **extra security**, you can add another hex nut on the outside of the cylinder mount. If you don't have a wrench, use **pliers** to tighten it!

VIDEO TUTORIAL

011

For a more in-depth walkthrough, simply click the video link below!

youtube link

THANK YOU!